This file is a merged representation of a subset of the codebase, containing files not matching ignore patterns, combined into a single document by Repomix. The content has been processed where content has been compressed (code blocks are separated by ⋯- delimiter).

File Summary

Purpose

This file contains a packed representation of a subset of the repository's contents that is considered the most important context. It is designed to be easily consumable by AI systems for analysis, code review, or other automated processes.

File Format

The content is organized as follows: 1. This summary section 2. Repository information 3. Directory structure 4. Repository files (if enabled) 5. Multiple file entries, each consisting of: a. A header with the file path (## File: path/to/file) b. The full contents of the file in a code block

Usage Guidelines

- This file should be treated as read-only. Any changes should be made to the original repository files, not this packed version.
- When processing this file, use the file path to distinguish between different files in the repository.
- Be aware that this file may contain sensitive information. Handle it with the same level of security as you would the original repository.

Notes

- Some files may have been excluded based on .gitignore rules and Repomix's configuration
- Binary files are not included in this packed representation. Please refer to the Repository Structure section for a complete list of file paths, including binary files
- Files matching these patterns are excluded: vendor/, parsers/, **/*.svg
- Files matching patterns in .gitignore are excluded
- Files matching default ignore patterns are excluded
- Content has been compressed - code blocks are separated by ⋯- delimiter
- Files are sorted by Git change count (files with more changes are at the bottom)

Directory Structure

```
.iide/
  lsp.json
data/
  icons/
    meson.build
  Adwaita-dark.xml
  Adwaita.xml
  meson.build
  org.github.kai66673.iide.desktop.in
  org.github.kai66673.iide.gschema.xml
  org.github.kai66673.iide.metainfo.xml.in
  org.github.kai66673.iide.service.in
po/
  LINGUAS
  meson.build
  POTFILES.in
scripts/
  install-local-settings.sh.in
src/
  Services/
    Actions/
      Action.vala
      ActionManager.vala
      ShortcutSettings.vala
    LSP/
      IdeLspClient.vala
      IdeLspManager.vala
      IdeLspService.vala
      LspClient.vala
      LspDiagnosticsMark.vala
    TreeSitter/
      cpp/
        CppTreeSitterHighlighter.vala
      model/
        AstItem.vala
        AstModel.vala
      python/
        higlights.scm
        PythonTreeSitterHighlighter.vala
      vala/
        ValaTreeSitterHighlighter.vala
      BaseTreeSitterHighlighter.vala
      TreeSitterManager.vala
    DocumentManager.vala
```

```
    IconProvider.vala
    LoggerService.vala
    PanelLayoutHelper.vala
    ProjectManager.vala
    SettingsService.vala
    Utils.vala
  Widgets/
    TextView/
      FontZoomer.vala
      GutterMarkRenderer.vala
      LspCompletionProvider.vala
      SourceView.vala
      TextView.vala
    ToolViews/
      LogView.vala
      ProjectView.vala
      TerminalView.vala
    FuzzyFinderDialog.vala
    PreferencesDialog.vala
    SearchInFilesDialog.vala
  application.vala
  config.vapi
  iide.gresource.xml
  main.vala
  meson.build
  shortcuts-dialog.ui
  style.css
  window.ui
  window.vala
test_temp/
  test.c
tests/
  test_lsp_service.vala
vapi/
  libtreesitter.vapi
.gitmodules
COPYING
meson.build
org.github.kai66673.iide.json
README.md
repomix-output.pdf
```

Files

File: data/icons/meson.build

```
application_id = 'org.github.kai66673.iide'

scalable_dir = 'hicolor' / 'scalable' / 'apps'
install_data(
  scalable_dir / ('@0@.svg').format(application_id),
  install_dir: get_option('datadir') / 'icons' / scalable_dir
)

symbolic_dir = 'hicolor' / 'symbolic' / 'apps'
install_data(
  symbolic_dir / ('@0@-symbolic.svg').format(application_id),
  install_dir: get_option('datadir') / 'icons' / symbolic_dir
)
```

File: data/meson.build

```
desktop_file = i18n.merge_file(
        input: 'org.github.kai66673.iide.desktop.in',
       output: 'org.github.kai66673.iide.desktop',
         type: 'desktop',
       po_dir: '../po',
      install: true,
  install_dir: get_option('datadir') / 'applications'
)

desktop_utils = find_program('desktop-file-validate', required: false)
if desktop_utils.found()
  test('Validate desktop file', desktop_utils, args: [desktop_file])
endif

appstream_file = i18n.merge_file(
        input: 'org.github.kai66673.iide.metainfo.xml.in',
       output: 'org.github.kai66673.iide.metainfo.xml',
       po_dir: '../po',
      install: true,
  install_dir: get_option('datadir') / 'metainfo'
)

appstreamcli = find_program('appstreamcli', required: false, disabler: true)
test('Validate appstream file', appstreamcli,
     args: ['validate', '--no-net', '--explain', appstream_file])
```

```
install_data('org.github.kai66673.iide.gschema.xml',
  install_dir: get_option('datadir') / 'glib-2.0' / 'schemas'
)

compile_schemas = find_program('glib-compile-schemas', required: false, disabler: true)
test('Validate schema file',
     compile_schemas,
     args: ['--strict', '--dry-run', meson.current_source_dir()])


service_conf = configuration_data()
service_conf.set('bindir', get_option('prefix') / get_option('bindir'))
configure_file(
  input: 'org.github.kai66673.iide.service.in',
  output: 'org.github.kai66673.iide.service',
  configuration: service_conf,
  install_dir: get_option('datadir') / 'dbus-1' / 'services'
)

subdir('icons')
```

File: data/org.github.kai66673.iide.desktop.in

```
[Desktop Entry]
Name=iide
Exec=iide
Icon=org.github.kai66673.iide
Terminal=false
Type=Application
Categories=Utility;
Keywords=GTK;
StartupNotify=true
DBusActivatable=true
```

File: data/org.github.kai66673.iide.metainfo.xml.in

```
<?xml version="1.0" encoding="UTF-8"?>
<component type="desktop-application">
  <id>org.github.kai66673.iide</id>
  <metadata_license>CC0-1.0</metadata_license>
  <project_license>GPL-3.0-or-later</project_license>

  <name>Iide</name>
  <summary>Keep the summary shorter, between 10 and 35 characters</summary>
  <description>
    <p>No description</p>
```

```
</description>

<developer id="tld.vendor">
  <name>Developer name</name>
</developer>

<!-- Required: Should be a link to the upstream homepage for the component -->
<url type="homepage">https://example.org/</url>
<!-- Recommended: It is highly recommended for open-source projects to display the source
<url type="vcs-browser">https://example.org/repository</url>
<!-- Should point to the software's bug tracking system, for users to report new bugs -->
<url type="bugtracker">https://example.org/issues</url>
<!-- Should link a FAQ page for this software, to answer some of the most-asked questions
<!-- URLs of this type should point to a webpage where users can submit or modify translat
<url type="translate">https://example.org/translate</url>
<url type="faq">https://example.org/faq</url>
<!-- Should provide a web link to an online user's reference, a software manual or help pa
<url type="help">https://example.org/help</url>
<!-- URLs of this type should point to a webpage showing information on how to donate to t
<url type="donation">https://example.org/donate</url>
<!-- This could for example be an HTTPS URL to an online form or a page describing how to
<url type="contact">https://example.org/contact</url>
<!-- URLs of this type should point to a webpage showing information on how to contribute
<url type="contribute">https://example.org/contribute</url>

<translation type="gettext">iide</translation>
<!-- All graphical applications having a desktop file must have this tag in the MetaInfo.
   If this is present, appstreamcli compose will pull icons, keywords and categories from
<launchable type="desktop-id">org.github.kai66673.iide.desktop</launchable>
<!-- Use the OARS website (https://hughsie.github.io/oars/generate.html) to generate these
<content_rating type="oars-1.1" />

<!-- Applications should set a brand color in both light and dark variants like so -->
<branding>
  <color type="primary" scheme_preference="light">#ff00ff</color>
  <color type="primary" scheme_preference="dark">#993d3d</color>
</branding>

<screenshots>
  <screenshot type="default">
    <image>https://example.org/example1.png</image>
    <caption>A caption</caption>
  </screenshot>
  <screenshot>
    <image>https://example.org/example2.png</image>
    <caption>A caption</caption>
```

```
    </screenshot>
  </screenshots>

  <releases>
    <release version="1.0.1" date="2024-01-18">
      <url type="details">https://example.org/changelog.html#version_1.0.1</url>
      <description translate="no">
        <p>Release description</p>
        <ul>
          <li>List of changes</li>
          <li>List of changes</li>
        </ul>
      </description>
    </release>
  </releases>

</component>
```

File: data/org.github.kai66673.iide.service.in

```
[D-BUS Service]
Name=org.github.kai66673.iide
Exec=@bindir@/iide --gapplication-service
```

File: po/LINGUAS

```
# Please keep this file sorted alphabetically.
```

File: po/meson.build

```
i18n.gettext('iide', preset: 'glib')
```

File: po/POTFILES.in

```
# List of source files containing translatable strings.
# Please keep this file sorted alphabetically.
data/org.github.kai66673.iide.desktop.in
data/org.github.kai66673.iide.metainfo.xml.in
data/org.github.kai66673.iide.gschema.xml
src/main.vala
src/window.vala
src/window.ui
```

File: src/config.vapi

```
[CCode (cprefix = "", lower_case_cprefix = "", cheader_filename = "config.h")]
```

```
namespace Config {
    public const string GETTEXT_PACKAGE;
    public const string LOCALEDIR;
    public const string PACKAGE_VERSION;
}
```

File: src/main.vala

```
/* main.vala
 *
 * Copyright 2026 kai
 *
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 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation, either version 3 of the License, or
 * (at your option) any later version.
 *
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 * along with this program.  If not, see <https://www.gnu.org/licenses/>.
 *
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 */

int main (string[] args) {
    Intl.bindtextdomain (Config.GETTEXT_PACKAGE, Config.LOCALEDIR);
    Intl.bind_textdomain_codeset (Config.GETTEXT_PACKAGE, "UTF-8");
    Intl.textdomain (Config.GETTEXT_PACKAGE);

    // GtkSource.init ();

    var app = new Iide.Application ();
    return app.run (args);
}
```

File: src/shortcuts-dialog.ui

```
<?xml version="1.0" encoding="UTF-8"?>
<interface>
  <object class="AdwShortcutsDialog" id="shortcuts_dialog">
    <child>
      <object class="AdwShortcutsSection">
```

```
        <property name="title" translatable="yes">Shortcuts</property>
        <child>
          <object class="AdwShortcutsItem">
            <property name="title" translatable="yes" context="shortcut window">Show Shortcu
            <property name="action-name">app.shortcuts</property>
          </object>
        </child>
        <child>
          <object class="AdwShortcutsItem">
            <property name="title" translatable="yes" context="shortcut window">Quit</proper
            <property name="action-name">app.quit</property>
          </object>
        </child>
      </object>
    </child>
  </object>
</interface>
```

File: src/window.ui

```
<?xml version="1.0" encoding="UTF-8" ?>
<interface>
  <template class="IideWindow" parent="PanelDocumentWorkspace">
    <child type="titlebar">
      <object class="AdwHeaderBar">
        <child type="title">
          <object class="AdwWindowTitle" id="window_title">
            <property name="title">IIDe Application</property>
          </object>
        </child>
        <child type="start">
          <object class="PanelToggleButton">
            <property name="area">start</property>
            <property name="dock">dock</property>
          </object>
        </child>
        <child type="start">
          <object class="GtkButton">
            <property name="icon-name">list-add-symbolic</property>
            <property name="action-name">workspace.add-page</property>
            <property name="tooltip-text">Add Page</property>
          </object>
        </child>
        <child type="end">
          <object class="PanelToggleButton">
            <property name="area">end</property>
```

```
        <property name="dock">dock</property>
      </object>
    </child>
    <child type="end">
      <object class="GtkMenuButton">
        <property name="icon-name">open-menu-symbolic</property>
        <property name="menu-model">primary_menu</property>
        <property name="tooltip-text">Primary Menu</property>
      </object>
    </child>
  </object>
</child>
<child internal-child="dock">
  <object class="PanelDock" id="dock">
  </object>
</child>
<child internal-child="start_area">
  <object class="PanelPaned" id="start_area">
    <child>
      <object class="PanelFrame">
        <child>
          <object class="PanelWidget">
            <property name="title">XXX</property>
            <property name="icon-name">folder-documents-symbolic</property>
            <property name="vexpand">true</property>
            <child>
              <object class="GtkScrolledWindow">
                <property name="hscrollbar-policy">never</property>
                <child>
                  <object class="GtkListView">
                  </object>
                </child>
              </object>
            </child>
          </object>
        </child>
      </object>
    </child>
  </object>
  <object class="PanelPaned">
    <child>
      <object class="PanelFrame">
        <child>
          <object class="PanelWidget">
            <property name="title">ZZZ ZZZ</property>
            <property name="icon-name">folder-symbolic</property>
```

```
            <property name="vexpand">true</property>
            <child>
              <object class="GtkScrolledWindow">
                <property name="hscrollbar-policy">never</property>
                <child>
                  <object class="GtkListView">
                  </object>
                </child>
              </object>
            </child>
          </object>
        </child>
      </object>
    </child>
  </object>
</child>
<child internal-child="bottom_area">
  <object class="PanelPaned">
    <child>
      <object class="PanelFrame">
        <child>
          <object class="PanelWidget">
            <property name="title">123</property>
            <property name="icon-name">folder-symbolic</property>
            <property name="vexpand">true</property>
            <child>
              <object class="GtkScrolledWindow">
                <property name="hscrollbar-policy">never</property>
                <child>
                  <object class="GtkListView">
                  </object>
                </child>
              </object>
            </child>
          </object>
        </child>
      </object>
    </child>
  </object>
  <object class="PanelPaned">
    <child>
      <object class="PanelFrame">
        <child>
          <object class="PanelWidget">
            <property name="title">OOP</property>
            <property name="icon-name">folder-symbolic</property>
```

```
                <property name="vexpand">true</property>
                <child>
                  <object class="GtkScrolledWindow">
                    <property name="hscrollbar-policy">never</property>
                    <child>
                      <object class="GtkListView">
                      </object>
                    </child>
                  </object>
                </child>
              </object>
            </child>
          </object>
        </child>
      </object>
    </child>
    <child internal-child="end_area">
      <object class="PanelPaned" id="end_area">
      </object>
    </child>
    <child internal-child="statusbar">
      <object class="PanelStatusbar">
        <child type="suffix">
          <object class="PanelToggleButton">
            <property name="area">bottom</property>
            <property name="dock">dock</property>
          </object>
        </child>
      </object>
      <object class="PanelStatusbar">
        <child type="suffix">
          <object class="PanelToggleButton">
            <property name="area">top</property>
            <property name="dock">dock</property>
          </object>
        </child>
      </object>
    </child>
  </template>
  <menu id="primary_menu">
    <section>
      <item>
        <attribute name="label">Quit</attribute>
        <attribute name="action">app.quit</attribute>
      </item>
    </section>
```

```
  </menu>
</interface>
```

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```
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                       Version 3, 29 June 2007

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```

```
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```

File: org.github.kai66673.iide.json

```
{
    "id" : "org.github.kai66673.iide",
    "runtime" : "org.gnome.Platform",
    "runtime-version" : "master",
    "sdk" : "org.gnome.Sdk",
    "sdk-extensions" : [
        "org.freedesktop.Sdk.Extension.vala"
    ],
```

```
    "command" : "iide",
    "finish-args" : [
        "--share=network",
        "--share=ipc",
        "--socket=fallback-x11",
        "--device=dri",
        "--socket=wayland"
    ],
    "build-options" : {
        "append-path" : "/usr/lib/sdk/vala/bin",
        "prepend-ld-library-path" : "/usr/lib/sdk/vala/lib"
    },
    "cleanup" : [
        "/include",
        "/lib/pkgconfig",
        "/man",
        "/share/doc",
        "/share/gtk-doc",
        "/share/man",
        "/share/pkgconfig",
        "/share/vala",
        "*.la",
        "*.a"
    ],
    "modules" : [
        {
            "name" : "iide",
            "builddir" : true,
            "buildsystem" : "meson",
            "sources" : [
                {
                    "type" : "git",
                    "url" : "file:///home/kai/Projects"
                }
            ]
        }
    ]
}
```

File: README.md

```
# iide

A description of this project.
```

File: .iide/lsp.json

```
{"languages": [{"id": "cpp", "server": ["clangd", "--header-insertion=never"], "patterns": [
```

File: data/Adwaita-dark.xml

```
<?xml version="1.0" encoding="UTF-8" ?>
<!--

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-->
<style-scheme id="Adwaita-dark" _name="Adwaita Dark" version="1.0"><author
    >Christian Hergert</author><_description
    >An style scheme for Adwaita</_description><metadata><property
            name="variant"
        >dark</property><property
            name="light-variant"
        >Adwaita</property></metadata><!-- Named Colors --><color
        name="blue_1"
        value="#99C1F1"
    /><color name="blue_2" value="#62A0EA" /><color
        name="blue_3"
        value="#3584E4"
    /><color name="blue_4" value="#1C71D8" /><color
        name="blue_5"
        value="#1A5FB4"
    /><color name="blue_6" value="#1B497E" /><color
        name="blue_7"
        value="#193D66"
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        name="brown_2"
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    value="#4E9A06"
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    name="dark_2"
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    name="dark_6"
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/><color name="dark_7" value="#000000" /><color
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/><color name="light_3" value="#F6F5F4" /><color
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    name="light_6"
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/><color name="light_7" value="#9A9996" /><color
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/><color name="orange_2" value="#FFA348" /><color
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    value="#FF7800"
/><color name="orange_4" value="#E66100" /><color
```

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    value="#D38B09"
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    background="#141414"
/><style name="bracket-match" bold="true" /><style
    name="current-line"
    background="libadwaita-dark-alt"
/><style
    name="current-line-number"
```

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    foreground="dark_1"
/><style name="cursor" foreground="light_5" /><style
    name="draw-spaces"
    foreground="dark_3"
/><style
    name="line-numbers"
    background="libadwaita-dark"
    foreground="dark_2"
/><style name="map-overlay" background="dark_1" /><style
    name="right-margin"
    background="dark_1"
    foreground="dark_1"
/><style
    name="search-match"
    background="#rgba(246,211,45,.5)"
    foreground="dark_5"
/><style
    name="text"
    background="libadwaita-dark"
    foreground="light_5"
/><!-- Defaults --><style
    name="def:base-n-integer"
    foreground="violet_2"
/><style name="def:boolean" foreground="violet_2" /><style
    name="def:comment"
    foreground="dark_1"
/><style name="def:constant" foreground="violet_2" /><style
    name="def:decimal"
    foreground="violet_2"
/><style name="def:deletion" strikethrough="true" /><style
    name="def:doc-comment-element"
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    underline="error"
    underline-color="red_4"
/><style name="def:floating-point" foreground="violet_2" /><style
    name="def:function"
    foreground="blue_2"
/><style name="def:heading" foreground="teal_3" bold="true" /><style
    name="def:identifier"
    foreground="chameleon_3"
/><style name="def:inline-code" foreground="violet_2" /><style
    name="def:link-destination"
    foreground="blue_2"
```

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    underline="low"
/><style name="def:link-text" foreground="red_2" /><style
    name="def:list-marker"
    foreground="orange_4"
    bold="true"
/><style name="def:net-address" foreground="blue_2" underline="low" /><style
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    foreground="dark_4"
    background="yellow_4"
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    foreground="light_7"
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/><style name="def:special-char" foreground="red_1" bold="false" /><style
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    bold="true"
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    bold="true"
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    name="def:underlined"
    underline="single"
/><style
    name="def:warning"
    underline="error"
    underline-color="yellow_4"
/><!-- C# --><style name="c-sharp:format" foreground="violet_4" /><style
    name="c-sharp:preprocessor"
    foreground="dark_2"
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    name="c:signal-name"
    foreground="red_1"
/><style name="c:storage-class" foreground="teal_2" bold="true" /><style
    name="c:type-keyword"
    foreground="teal_2"
    bold="true"
/><!-- CSS --><style
    name="css:id-selector"
    foreground="teal_3"
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```

```
/><style name="css:property-name" foreground="orange_3" /><style
    name="css:pseudo-selector"
    foreground="violet_2"
    bold="true"
/><style
    name="css:selector-symbol"
    foreground="orange_3"
    bold="true"
/><style name="css:type-selector" foreground="teal_3" bold="true" /><style
    name="css:vendor-specific"
    foreground="yellow_5"
/><!-- Diff --><style name="diff:added-line" foreground="teal_3" /><style
    name="diff:changed-line"
    foreground="orange_3"
/><style name="diff:diff-file" foreground="violet_2" /><style
    name="diff:location"
    foreground="yellow_4"
/><style name="diff:removed-line" foreground="red_1" /><!-- Go --><style
    name="go:printf"
    foreground="violet_4"
/><!-- Python 2 --><style
    name="python:builtin-function"
    foreground="blue_2"
/><style name="python:class-name" foreground="teal_2" bold="true" /><style
    name="python:module-handler"
    foreground="red_1"
/><!-- Rust --><style name="rust:attribute" foreground="violet_2" /><style
    name="rust:lifetime"
    foreground="orange_2"
    bold="false"
    italic="false"
/><style name="rust:macro" foreground="violet_2" bold="false" /><style
    name="rust:scope"
    foreground="orange_2"
/><!-- Vala --><style
    name="vala:attributes"
    foreground="light_5"
    bold="false"
/><!-- XML --><style
    name="xml:attribute-name"
    foreground="orange_3"
/><style name="xml:attribute-value" foreground="violet_2" /><style
    name="xml:element-name"
    foreground="teal_3"
/><style name="xml:namespace" foreground="yellow_4" /><style
    name="xml:processing-instruction"
```

```
        foreground="yellow_4"
        bold="true"
    /></style-scheme>
```

File: data/Adwaita.xml

```
<?xml version="1.0" encoding="UTF-8" ?>
<!--

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-->
<style-scheme id="Adwaita" _name="Adwaita" version="1.0"><author
    >Christian Hergert</author><_description
    >An style scheme for Adwaita</_description><metadata><property
            name="variant"
        >light</property><property
            name="dark-variant"
        >Adwaita-dark</property></metadata><!-- Named Colors --><color
        name="blue_1"
        value="#99C1F1"
    /><color name="blue_2" value="#62A0EA" /><color
        name="blue_3"
        value="#3584E4"
    /><color name="blue_4" value="#1C71D8" /><color
        name="blue_5"
        value="#1A5FB4"
    /><color name="blue_6" value="#1B497E" /><color
        name="blue_7"
        value="#193D66"
    /><color name="brown_1" value="#CDAB8F" /><color
        name="brown_2"
        value="#B5835A"
```

```
/><color name="brown_3" value="#986A44" /><color
    name="brown_4"
    value="#865E3C"
/><color name="brown_5" value="#63452C" /><color
    name="chameleon_3"
    value="#4E9A06"
/><color name="dark_1" value="#77767B" /><color
    name="dark_2"
    value="#5E5C64"
/><color name="dark_3" value="#504E55" /><color
    name="dark_4"
    value="#3D3846"
/><color name="dark_5" value="#241F31" /><color
    name="dark_6"
    value="#000000"
/><color name="green_1" value="#8FF0A4" /><color
    name="green_2"
    value="#57E389"
/><color name="green_3" value="#33D17A" /><color
    name="green_4"
    value="#2EC27E"
/><color name="green_5" value="#26A269" /><color
    name="green_6"
    value="#1F7F56"
/><color name="green_7" value="#1C6849" /><color
    name="light_1"
    value="#FFFFFF"
/><color name="light_2" value="#FCFCFC" /><color
    name="light_3"
    value="#F6F5F4"
/><color name="light_4" value="#DEDDDA" /><color
    name="light_5"
    value="#C0BFBC"
/><color name="light_6" value="#B0AFAC" /><color
    name="light_7"
    value="#9A9996"
/><color name="orange_1" value="#FFBE6F" /><color
    name="orange_2"
    value="#FFA348"
/><color name="orange_3" value="#FF7800" /><color
    name="orange_4"
    value="#E66100"
/><color name="orange_5" value="#C64600" /><color
    name="purple_1"
    value="#DC8ADD"
/><color name="purple_2" value="#C061CB" /><color
```

```
    name="purple_3"
    value="#9141AC"
/><color name="purple_4" value="#813D9C" /><color
    name="purple_5"
    value="#613583"
/><color name="red_1" value="#F66151" /><color
    name="red_2"
    value="#ED333B"
/><color name="red_3" value="#E01B24" /><color
    name="red_4"
    value="#C01C28"
/><color name="red_5" value="#A51D2D" /><color
    name="teal_1"
    value="#93DDC2"
/><color name="teal_2" value="#5BC8AF" /><color
    name="teal_3"
    value="#33B2A4"
/><color name="teal_4" value="#26A1A2" /><color
    name="teal_5"
    value="#218787"
/><color name="violet_2" value="#7D8AC7" /><color
    name="violet_3"
    value="#6362C8"
/><color name="violet_4" value="#4E57BA" /><color
    name="yellow_1"
    value="#F9F06B"
/><color name="yellow_2" value="#F8E45C" /><color
    name="yellow_3"
    value="#F6D32D"
/><color name="yellow_4" value="#F5C211" /><color
    name="yellow_5"
    value="#E5A50A"
/><color name="yellow_6" value="#D38B09" /><!-- Global Styles --><style
    name="background-pattern"
    background="#FAFAFA"
/><style name="bracket-match" bold="true" /><style
    name="current-line"
    background="light_3"
/><style
    name="current-line-number"
    background="light_3"
    foreground="light_7"
/><style name="cursor" foreground="dark_1" /><style
    name="draw-spaces"
    foreground="light_6"
/><style
```

```
    name="line-numbers"
    background="light_1"
    foreground="light_6"
/><style name="map-overlay" background="dark_1" /><style
    name="right-margin"
    background="dark_1"
    foreground="dark_1"
/><style
    name="search-match"
    background="yellow_2"
    foreground="dark_4"
/><style
    name="text"
    background="light_1"
    foreground="dark_3"
/><!-- Defaults --><style
    name="def:base-n-integer"
    foreground="violet_4"
/><style name="def:boolean" foreground="violet_4" /><style
    name="def:comment"
    foreground="dark_1"
/><style name="def:constant" foreground="violet_4" /><style
    name="def:decimal"
    foreground="violet_4"
/><style name="def:deletion" strikethrough="true" /><style
    name="def:doc-comment-element"
    foreground="dark_3"
/><style name="def:emphasis" italic="true" /><style
    name="def:error"
    underline="error"
    underline-color="red_4"
/><style name="def:floating-point" foreground="violet_4" /><style
    name="def:function"
    foreground="blue_4"
/><style name="def:heading" foreground="teal_5" bold="true" /><style
    name="def:identifier"
    foreground="chameleon_3"
/><style name="def:inline-code" foreground="violet_4" /><style
    name="def:link-destination"
    foreground="blue_3"
    italic="true"
    underline="low"
/><style name="def:link-text" foreground="red_3" /><style
    name="def:list-marker"
    foreground="orange_5"
    bold="true"
```

```
/><style name="def:net-address" foreground="blue_3" underline="low" /><style
    name="def:note"
    foreground="dark_4"
    background="#FCF7B5"
    bold="true"
/><style name="def:number" foreground="violet_4" /><style
    name="def:preformatted-section"
    foreground="violet_4"
/><style name="def:preprocessor" foreground="orange_5" /><style
    name="def:shebang"
    foreground="dark_1"
    bold="true"
/><style name="def:special-char" foreground="red_2" bold="false" /><style
    name="def:statement"
    foreground="orange_5"
    bold="true"
/><style name="def:string" foreground="teal_5" /><style
    name="def:strong-emphasis"
    bold="true"
/><style name="def:type" foreground="teal_5" bold="true" /><style
    name="def:underlined"
    underline="single"
/><style
    name="def:warning"
    underline="error"
    underline-color="yellow_4"
/><!-- C# --><style name="c-sharp:format" foreground="violet_4" /><style
    name="c-sharp:preprocessor"
    foreground="dark_2"
/><!-- C --><style name="c:printf" foreground="violet_4" /><style
    name="c:signal-name"
    foreground="red_4"
/><style name="c:storage-class" foreground="teal_5" bold="true" /><style
    name="c:type-keyword"
    foreground="teal_5"
    bold="true"
/><!-- CSS --><style
    name="css:id-selector"
    foreground="teal_5"
    bold="true"
/><style name="css:property-name" foreground="orange_5" /><style
    name="css:pseudo-selector"
    foreground="violet_4"
    bold="true"
/><style
    name="css:selector-symbol"
```

```
    foreground="orange_5"
    bold="true"
/><style name="css:type-selector" foreground="teal_5" bold="true" /><style
    name="css:vendor-specific"
    foreground="yellow_6"
/><!-- Diff --><style name="diff:added-line" foreground="teal_4" /><style
    name="diff:changed-line"
    foreground="orange_4"
/><style name="diff:diff-file" foreground="violet_4" /><style
    name="diff:location"
    foreground="yellow_6"
/><style name="diff:removed-line" foreground="red_1" /><!-- Go --><style
    name="go:printf"
    foreground="violet_4"
/><!-- Python 2 --><style
    name="python:builtin-function"
    foreground="blue_4"
/><style name="python:class-name" foreground="teal_5" bold="true" /><style
    name="python:module-handler"
    foreground="red_3"
/><!-- Rust --><style name="rust:attribute" foreground="violet_4" /><style
    name="rust:lifetime"
    foreground="orange_5"
    bold="false"
    italic="false"
/><style name="rust:macro" foreground="violet_4" bold="false" /><style
    name="rust:scope"
    foreground="orange_5"
/><!-- Vala --><style
    name="vala:attributes"
    foreground="dark_2"
    bold="false"
/><!-- XML --><style
    name="xml:attribute-name"
    foreground="orange_5"
/><style name="xml:attribute-value" foreground="violet_4" /><style
    name="xml:element-name"
    foreground="teal_5"
/><style name="xml:namespace" foreground="yellow_6" /><style
    name="xml:processing-instruction"
    foreground="yellow_6"
    bold="true"
/></style-scheme>
```

File: scripts/install-local-settings.sh.in

```
#!/bin/bash
set -e

PREFIX="@PREFIX@"
LOCAL_SCHEMA_DIR="$HOME/.local/share/glib-2.0/schemas"
LOCAL_ICON_DIR="$HOME/.local/share/icons/hicolor"
CURRENT_DIR="$(cd "$(dirname "$0")" && pwd)"

install_schemas() {
    mkdir -p "$LOCAL_SCHEMA_DIR"
    if [ -f "@PREFIX@/share/glib-2.0/schemas/org.github.kai66673.iide.gschema.xml" ]; then
        cp "@PREFIX@/share/glib-2.0/schemas/org.github.kai66673.iide.gschema.xml" "$LOCAL_SC
    elif [ -f "$CURRENT_DIR/data/org.github.kai66673.iide.gschema.xml" ]; then
        cp "$CURRENT_DIR/data/org.github.kai66673.iide.gschema.xml" "$LOCAL_SCHEMA_DIR/"
    fi
    glib-compile-schemas "$LOCAL_SCHEMA_DIR"
    echo "GSettings schemas installed to $LOCAL_SCHEMA_DIR"
}

install_icons() {
    if [ -d "$CURRENT_DIR/data/icons" ]; then
        mkdir -p "$LOCAL_ICON_DIR"
        cp -r "$CURRENT_DIR/data/icons/"* "$LOCAL_ICON_DIR/" 2>/dev/null || true
        gtk-update-icon-cache "$LOCAL_ICON_DIR" 2>/dev/null || true
        echo "Icons installed to $LOCAL_ICON_DIR"
    fi
}

echo "Installing iide settings locally for development..."
install_schemas
install_icons
echo "Done! If needed, set:"
echo "  export XDG_DATA_DIRS=\"\$HOME/.local/share:\$XDG_DATA_DIRS\""
```

File: src/Services/Actions/Action.vala

```
public abstract class Iide.Action : Object {
    public abstract string id { get; }
    public abstract string name { get; }
    public abstract string? description { get; }
    public abstract string? icon_name { get; }
    public string? shortcut { get; set; }

    public signal void shortcut_changed (string? new_shortcut);
```

```
    public signal void state_changed (bool new_state);

    public virtual bool is_toggle { get; default = false; }
    public virtual bool state { get; protected set; default = false; }

    public void update_state (bool new_state) {
        state = new_state;
        state_changed (new_state);
    }

    public abstract bool can_execute ();
    public abstract void execute ();

    public virtual string? category { get; default = null; }
}
```

File: src/Services/Actions/ActionManager.vala

```
public class Iide.ActionManager : Object {
    private static ActionManager? _instance;
    private Gee.HashMap<string, Iide.Action> actions;
    private Gee.HashMap<string, string> shortcuts;
    private Iide.ShortcutSettings shortcut_settings;

    public static ActionManager get_instance () {
        if (_instance == null) {
            _instance = new ActionManager ();
        }
        return _instance;
    }

    private ActionManager () {
        actions = new Gee.HashMap<string, Iide.Action> ();
        shortcuts = new Gee.HashMap<string, string> ();
        shortcut_settings = Iide.ShortcutSettings.get_instance ();
        shortcut_settings.shortcut_changed.connect (on_shortcut_changed);
    }

    public void register_action (Iide.Action action) {
        actions.set (action.id, action);
        var saved_shortcut = shortcut_settings.get_shortcut (action.id);
        action.shortcut = saved_shortcut;
        shortcuts.set (action.id, saved_shortcut);

        if (action.is_toggle) {
            var saved_state = shortcut_settings.get_toggle_state (action.id);
```

```
        action.update_state (saved_state);
    }
}

public void unregister_action (string action_id) {
    actions.unset (action_id);
    shortcuts.unset (action_id);
}

public Iide.Action? get_action (string action_id) {
    return actions.get (action_id);
}

public Gee.Collection<Iide.Action> get_all_actions () {
    return actions.values;
}

public Gee.ArrayList<Iide.Action> get_actions_by_category (string category) {
    var result = new Gee.ArrayList<Iide.Action> ();
    foreach (var action in actions.values) {
        if (action.category == category) {
            result.add (action);
        }
    }
    return result;
}

public void execute (string action_id) {
    var action = actions.get (action_id);
    if (action != null && action.can_execute ()) {
        action.execute ();
    }
}

public bool can_execute (string action_id) {
    var action = actions.get (action_id);
    return action != null && action.can_execute ();
}

public string? get_shortcut (string action_id) {
    return shortcuts.get (action_id);
}

public void set_shortcut (string action_id, string? shortcut) {
    var action = actions.get (action_id);
    if (action != null) {
```

```
        action.shortcut = shortcut;
        shortcuts.set (action_id, shortcut);
        shortcut_settings.set_shortcut (action_id, shortcut);
        action.shortcut_changed (shortcut);
    }
}

public void set_toggle_state (string action_id, bool state) {
    var action = actions.get (action_id);
    if (action != null && action.is_toggle) {
        action.update_state (state);
        shortcut_settings.set_toggle_state (action_id, state);
    }
}

private void on_shortcut_changed (string action_id, string? new_shortcut) {
    var action = actions.get (action_id);
    if (action != null) {
        action.shortcut = new_shortcut;
        shortcuts.set (action_id, new_shortcut);
        action.shortcut_changed (new_shortcut);
    }
}

public void apply_shortcuts_to_application (Gtk.Application app) {
    foreach (var action in actions.values) {
        if (action.is_toggle) {
            var toggle_action = new SimpleAction.stateful (
                action.id,
                null,
                new Variant.boolean (action.state)
            );
            toggle_action.activate.connect (() => {
                execute (action.id);
            });
            action.state_changed.connect ((new_state) => {
                toggle_action.set_state (new Variant.boolean (new_state));
            });
            app.add_action (toggle_action);
        } else {
            var simple_action = new SimpleAction (action.id, null);
            simple_action.activate.connect (() => {
                execute (action.id);
            });
            app.add_action (simple_action);
        }
```

```
            if (action.shortcut != null && action.shortcut != "") {
                app.set_accels_for_action ("app." + action.id, { action.shortcut });
            }
        }
    }
}
```

File: src/Services/LSP/LspDiagnosticsMark.vala

```
public class Iide.LspDiagnosticsMark : GtkSource.Mark {
    public int severity { get; construct set; }
    public string diagnostic_message { get; construct set; }

    public LspDiagnosticsMark(string lsp_category, int lsp_severity, string lsp_message) {
        Object(name: null,
               category: lsp_category,
               severity: lsp_severity,
               diagnostic_message: lsp_message);
    }

    private static string category_from_severity(int severity) {
        switch (severity) {
        case 1:
            return "lsp_error";
        case 2:
            return "lsp_warning";
        case 3:
        case 4:
            return "lsp_info";
        }
        return "lsp_error";
    }

    public LspDiagnosticsMark.from_lsp_diagnostic(IdeLspDiagnostic diagnostics) {
        Object(name: null,
               category: category_from_severity(diagnostics.severity),
               severity: diagnostics.severity,
               diagnostic_message: diagnostics.message);
    }

    public static void set_mark_attributes(GtkSource.View text_view) {
        var error_mark_attrs = new GtkSource.MarkAttributes();
        var err_bg = Gdk.RGBA();
        err_bg.parse("#e01b2430");
        error_mark_attrs.set_background(err_bg);
```

```
        text_view.set_mark_attributes("lsp_error", error_mark_attrs, 100);

        var warning_mark_attrs = new GtkSource.MarkAttributes();
        var warn_bg = Gdk.RGBA();
        warn_bg.parse("#f5c21130");
        warning_mark_attrs.set_background(warn_bg);
        text_view.set_mark_attributes("lsp_warning", warning_mark_attrs, 90);

        var info_mark_attrs = new GtkSource.MarkAttributes();
        text_view.set_mark_attributes("lsp_info", info_mark_attrs, 80);
    }

    public static void clear_mark_attributes(GtkSource.View text_view) {
        var buffer = (GtkSource.Buffer) text_view.buffer;
        Gtk.TextIter start, end;
        buffer.get_start_iter(out start);
        buffer.get_end_iter(out end);

        buffer.remove_source_marks(start, end, "lsp_error");
        buffer.remove_source_marks(start, end, "lsp_warning");
        buffer.remove_source_marks(start, end, "lsp_info");
    }

    public string ? get_icon_name() {
        switch (severity) {
        case 1:
            return "dialog-error";
        case 2:
            return "dialog-warning";
        case 3:
        case 4:
            return "dialog-information";
        }
        return null;
    }
}
```

File: src/Services/TreeSitter/cpp/CppTreeSitterHighlighter.vala

```
using Gtk;
using GtkSource;

public class Iide.CppTreeSitterHighlighter : BaseTreeSitterHighlighter {
    public CppTreeSitterHighlighter (View view) {
        base (view);
        view.set_insert_spaces_instead_of_tabs (true);
```

```
    view.set_tab_width (4);
    view.set_smart_backspace (true);
}

protected override unowned TreeSitter.Language language () {
    return get_language_cpp ();
}

protected override void highlight_node (TreeSitter.Node node) {
    switch (node.type ()) {
    case "identifier": {
        TextIter s_iter, e_iter;
        get_iters_from_ts_node (buffer, node, out s_iter, out e_iter);
        string node_text = buffer.get_text (s_iter, e_iter, false);
        switch (node_text) {
        case "auto":
        case "bool":
        case "char":
        case "double":
        case "float":
        case "int":
        case "long":
        case "short":
        case "signed":
        case "unsigned":
        case "void":
        case "wchar_t":
            highlight_range (node, "c:type-keyword");
            break;
        default:
            highlight_range (node, "def:identifier");
            break;
        }
        break;
    }
    case "and":
    case "and_eq":
    case "bitand":
    case "bitor":
    case "not":
    case "not_eq":
    case "or":
    case "or_eq":
    case "xor":
    case "xor_eq":
        highlight_range (node, "def:special-char");
```

```
    break;
case "true":
case "false":
    highlight_range (node, "def:boolean");
    break;
case "string_literal":
    highlight_range (node, "def:string");
    break;
case "number_literal":
    highlight_range (node, "def:floating-point");
    break;
case "comment":
    highlight_range (node, "def:comment");
    break;
case "preproc_directive":
case "preproc_arg":
    highlight_range (node, "def:preprocessor");
    break;
case "if":
case "else":
case "elif":
case "while":
case "for":
case "do":
case "switch":
case "case":
case "default":
case "break":
case "continue":
case "return":
case "goto":
    highlight_range (node, "c:type-keyword");
    break;
case "class":
case "struct":
case "union":
case "enum":
case "namespace":
case "typedef":
case "using":
    highlight_range (node, "c:type-keyword");
    break;
case "public":
case "private":
case "protected":
    highlight_range (node, "c:type-keyword");
```

```
            break;
        case "new":
        case "delete":
        case "this":
        case "operator":
        case "template":
        case "typename":
            highlight_range (node, "c:type-keyword");
            break;
        default:
            break;
        }
    }
}
```

File: src/Services/TreeSitter/model/AstItem.vala

```
/*
 * Item for Abstract Syntax tree model
 *
 * Author: Alberto Fanjul <albfan@gnome.org>
 */
public class ASTItem : GLib.Object {
    public int start { get; set; }
    public int start_offset { get; set; }
    public int end { get; set; }
    public int end_offset { get; set; }
    public string node_type { get; set; }
    public bool fold { get; set; }
    public bool foldable { get; set; }

    public ASTItem(int start, int start_offset, int end, int end_offset,
        string node_type, bool fold) {
        this.start = start;
        this.start_offset = start_offset;
        this.end = end;
        this.end_offset = end_offset;
        this.node_type = node_type;
        this.fold = fold;
        this.foldable = false;
    }

    public bool contains_text_iter(Gtk.TextIter iter) {
        int line = iter.get_line();
        int offset = iter.get_line_offset();
```

```
        return (start < line || (start == line && start_offset <= offset))
               && (line < end || (line == end && offset <= end_offset));
    }
}
```

File: src/Services/TreeSitter/model/AstModel.vala

```
/*
 * Model for Abstract Syntax Tree
 *
 * Author: Alberto Fanjul <albfan@gnome.org>
 */
public class ASTModel : Gee.ArrayList<ASTItem> {

    public enum Style {
        INTELLIJ,
        ATOM;

        public string to_string() {
            switch (this) {
            case INTELLIJ:
                return "Intellij";
            case ATOM:
                return "Atom";
            default:
                assert_not_reached();
            }
        }
    }

    public Array<ASTItem> highlighted { get; set; default = new Array<ASTItem> (); }
    public bool show_symbols { get; set; }
    public string strategy { get; set; }
    public Style style { get; set; }
    public char whitespace { get; set; }
    public int whitespace_count { get; set; }

    public ASTModel() {
        whitespace = ' ';
        whitespace_count = 2;
    }

    public void add_item_from_iter(Gtk.TextIter start, Gtk.TextIter end,
                                   string type, bool fold) {
        add_item(start.get_line(), start.get_line_offset(), end.get_line(),
                 end.get_line_offset(), type, fold);
```

```
    }

    public void add_item(int start, int start_offset, int end, int end_offset,
                         string type, bool fold) {
        add(new ASTItem(start, start_offset, end, end_offset, type, fold));
    }

    public Array<ASTItem> get_folds(int line) {
        var result = new Array<ASTItem> ();
        foreach (ASTItem item in this) {
            if (line == item.start || (!item.fold && line == item.end)) {
                result.append_val(item);
            }
        }
        return result;
    }
}
```

File: src/Services/TreeSitter/python/higlights.scm

```
; Identifier naming conventions
(identifier) @variable

; Reset highlighting in f-string interpolations
(interpolation) @none

; Identifier naming conventions
((identifier) @type
  (#match? @type "^[A-Z].*[a-z]"))

((identifier) @constant
  (#match? @constant "^[A-Z][A-Z_0-9]*$"))

((identifier) @constant.builtin
  (#match? @constant.builtin "^__[a-zA-Z0-9_]*__$"))

((identifier) @constant.builtin
  (#any-of? @constant.builtin
    ; https://docs.python.org/3/library/constants.html
    "NotImplemented" "Ellipsis" "quit" "exit" "copyright" "credits" "license"))

"_" @character.special ; match wildcard

((assignment
  left: (identifier) @type.definition
  (type
```

```
    (identifier) @_annotation))
  (#eq? @_annotation "TypeAlias"))

((assignment
  left: (identifier) @type.definition
  right: (call
    function: (identifier) @_func))
  (#any-of? @_func "TypeVar" "NewType"))

; Function definitions
(function_definition
  name: (identifier) @function)

(type
  (identifier) @type)

(type
  (subscript
    (identifier) @type)) ; type subscript: Tuple[int]

((call
  function: (identifier) @_isinstance
  arguments: (argument_list
    (_)
    (identifier) @type))
  (#eq? @_isinstance "isinstance"))

; Literals
(none) @constant.builtin

[
  (true)
  (false)
] @boolean

(integer) @number

(float) @number.float

(comment) @comment @spell

((module
  .
  (comment) @keyword.directive @nospell)
  (#match? @keyword.directive "^#!/"))
```

```
(string) @string

[
  (escape_sequence)
  (escape_interpolation)
] @string.escape

; doc-strings
(expression_statement
  (string
    (string_content) @spell) @string.documentation)

; Tokens
[
  "-"
  "-="
  ":="
  "!="
  "*"
  "**"
  "**="
  "*="
  "/"
  "//"
  "//="
  "/="
  "&"
  "&="
  "%"
  "%="
  "^"
  "^="
  "+"
  "+="
  "<"
  "<<"
  "<<="
  "<="
  "<>"
  "="
  "=="
  ">"
  ">="
  ">>"
  ">>="
  "@"
```

```
  "@="
  "|"
  "|="
  "~"
  "->"
] @operator

; Keywords
[
  "and"
  "in"
  "is"
  "not"
  "or"
  "is not"
  "not in"
  "del"
] @keyword.operator

[
  "def"
  "lambda"
] @keyword.function

[
  "assert"
  "exec"
  "global"
  "nonlocal"
  "pass"
  "print"
  "with"
  "as"
] @keyword

[
  "type"
  "class"
] @keyword.type

[
  "async"
  "await"
] @keyword.coroutine

[
```

```
  "return"
  "yield"
] @keyword.return

(yield
  "from" @keyword.return)

(future_import_statement
  "from" @keyword.import
  "__future__" @module.builtin)

(import_from_statement
  "from" @keyword.import)

"import" @keyword.import

(aliased_import
  "as" @keyword.import)

(wildcard_import
  "*" @character.special)

(import_statement
  name: (dotted_name
    (identifier) @module))

(import_statement
  name: (aliased_import
    name: (dotted_name
      (identifier) @module)
    alias: (identifier) @module))

(import_from_statement
  module_name: (dotted_name
    (identifier) @module))

(import_from_statement
  module_name: (relative_import
    (dotted_name
      (identifier) @module)))

[
  "if"
  "elif"
  "else"
  "match"
```

```
  "case"
] @keyword.conditional

[
  "for"
  "while"
  "break"
  "continue"
] @keyword.repeat

[
  "try"
  "except"
  "raise"
  "finally"
] @keyword.exception

(raise_statement
  "from" @keyword.exception)

(try_statement
  (else_clause
    "else" @keyword.exception))

[
  "("
  ")"
  "["
  "]"
  "{"
  "}"
] @punctuation.bracket

(interpolation
  "{" @punctuation.special
  "}" @punctuation.special)

(type_conversion) @function.macro

[
  ","
  "."
  ":"
  ";"
  (ellipsis)
] @punctuation.delimiter
```

```
((identifier) @type.builtin
  (#any-of? @type.builtin
    ; https://docs.python.org/3/library/exceptions.html
    "BaseException" "Exception" "ArithmeticError" "BufferError" "LookupError" "AssertionErro
    "AttributeError" "EOFError" "FloatingPointError" "GeneratorExit" "ImportError"
    "ModuleNotFoundError" "IndexError" "KeyError" "KeyboardInterrupt" "MemoryError" "NameErr
    "NotImplementedError" "OSError" "OverflowError" "RecursionError" "ReferenceError" "Runti
    "StopIteration" "StopAsyncIteration" "SyntaxError" "IndentationError" "TabError" "Systen
    "SystemExit" "TypeError" "UnboundLocalError" "UnicodeError" "UnicodeEncodeError"
    "UnicodeDecodeError" "UnicodeTranslateError" "ValueError" "ZeroDivisionError" "Environme
    "IOError" "WindowsError" "BlockingIOError" "ChildProcessError" "ConnectionError"
    "BrokenPipeError" "ConnectionAbortedError" "ConnectionRefusedError" "ConnectionResetErro
    "FileExistsError" "FileNotFoundError" "InterruptedError" "IsADirectoryError"
    "NotADirectoryError" "PermissionError" "ProcessLookupError" "TimeoutError" "Warning"
    "UserWarning" "DeprecationWarning" "PendingDeprecationWarning" "SyntaxWarning" "RuntimeW
    "FutureWarning" "ImportWarning" "UnicodeWarning" "BytesWarning" "ResourceWarning"
    ; https://docs.python.org/3/library/stdtypes.html
    "bool" "int" "float" "complex" "list" "tuple" "range" "str" "bytes" "bytearray" "memoryv
    "set" "frozenset" "dict" "type" "object"))

; Normal parameters
(parameters
  (identifier) @variable.parameter)

; Lambda parameters
(lambda_parameters
  (identifier) @variable.parameter)

(lambda_parameters
  (tuple_pattern
    (identifier) @variable.parameter))

; Default parameters
(keyword_argument
  name: (identifier) @variable.parameter)

; Naming parameters on call-site
(default_parameter
  name: (identifier) @variable.parameter)

(typed_parameter
  (identifier) @variable.parameter)

(typed_default_parameter
  name: (identifier) @variable.parameter)
```

```
; Variadic parameters *args, **kwargs
(parameters
  (list_splat_pattern ; *args
    (identifier) @variable.parameter))

(parameters
  (dictionary_splat_pattern ; **kwargs
    (identifier) @variable.parameter))

; Typed variadic parameters
(parameters
  (typed_parameter
    (list_splat_pattern ; *args: type
      (identifier) @variable.parameter)))

(parameters
  (typed_parameter
    (dictionary_splat_pattern ; *kwargs: type
      (identifier) @variable.parameter)))

; Lambda parameters
(lambda_parameters
  (list_splat_pattern
    (identifier) @variable.parameter))

(lambda_parameters
  (dictionary_splat_pattern
    (identifier) @variable.parameter))

((identifier) @variable.builtin
  (#eq? @variable.builtin "self"))

((identifier) @variable.builtin
  (#eq? @variable.builtin "cls"))

; After @type.builtin bacause builtins (such as `type`) are valid as attribute name
((attribute
  attribute: (identifier) @variable.member)
  (#match? @variable.member "^[%l_].*$"))

; Class definitions
(class_definition
  name: (identifier) @type)

(class_definition
```

```
  body: (block
    (function_definition
      name: (identifier) @function.method)))

(class_definition
  superclasses: (argument_list
    (identifier) @type))

((class_definition
  body: (block
    (expression_statement
      (assignment
        left: (identifier) @variable.member))))
  (#match? @variable.member "^[%l_].*$"))

((class_definition
  body: (block
    (expression_statement
      (assignment
        left: (_
          (identifier) @variable.member)))))
  (#match? @variable.member "^[%l_].*$"))

((class_definition
  (block
    (function_definition
      name: (identifier) @constructor)))
  (#any-of? @constructor "__new__" "__init__"))

; Function calls
(call
  function: (identifier) @function.call)

(call
  function: (attribute
    attribute: (identifier) @function.method.call))

((call
  function: (identifier) @constructor)
  (#match? @constructor "^%u"))

((call
  function: (attribute
    attribute: (identifier) @constructor))
  (#match? @constructor "^%u"))
```

```
; Builtin functions
((call
  function: (identifier) @function.builtin)
  (#any-of? @function.builtin
    "abs" "all" "any" "ascii" "bin" "bool" "breakpoint" "bytearray" "bytes" "callable" "chr"
    "classmethod" "compile" "complex" "delattr" "dict" "dir" "divmod" "enumerate" "eval" "ex
    "filter" "float" "format" "frozenset" "getattr" "globals" "hasattr" "hash" "help" "hex"
    "input" "int" "isinstance" "issubclass" "iter" "len" "list" "locals" "map" "max" "memory
    "min" "next" "object" "oct" "open" "ord" "pow" "print" "property" "range" "repr" "revers
    "round" "set" "setattr" "slice" "sorted" "staticmethod" "str" "sum" "super" "tuple" "typ
    "vars" "zip" "__import__"))

; Regex from the `re` module
(call
  function: (attribute
    object: (identifier) @_re)
  arguments: (argument_list
    .
    (string
      (string_content) @string.regexp))
  (#eq? @_re "re"))

; Decorators
((decorator
  "@" @attribute)
  (#set! priority 101))

(decorator
  (identifier) @attribute)

(decorator
  (attribute
    attribute: (identifier) @attribute))

(decorator
  (call
    (identifier) @attribute))

(decorator
  (call
    (attribute
      attribute: (identifier) @attribute)))

((decorator
  (identifier) @attribute.builtin)
  (#any-of? @attribute.builtin "classmethod" "property" "staticmethod"))
```

File: src/Services/TreeSitter/vala/ValaTreeSitterHighlighter.vala

```
using Gtk;
using GtkSource;

public class Iide.ValaTreeSitterHighlighter : BaseTreeSitterHighlighter {
    public ValaTreeSitterHighlighter(View view) {
        base (view);
        view.set_insert_spaces_instead_of_tabs (true);
        view.set_tab_width (4);
        view.set_smart_backspace (true);
    }

    protected override unowned TreeSitter.Language language () {
        return get_language_vala ();
    }

    protected override void highlight_node (TreeSitter.Node node) {
        switch (node.type ()) {
        case "comment":
            highlight_range (node, "def:comment");
            break;
        case "string":
        case "template_string":
            highlight_range (node, "def:string");
            break;
        case "integer":
            highlight_range (node, "def:floating-point");
            break;
        case "boolean":
            highlight_range (node, "def:boolean");
            break;
        case "identifier": {
            TextIter s_iter, e_iter;
            get_iters_from_ts_node (buffer, node, out s_iter, out e_iter);
            string node_text = buffer.get_text (s_iter, e_iter, false);
            if (node_text == "this" || node_text == "base") {
                highlight_range (node, "def:special-char");
            } else {
                highlight_range (node, "def:identifier");
            }
            break;
        }
        case "if":
        case "elif":
        case "else":
```

```
case "switch":
case "case":
case "default":
    highlight_range (node, "c:type-keyword");
    break;
case "for":
case "foreach":
case "while":
case "do":
case "break":
case "continue":
    highlight_range (node, "c:type-keyword");
    break;
case "class":
case "interface":
case "enum":
case "struct":
    highlight_range (node, "c:type-keyword");
    break;
case "public":
case "private":
case "protected":
case "internal":
case "static":
case "virtual":
case "override":
case "abstract":
case "sealed":
case "async":
case "const":
case "ref":
case "out":
case "owned":
case "unowned":
case "weak":
    highlight_range (node, "c:type-keyword");
    break;
case "void":
case "bool":
case "int":
case "uint":
case "int8":
case "uint8":
case "int16":
case "uint16":
case "int32":
```

```
case "uint32":
case "int64":
case "uint64":
case "size_t":
case "ssize_t":
case "float":
case "double":
case "var":
    highlight_range (node, "c:type-keyword");
    break;
case "return":
case "yield":
case "try":
case "catch":
case "finally":
case "throw":
    highlight_range (node, "c:type-keyword");
    break;
case "namespace":
case "using":
    highlight_range (node, "def:preprocessor");
    break;
case "signal":
case "construct":
case "property":
    highlight_range (node, "c:printf");
    break;
case "get":
case "set":
case "will":
case "notify":
    highlight_range (node, "c:printf");
    break;
case "in":
case "is":
case "as":
case "typeof":
case "sizeof":
case "alignof":
case "lock":
case "throws":
case "await":
case "new":
case "delete":
    highlight_range (node, "def:special-char");
    break;
```

```
        default:
            break;
        }
    }
}
```

File: test_temp/test.c

```
int main() {
```

File: src/Services/TreeSitter/python/PythonTreeSitterHighlighter.vala

```
using Gtk;
using GtkSource;

[CCode(cname = "tree_sitter_python")]
extern unowned TreeSitter.Language ? get_language_python();

public class Iide.PythonTreeSitterHighlighter : BaseTreeSitterHighlighter {
    public PythonTreeSitterHighlighter(View view) {
        base(view);
        view.set_insert_spaces_instead_of_tabs(true);
        view.set_tab_width(4);
        view.set_smart_backspace(true);
    }

    protected override unowned TreeSitter.Language language() {
        return get_language_python();
    }

    protected override void highlight_node(TreeSitter.Node node) {
        // message(depth.to_string() + " " + parent_node?.type() + " -> " + node.type());
        switch (node.type()) {
        case "identifier": {
            TextIter s_iter, e_iter;
            get_iters_from_ts_node(buffer, node, out s_iter, out e_iter);
            string node_text = buffer.get_text(s_iter, e_iter, false);
            switch (node_text) {
            case "bool":
            case "int":
            case "float":
            case "complex":
            case "list":
            case "tuple":
            case "range":
            case "str":
```

```
        case "bytes":
        case "bytearray":
        case "memoryview":
        case "set":
        case "frozenset":
        case "dict":
        case "type":
        case "object":
            highlight_range(node, "python:builtin-function");
            break;
        default:
            highlight_range(node, "def:identifier");
            break;
        }
        break;
    }
    case "and":
    case "in":
    case "is":
    case "not":
    case "or":
    case "is not":
    case "not in":
    case "del":
        highlight_range(node, "def:special-char");
        break;
    case "none":
    case "true":
    case "false":
        highlight_range(node, "def:boolean");
        break;
    case "string":
        highlight_range(node, "def:string");
        break;
    case "integer":
        highlight_range(node, "def:floating-point");
        break;
    case "float":
        highlight_range(node, "def:floating-point");
        break;
    case "comment":
        highlight_range(node, "def:comment");
        break;
    case "import":
    case "from":
        highlight_range(node, "def:preprocessor");
```

```
            break;
        case "def":
        case "class":
        case "return":
        case "break":
        case "continue":
        case "if":
        case "else":
        case "elif":
        case "raise":
        case "while":
            highlight_range(node, "c:type-keyword");
            break;
        case "assert":
        case "exec":
        case "global":
        case "nonlocal":
        case "pass":
        case "print":
        case "with":
        case "as":
            highlight_range(node, "c:printf");
            break;
        default:
            break;
        }
    }
}
```

File: src/Services/Utils.vala

```
using GLib;

namespace Iide {
    public string mime_type_for_file(File file) {
        string mime_type = "text-x";
        try {
            var info = file.query_info("standard::*", FileQueryInfoFlags.NONE, null);
            mime_type = ContentType.get_mime_type(info.get_attribute_as_string(FileAttribute
        } catch (Error e) {
        }
        return mime_type;
    }

    public void copy_resource_to_file(string resource_path, string local_path) {
        // 1.          File        (               resource:///)
```

```
        var resource_file = File.new_for_uri (resource_path);

        // 2.          File
        var local_file = File.new_for_path (local_path);

        try {
            // 3.             .    OVERWRITE          ,            .
            resource_file.copy (local_file, FileCopyFlags.OVERWRITE, null, null);
            print ("               : %s\n", local_path);
        } catch (Error e) {
            stderr.printf ("             : %s\n", e.message);
        }
    }
}
```

File: src/Widgets/ToolViews/LogView.vala

```
/*
 * logview.vala
 *
 * Copyright 2026 kai
 *
 * This program is free software: you can redistribute it and/or modify
 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation, either version 3 of the License, or
 * (at your option) any later version.
 *
 * This program is distributed in the hope that it will be useful,
 * but WITHOUT ANY WARRANTY; without even the implied warranty of
 * MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
 * GNU General Public License for more details.
 *
 * You should have received a copy of the GNU General Public License
 * along with this program.  If not, see <https://www.gnu.org/licenses/>.
 *
 * SPDX-License-Identifier: GPL-3.0-or-later
 */

public class Iide.LogView : Gtk.Box {
    private Gtk.TextView text_view;
    private Gtk.TextBuffer buffer;
    private Gtk.ScrolledWindow scrolled_window;
    private Gtk.Button clear_button;
    private Gtk.ToggleButton wrap_button;
    private Gtk.DropDown level_filter;
    private Gtk.SearchEntry search_entry;
```

```
private Iide.LoggerService logger;

private Gtk.TextTag tag_debug;
private Gtk.TextTag tag_info;
private Gtk.TextTag tag_warning;
private Gtk.TextTag tag_error;
private Gtk.TextTag tag_critical;
private Gtk.TextTag tag_bold;
private Gtk.TextTag tag_dim;

private GLib.Mutex buffer_lock;
private bool auto_scroll = true;
private const int MAX_VISIBLE_LINES = 5000;
private int line_count = 0;

public LogView () {
    Object (orientation: Gtk.Orientation.VERTICAL, spacing: 0);
}

construct {
    logger = Iide.LoggerService.get_instance ();

    var toolbar = new Gtk.Box (Gtk.Orientation.HORIZONTAL, 6);
    toolbar.add_css_class ("toolbar");
    toolbar.margin_start = 6;
    toolbar.margin_end = 6;
    toolbar.margin_top = 4;
    toolbar.margin_bottom = 4;

    clear_button = new Gtk.Button () {
        icon_name = "edit-clear-symbolic",
        tooltip_text = _("Clear logs")
    };
    clear_button.clicked.connect (on_clear_clicked);

    wrap_button = new Gtk.ToggleButton () {
        icon_name = "preferences-desktop-text-wrap-symbolic",
        tooltip_text = _("Word wrap")
    };
    wrap_button.active = true;
    wrap_button.toggled.connect (on_wrap_toggled);

    var level_label = new Gtk.Label (_("Level:")) {
        margin_start = 12
    };
```

```
var level_list = new Gtk.StringList ({
    _("All"),
    "DEBUG",
    "INFO",
    "WARN",
    "ERROR",
    "CRITICAL"
});
level_filter = new Gtk.DropDown (level_list, null);
level_filter.selected = 0;
level_filter.notify["selected"].connect (on_level_changed);

search_entry = new Gtk.SearchEntry () {
    placeholder_text = _("Search logs..."),
    hexpand = true
};
search_entry.search_changed.connect (on_search_changed);

toolbar.append (clear_button);
toolbar.append (wrap_button);
toolbar.append (new Gtk.Separator (Gtk.Orientation.VERTICAL));
toolbar.append (level_label);
toolbar.append (level_filter);
toolbar.append (search_entry);

buffer = new Gtk.TextBuffer (null);
tag_debug = buffer.create_tag (null, foreground: "#888888");
tag_info = buffer.create_tag (null, foreground: "#4a9eff", weight: Pango.Weight.BOLD
tag_warning = buffer.create_tag (null, foreground: "#f5a623", weight: Pango.Weight.B
tag_error = buffer.create_tag (null, foreground: "#e74c3c", weight: Pango.Weight.BOL
tag_critical = buffer.create_tag (null, foreground: "#ff4757", weight: Pango.Weight.
tag_bold = buffer.create_tag (null, weight: Pango.Weight.BOLD);
tag_dim = buffer.create_tag (null, foreground: "#666666", style: Pango.Style.ITALIC)

text_view = new Gtk.TextView () {
    buffer = buffer,
    editable = false,
    monospace = true,
    wrap_mode = Gtk.WrapMode.WORD_CHAR,
    hscroll_policy = Gtk.ScrollablePolicy.NATURAL,
    vscroll_policy = Gtk.ScrollablePolicy.NATURAL
};
text_view.add_css_class ("log-view");

scrolled_window = new Gtk.ScrolledWindow () {
    child = text_view,
```

```
        hexpand = true,
        vexpand = true
    };

    append (toolbar);
    append (scrolled_window);

    logger.log_added.connect (on_log_added);
    logger.log_cleared.connect (on_log_cleared);

    foreach (var entry in logger.get_entries ()) {
        append_entry (entry);
    }
}

private void on_log_added (Iide.LogEntry entry) {
    append_entry (entry);
}

private void append_entry (Iide.LogEntry entry) {
    buffer_lock.lock ();
    try {
        var timestamp = entry.get_timestamp_string ();
        var level_str = entry.level.to_string ();
        var domain = entry.domain;
        var message = entry.message;

        Gtk.TextTag tag;
        switch (entry.level) {
        case Iide.LogLevel.DEBUG:
            tag = tag_debug;
            break;
        case Iide.LogLevel.INFO:
            tag = tag_info;
            break;
        case Iide.LogLevel.WARNING:
            tag = tag_warning;
            break;
        case Iide.LogLevel.ERROR:
            tag = tag_error;
            break;
        case Iide.LogLevel.CRITICAL:
            tag = tag_critical;
            break;
        default:
            tag = tag_dim;
```

```
    break;
}

var ts_text = "[%s] ".printf (timestamp);
var level_text = "[%s] ".printf (level_str);
var domain_text = "[%s] ".printf (domain);

int offset = buffer.get_char_count ();

Gtk.TextIter iter;
buffer.get_end_iter (out iter);
buffer.insert (ref iter, ts_text, ts_text.length);
buffer.get_end_iter (out iter);
buffer.insert (ref iter, level_text, level_text.length);
buffer.get_end_iter (out iter);
buffer.insert (ref iter, domain_text, domain_text.length);
buffer.get_end_iter (out iter);
buffer.insert (ref iter, message, message.length);

if (entry.details != null) {
    buffer.get_end_iter (out iter);
    buffer.insert (ref iter, "\n  ", 3);
    buffer.get_end_iter (out iter);
    buffer.insert (ref iter, entry.details, entry.details.length);
}

buffer.get_end_iter (out iter);
buffer.insert (ref iter, "\n", 1);

Gtk.TextIter start, end;
buffer.get_iter_at_offset (out start, offset);
buffer.get_iter_at_offset (out end, offset + (int) ts_text.length);
buffer.apply_tag (tag_dim, start, end);

buffer.get_iter_at_offset (out start, offset + (int) ts_text.length);
buffer.get_iter_at_offset (out end, offset + (int) ts_text.length + (int) level_
buffer.apply_tag (tag, start, end);

buffer.get_iter_at_offset (out start, offset + (int) ts_text.length + (int) leve
buffer.get_iter_at_offset (out end, offset + (int) ts_text.length + (int) level_
buffer.apply_tag (tag_bold, start, end);

if (entry.details != null) {
    int detail_start = offset + (int) ts_text.length + (int) level_text.length +
    buffer.get_iter_at_offset (out start, detail_start);
    buffer.get_iter_at_offset (out end, detail_start + 3);
```

```
            buffer.apply_tag (tag_dim, start, end);
        }

        line_count++;
        if (auto_scroll) {
            scroll_to_end ();
        }

        trim_buffer ();
    } finally {
        buffer_lock.unlock ();
    }
}

private void scroll_to_end () {
    Gtk.TextIter end;
    buffer.get_end_iter (out end);
    text_view.scroll_to_iter (end, 0.0, true, 0.0, 1.0);
}

private void trim_buffer () {
    if (line_count > MAX_VISIBLE_LINES) {
        buffer_lock.lock ();
        try {
            Gtk.TextIter start;
            buffer.get_iter_at_line (out start, line_count - MAX_VISIBLE_LINES);
            Gtk.TextIter end = start;
            buffer.get_iter_at_line (out end, line_count - MAX_VISIBLE_LINES + 100);
            buffer.delete (ref start, ref end);
            line_count -= 100;
        } finally {
            buffer_lock.unlock ();
        }
    }
}

private void on_clear_clicked () {
    logger.clear ();
}

private void on_log_cleared () {
    buffer.set_text ("");
    line_count = 0;
}

private void on_wrap_toggled () {
```

```
    text_view.wrap_mode = wrap_button.active ? Gtk.WrapMode.WORD_CHAR : Gtk.WrapMode.NON
}

private void on_level_changed () {
    rebuild_log_view ();
}

private void on_search_changed () {
    rebuild_log_view ();
}

private bool matches_search (Iide.LogEntry entry, string search_text) {
    if (search_text == "") {
        return true;
    }
    var lower_search = search_text.down ();
    return entry.message.down ().contains (lower_search) ||
           entry.domain.down ().contains (lower_search) ||
           (entry.details != null && entry.details.down ().contains (lower_search));
}

private void rebuild_log_view () {
    buffer_lock.lock ();
    try {
        buffer.set_text ("");
        line_count = 0;

        var filter_level = (int) level_filter.selected;
        var search_text = search_entry.text;

        foreach (var entry in logger.get_entries ()) {
            if (filter_level > 0) {
                var entry_level = (int) entry.level + 1;
                if (entry_level != filter_level) {
                    continue;
                }
            }
            if (!matches_search (entry, search_text)) {
                continue;
            }
            append_entry_unlocked (entry);
        }
    } finally {
        buffer_lock.unlock ();
    }
}
```

```
private void append_entry_unlocked (Iide.LogEntry entry) {
    Gtk.TextIter end;
    buffer.get_end_iter (out end);

    var timestamp = entry.get_timestamp_string ();
    var level_str = entry.level.to_string ();
    var domain = entry.domain;
    var message = entry.message;

    Gtk.TextTag tag;
    switch (entry.level) {
    case Iide.LogLevel.DEBUG:
        tag = tag_debug;
        break;
    case Iide.LogLevel.INFO:
        tag = tag_info;
        break;
    case Iide.LogLevel.WARNING:
        tag = tag_warning;
        break;
    case Iide.LogLevel.ERROR:
        tag = tag_error;
        break;
    case Iide.LogLevel.CRITICAL:
        tag = tag_critical;
        break;
    default:
        tag = tag_dim;
        break;
    }

    var ts_text = "[%s] ".printf (timestamp);
    var level_text = "[%s] ".printf (level_str);
    var domain_text = "[%s] ".printf (domain);

    var ts_start = end;
    buffer.insert (ref end, ts_text, ts_text.length);
    var ts_end = end;
    buffer.apply_tag (tag_dim, ts_start, ts_end);

    var level_start = end;
    buffer.insert (ref end, level_text, level_text.length);
    var level_end = end;
    buffer.apply_tag (tag, level_start, level_end);
```

```
        var domain_start = end;
        buffer.insert (ref end, domain_text, domain_text.length);
        var domain_end = end;
        buffer.apply_tag (tag_bold, domain_start, domain_end);

        buffer.insert (ref end, message, message.length);

        if (entry.details != null) {
            var detail_start = end;
            buffer.insert (ref end, "\n  ", 3);
            var detail_end = end;
            buffer.apply_tag (tag_dim, detail_start, detail_end);

            buffer.insert (ref end, entry.details, entry.details.length);
        }

        buffer.insert (ref end, "\n", 1);

        line_count++;
    }
}
```

File: src/Widgets/ToolViews/TerminalView.vala

```
/*
 * SPDX-License-Identifier: LGPL-3.0-or-later
 * SPDX-FileCopyrightText: 2022 elementary, Inc. (https://elementary.io)
 *                         2011-2013 Mario Guerriero <mario@elementaryos.org>
 */

public class Iide.Terminal : Gtk.Box {
    private const string ACTION_GROUP = "term";
    private const string ACTION_PREFIX = ACTION_GROUP + ".";
    private const string ACTION_COPY = "action-copy";
    private const string ACTION_PASTE = "action-paste";
    private const string ACTION_INCREASE_FONT = "increase-font";
    private const string ACTION_DECREASE_FONT = "decrease-font";
    private const string ACTION_RESET_FONT = "reset-font";

    private const double MAX_SCALE = 5.0;
    private const double MIN_SCALE = 0.2;
    private const string LEGACY_SETTINGS_SCHEMA = "org.pantheon.terminal.settings";
    private const string SETTINGS_SCHEMA = "io.elementary.terminal.settings";
    private const string GNOME_DESKTOP_INTERFACE_SCHEMA = "org.gnome.desktop.interface";
    private const string GNOME_DESKTOP_WM_PREFERENCES_SCHEMA = "org.gnome.desktop.wm.prefere
    private const string TERMINAL_FONT_KEY = "font";
```

```
private const string TERMINAL_BELL_KEY = "audible-bell";
private const string TERMINAL_CURSOR_KEY = "cursor-shape";
private const string TERMINAL_FOREGROUND_KEY = "foreground";
private const string TERMINAL_BACKGROUND_KEY = "background";
private const string TERMINAL_PALETTE_KEY = "palette";
private const string GNOME_FONT_KEY = "monospace-font-name";
private const string GNOME_BELL_KEY = "audible-bell";

public Vte.Terminal terminal { get; construct; }
private Gtk.EventControllerKey key_controller;
private Gtk.GestureClick gesture_click;
private Gtk.EventControllerScroll scroll_controller;
private Gdk.Clipboard current_clipboard;
private Gtk.ScrolledWindow scrolled_window;

private const double FONT_SCALE_STEP = 0.1;

private Settings? terminal_settings = null;
private Settings? gnome_interface_settings = null;
private Settings? gnome_wm_settings = null;

public SimpleActionGroup actions { get; construct; }

private GLib.Pid child_pid;
// private Gtk.Clipboard current_clipboard;

construct {
    terminal = new Vte.Terminal () {
        hexpand = true,
        vexpand = true,
        scrollback_lines = -1,
        cursor_blink_mode = SYSTEM // There is no Terminal setting so follow Gnome
    };

    // Set font, allow-bold, audible-bell, background, foreground, and palette of panthe
    var schema_source = SettingsSchemaSource.get_default ();
    var terminal_schema = schema_source.lookup (SETTINGS_SCHEMA, true);
    if (terminal_schema == null) {
        terminal_schema = schema_source.lookup (LEGACY_SETTINGS_SCHEMA, true);
    }

    if (terminal_schema != null) {
        terminal_settings = new Settings.full (terminal_schema, null, null);
        terminal_settings.changed.connect ((key) => {
            switch (key) {
                case TERMINAL_FONT_KEY :
```

```
                update_font ();
                break;
            case TERMINAL_BELL_KEY :
                update_audible_bell ();
                break;
            case TERMINAL_CURSOR_KEY :
                update_cursor ();
                break;
            case TERMINAL_FOREGROUND_KEY:
            case TERMINAL_BACKGROUND_KEY:
            case TERMINAL_PALETTE_KEY:
                update_colors ();
                break;
            default:
                // TODO Handle other relevant terminal settings?
                // "theme"
                // "prefer-dark-style"
                break;
        }
    });
} else {
    var gnome_wm_settings_schema = schema_source.lookup (GNOME_DESKTOP_WM_PREFERENCE
    if (gnome_wm_settings_schema != null) {
        gnome_wm_settings = new Settings.full (gnome_wm_settings_schema, null, null)
        gnome_wm_settings.changed.connect ((key) => {
            switch (key) {
                case GNOME_BELL_KEY:
                    update_audible_bell ();
                    break;
                default:
                    break;
            }
        });
    }
    // TODO monitor changes in relevant system settings?
    // "org.gnome.desktop.interface.color-scheme"
}

// Always monitor changes in default font as that is what Terminal usually follows
var gnome_interface_settings_schema = schema_source.lookup (GNOME_DESKTOP_INTERFACE_
if (gnome_interface_settings_schema != null) {
    gnome_interface_settings = new Settings.full (gnome_interface_settings_schema, n
    gnome_interface_settings.changed.connect ((key) => {
        switch (key) {
            case GNOME_FONT_KEY:
                update_font ();
```

```
                break;
            default:
                break;
        }
    });
}

update_font ();
update_audible_bell ();
update_cursor ();
update_colors ();

var adw_style_manager = Adw.StyleManager.get_default ();
adw_style_manager.notify["color-scheme"].connect (() => {
    update_colors ();
});

terminal.child_exited.connect (() => {
    // Hide the exited terminal
    // var win_group = get_action_group (Scratch.MainWindow.ACTION_GROUP);
    // win_group.activate_action (Scratch.MainWindow.ACTION_TOGGLE_TERMINAL, null);
    //// Get ready to resume at last saved location
    // spawn_shell (Scratch.saved_state.get_string ("last-opened-path"));
    //// Clear screen of new shell
    terminal.feed_child ("clear -x\n".data);
});

var copy_action = new SimpleAction (ACTION_COPY, null);
copy_action.set_enabled (false);
copy_action.activate.connect (() => terminal.copy_clipboard ());

var paste_action = new SimpleAction (ACTION_PASTE, null);
paste_action.activate.connect (() => terminal.paste_clipboard ());

var increase_font_action = new SimpleAction (ACTION_INCREASE_FONT, null);
increase_font_action.activate.connect (() => increment_size ());

var decrease_font_action = new SimpleAction (ACTION_DECREASE_FONT, null);
decrease_font_action.activate.connect (() => decrement_size ());

var reset_font_action = new SimpleAction (ACTION_RESET_FONT, null);
reset_font_action.activate.connect (() => set_default_font_size ());

actions = new SimpleActionGroup ();
actions.add_action (copy_action);
actions.add_action (paste_action);
```

```
actions.add_action (increase_font_action);
actions.add_action (decrease_font_action);
actions.add_action (reset_font_action);

var menu_model = new GLib.Menu ();
var clipboard_section = new GLib.Menu ();
clipboard_section.append (_("Copy"), ACTION_PREFIX + ACTION_COPY);
clipboard_section.append (_("Paste"), ACTION_PREFIX + ACTION_PASTE);
menu_model.append_section (null, clipboard_section);
var zoom_section = new GLib.Menu ();
zoom_section.append (_("Zoom In"), ACTION_PREFIX + ACTION_INCREASE_FONT);
zoom_section.append (_("Zoom Out"), ACTION_PREFIX + ACTION_DECREASE_FONT);
zoom_section.append (_("Reset Zoom"), ACTION_PREFIX + ACTION_RESET_FONT);
menu_model.append_section (null, zoom_section);

scrolled_window = new Gtk.ScrolledWindow ();
scrolled_window.hexpand = true;
scrolled_window.set_child (terminal);

var popover_menu = new Gtk.PopoverMenu.from_model (menu_model);
popover_menu.insert_action_group (ACTION_GROUP, actions);
popover_menu.set_parent (scrolled_window);
// menu.show_all ();

key_controller = new Gtk.EventControllerKey ();
key_controller.propagation_phase = BUBBLE;
terminal.add_controller (key_controller);
key_controller.key_pressed.connect (key_pressed);

gesture_click = new Gtk.GestureClick ();
gesture_click.button = 3; // Right click
scrolled_window.add_controller (gesture_click);
gesture_click.pressed.connect ((n_press, x, y) => {
    if (n_press == 1) {
        paste_action.set_enabled (current_clipboard.content != null);
        var rect = Gdk.Rectangle () { x = (int) x, y = (int) y, width = 1, height =
        popover_menu.set_pointing_to (rect);
        popover_menu.popup ();
    }
});

scroll_controller = new Gtk.EventControllerScroll (Gtk.EventControllerScrollFlags.VE
scroll_controller.propagation_phase = CAPTURE;
scrolled_window.add_controller (scroll_controller);
scroll_controller.scroll.connect ((dx, dy) => {
    var state = scroll_controller.get_current_event_state ();
```

```
        if (Gdk.ModifierType.CONTROL_MASK in state) {
            if (dy > 0) {
                decrement_size ();
            } else {
                increment_size ();
            }
            return Gdk.EVENT_STOP;
        }
        return Gdk.EVENT_PROPAGATE;
    });

    // terminal.button_press_event.connect ((event) => {
    // if (event.button == 3) {
    // paste_action.set_enabled (current_clipboard.wait_is_text_available ());
    // menu.select_first (false);
    // menu.popup_at_pointer (event);
    // }
    // return false;
    // });

    // realize.connect (() => {
    // current_clipboard = terminal.get_clipboard (Gdk.SELECTION_CLIPBOARD);
    // copy_action.set_enabled (terminal.get_has_selection ());
    // });

    spawn_shell ();

    terminal.selection_changed.connect (() => {
        copy_action.set_enabled (terminal.get_has_selection ());
    });

    realize.connect (() => {
        current_clipboard = scrolled_window.get_clipboard ();
        copy_action.set_enabled (terminal.get_has_selection ());
    });

    append (scrolled_window);
}

private void spawn_shell (string dir = GLib.Environment.get_current_dir ()) {
    terminal.spawn_async (
                          Vte.PtyFlags.DEFAULT,
                          dir,
                          { Vte.get_user_shell () },
                          null,
                          SpawnFlags.SEARCH_PATH,
```

```
                            null,
                            100,
                            null,
                            (source, pid, error) => {
        if (error != null) {
            stderr.printf ("           : %s\n", error.message);
        } else {
            this.child_pid = pid;
            stdout.printf ("              . PID: %d\n", pid);
        }
    });
}

// public void change_location (string dir) {
// Posix.kill (child_pid, Posix.Signal.TERM);
// terminal.reset (true, true);
// spawn_shell (dir);
// Scratch.saved_state.set_string ("last-opened-path", dir);
// }

// private string get_shell_location () {
// int pid = (!) (this.child_pid);
// try {
// return GLib.FileUtils.read_link ("/proc/%d/cwd".printf (pid));
// } catch (GLib.FileError error) {
// warning ("An error occurred while fetching the current dir of shell: %s", error.messa
// return "";
// }
// }

private void update_font () {
    var font_name = "";
    if (terminal_settings != null) {
        font_name = terminal_settings.get_string (TERMINAL_FONT_KEY);
    }

    if (font_name == "" && gnome_interface_settings != null) {
        font_name = gnome_interface_settings.get_string (GNOME_FONT_KEY);
    }

    var fd = Pango.FontDescription.from_string (font_name);
    terminal.set_font (fd);
}

private void update_audible_bell () {
    var audible_bell = false;
```

```
    if (terminal_settings != null) {
        audible_bell = terminal_settings.get_boolean (TERMINAL_BELL_KEY);
    } else if (gnome_wm_settings != null) {
        audible_bell = gnome_wm_settings.get_boolean (GNOME_BELL_KEY);
    }

    terminal.set_audible_bell (audible_bell);
}

private void update_cursor () {
    if (terminal_settings != null) {
        var cursor_shape_setting = terminal_settings.get_string (TERMINAL_CURSOR_KEY);
        switch (cursor_shape_setting) {
        case "Block":
            terminal.cursor_shape = Vte.CursorShape.BLOCK;
            break;
        case "I-Beam":
            terminal.cursor_shape = Vte.CursorShape.IBEAM;
            break;
        case "Underline":
            terminal.cursor_shape = Vte.CursorShape.UNDERLINE;
            break;
        }
    } // No corresponding system keymap
}

private void update_colors () {
    if (terminal_settings != null) {
        string background_setting = terminal_settings.get_string (TERMINAL_BACKGROUND_KE
        Gdk.RGBA background_color = Gdk.RGBA ();
        background_color.parse (background_setting);

        string foreground_setting = terminal_settings.get_string (TERMINAL_FOREGROUND_KE
        Gdk.RGBA foreground_color = Gdk.RGBA ();
        foreground_color.parse (foreground_setting);

        string palette_setting = terminal_settings.get_string (TERMINAL_PALETTE_KEY);

        string[] hex_palette = { "#000000", "#FF6C60", "#A8FF60", "#FFFFCC", "#96CBFE",
                                 "#FF73FE", "#C6C5FE", "#EEEEEE", "#000000", "#FF6C60",
                                 "#A8FF60", "#FFFFB6", "#96CBFE", "#FF73FE", "#C6C5FE",
                                 "#EEEEEE" };

        string current_string = "";
        int current_color = 0;
        for (var i = 0; i < palette_setting.length; i++) {
```

```
            if (palette_setting[i] == ':') {
                hex_palette[current_color] = current_string;
                current_string = "";
                current_color++;
            } else {
                current_string += palette_setting[i].to_string ();
            }
        }

        Gdk.RGBA[] palette = new Gdk.RGBA[16];

        for (int i = 0; i < hex_palette.length; i++) {
            Gdk.RGBA new_color = Gdk.RGBA ();
            new_color.parse (hex_palette[i]);
            palette[i] = new_color;
        }

        terminal.set_colors (foreground_color, background_color, palette);
    } else {
        update_colors_for_theme (Adw.StyleManager.get_default ().color_scheme);
    }
}

private void update_colors_for_theme (Adw.ColorScheme scheme) {
    Gdk.RGBA foreground_color;
    Gdk.RGBA background_color;
    string[] hex_palette = { "#000000", "#cc0000", "#4e9a06", "#c4a000", "#3465a4", "#75

    Gdk.RGBA[] palette = new Gdk.RGBA[16];

    for (int i = 0; i < hex_palette.length; i++) {
        Gdk.RGBA new_color = Gdk.RGBA ();
        new_color.parse (hex_palette[i]);
        palette[i] = new_color;
    }

    if (scheme == Adw.ColorScheme.FORCE_LIGHT) {
        foreground_color = { 0.0f, 0.0f, 0.0f, 1.0f };
        background_color = { 1.0f, 1.0f, 1.0f, 1.0f };
    } else {
        foreground_color = { 1.0f, 1.0f, 1.0f, 1.0f };
        background_color = { 0.0f, 0.0f, 0.0f, 1.0f };
    }

    terminal.set_colors (foreground_color, background_color, palette);
}
```

```
    public void save_settings () {
        // Scratch.saved_state.set_string ("last-opened-path", get_shell_location ());
    }

    private void increment_size () {
        terminal.font_scale = (terminal.font_scale + FONT_SCALE_STEP).clamp (MIN_SCALE, MAX_
    }

    private void decrement_size () {
        terminal.font_scale = (terminal.font_scale - FONT_SCALE_STEP).clamp (MIN_SCALE, MAX_
    }

    public void set_default_font_size () {
        terminal.font_scale = 1.0;
    }

    private bool key_pressed (uint keyval, uint keycode, Gdk.ModifierType modifiers) {
        if (Gdk.ModifierType.CONTROL_MASK in modifiers && keyval == Gdk.Key.d) {
            terminal.reset (true, true);
            return Gdk.EVENT_STOP;
        }

        if (Gdk.ModifierType.CONTROL_MASK in modifiers &&
            (Gdk.ModifierType.SHIFT_MASK in modifiers ||
             terminal_settings != null && terminal_settings.get_boolean ("natural-copy-paste

            if (keyval == Gdk.Key.c && terminal.get_has_selection ()) {
                actions.activate_action (ACTION_COPY, null);
                return Gdk.EVENT_STOP;
                // } else if (keyval == Gdk.Key.v && current_clipboard.wait_is_text_availabl
                // actions.activate_action (ACTION_PASTE, null);
                // return Gdk.EVENT_STOP;
            }
        }

        return Gdk.EVENT_PROPAGATE;
    }
}
```

File: src/Services/Actions/ShortcutSettings.vala

```
public class Iide.ShortcutSettings : Object {
    private static ShortcutSettings? _instance;
    private GLib.Settings settings;
    private Gee.HashMap<string, string> shortcuts_cache;
```

```
private Gee.HashMap<string, bool> toggle_states_cache;

public static ShortcutSettings get_instance () {
    if (_instance == null) {
        _instance = new ShortcutSettings ();
    }
    return _instance;
}

private ShortcutSettings () {
    settings = new GLib.Settings ("org.github.kai66673.iide");
    shortcuts_cache = new Gee.HashMap<string, string> ();
    toggle_states_cache = new Gee.HashMap<string, bool> ();
    set_default_shortcuts ();
    set_default_toggle_states ();
    load_from_gsettings ();
}

private void set_default_shortcuts () {
    shortcuts_cache.set ("save", "<primary>s");
    shortcuts_cache.set ("open_project", "<primary>o");
    shortcuts_cache.set ("preferences", "<primary>comma");
    shortcuts_cache.set ("toggle_minimap", "<primary>m");
    shortcuts_cache.set ("fuzzy_finder", "<primary>p");
    shortcuts_cache.set ("search_in_files", "<primary><shift>f");
    shortcuts_cache.set ("zoom_in", "<primary>plus");
    shortcuts_cache.set ("zoom_out", "<primary>minus");
    shortcuts_cache.set ("zoom_reset", "<primary>0");
    shortcuts_cache.set ("quit", "<primary>q");
}

private void set_default_toggle_states () {
    toggle_states_cache.set ("toggle_minimap", true);
}

private void load_from_gsettings () {
    var json_str = settings.get_string ("shortcuts");
    if (json_str == null || json_str == "") {
        save_to_gsettings ();
        return;
    }

    try {
        var parser = new Json.Parser ();
        parser.load_from_data (json_str);
        var root = parser.get_root ();
```

```
        if (root.get_node_type () == Json.NodeType.OBJECT) {
            var obj = root.get_object ();
            obj.foreach_member ((obj, name, node) => {
                shortcuts_cache.set (name, node.get_string ());
            });
        }
    } catch (Error e) {
        warning ("Failed to parse shortcuts: %s", e.message);
    }
}

private void save_to_gsettings () {
    var root = new Json.Node (Json.NodeType.OBJECT);
    var obj = new Json.Object ();
    foreach (var entry in shortcuts_cache.entries) {
        obj.set_string_member (entry.key, entry.value);
    }
    root.set_object (obj);

    var generator = new Json.Generator ();
    generator.set_root (root);
    var json_str = generator.to_data (null);
    settings.set_string ("shortcuts", json_str);
}

public signal void shortcut_changed (string action_id, string? new_shortcut);

public string? get_shortcut (string action_id) {
    return shortcuts_cache.get (action_id);
}

public void set_shortcut (string action_id, string? shortcut) {
    if (shortcut == null || shortcut == "") {
        shortcuts_cache.unset (action_id);
    } else {
        shortcuts_cache.set (action_id, shortcut);
    }
    save_to_gsettings ();
    shortcut_changed (action_id, shortcut);
}

public Gee.Map<string, string?> get_all_shortcuts () {
    var result = new Gee.HashMap<string, string?> ();
    foreach (var entry in shortcuts_cache.entries) {
        result.set (entry.key, entry.value);
    }
```

```
        return result;
    }

    public void reset_shortcut (string action_id) {
        shortcuts_cache.unset (action_id);
        save_to_gsettings ();
        shortcut_changed (action_id, null);
    }

    public void reset_all_shortcuts () {
        shortcuts_cache.clear ();
        save_to_gsettings ();
        foreach (var action_id in shortcuts_cache.keys) {
            shortcut_changed (action_id, null);
        }
    }

    public bool get_toggle_state (string action_id) {
        return toggle_states_cache.get (action_id);
    }

    public void set_toggle_state (string action_id, bool state) {
        toggle_states_cache.set (action_id, state);
    }
}
```

File: src/Services/LSP/IdeLspManager.vala

```
using GLib;
using Gee;
using GtkSource;

namespace Iide {

    public class IdeLspManager : GLib.Object {
        private static IdeLspManager? _instance;
        private IdeLspService lsp_service;

        public static unowned IdeLspManager get_instance () {
            if (_instance == null) {
                _instance = new IdeLspManager ();
            }
            return _instance;
        }

        construct {
```

```
    lsp_service = IdeLspService.get_instance ();
}

public async void open_document (string uri, string language_id, string content, str
    yield lsp_service.open_document (uri, language_id, content, workspace_root, view
}

public async void change_document (string uri, string content, int? change_start = n
    yield lsp_service.change_document (uri, content, change_start, change_end);
}

public async void close_document (string uri) {
    yield lsp_service.close_document (uri);
}

public void set_language_config (string language_id, string command, string[] args,
    lsp_service.set_language_config (language_id, command, args, workspace_root);
}

public LspClient ? get_client_for_uri (string uri) {
    return lsp_service.get_client_for_uri (uri);
}

public void connect_diagnostics (DiagnosticsCallback diagnostics_callback) {
    lsp_service.diagnostics_updated.connect ((uri, diagnostics) => {
        diagnostics_callback (uri, diagnostics);
    });
}

public string ? get_language_id_for_file (GLib.File file) {
    string filename = file.get_basename () ?? "";

    switch (filename) {
    case "CMakeLists.txt" :
        return "cmake";
    case ".gitignore" :
        return "git-config";
    case "meson.build":
        return "meson";
    case "PKGBUILD":
        return "bash";
    default:
        break;
    }

    string path = file.get_path () ?? "";
```

```
int dot_pos = path.last_index_of (".");
if (dot_pos >= 0 && dot_pos < path.length - 1) {
    string ext = path[dot_pos + 1 : path.length].down ();
    switch (ext) {
    case "py":
        return "python";
    case "c":
    case "h":
        return "c";
    case "cpp":
    case "cc":
    case "cxx":
    case "hpp":
    case "hxx":
        return "cpp";
    case "vala":
    case "vapi":
        return "vala";
    case "rs":
        return "rust";
    case "go":
        return "go";
    case "js":
    case "ts":
        return "javascript";
    case "json":
        return "json";
    case "xml":
        return "xml";
    case "html":
    case "htm":
        return "html";
    case "css":
        return "css";
    case "md":
    case "markdown":
        return "markdown";
    case "sh":
    case "bash":
    case "zsh":
        return "bash";
    case "yaml":
    case "yml":
        return "yaml";
    }
}
```

```
            return null;
        }

        public delegate void DiagnosticsCallback (string uri, Gee.ArrayList<IdeLspDiagnostic
    }
}
```

File: src/Services/LSP/LspClient.vala

```
using GLib;
using Gee;

[CCode (cheader_filename = "unistd.h")]
extern int getpid ();

public class Iide.LspPromise : Object {
    public SourceFunc callback;
    //
    public int64 creation_time;

    public LspPromise (owned SourceFunc cb) {
        this.callback = (owned) cb;
        this.creation_time = get_monotonic_time ();
    }
}

public enum Iide.TextDocumentSyncKind {
    NONE = 0,
    FULL = 1,
    INCREMENTAL = 2
}

public class Iide.ServerCapabilities : Object {
    public TextDocumentSyncKind sync_kind = TextDocumentSyncKind.FULL;
    public HashSet<string> completion_triggers = new HashSet<string> ();
    public bool hover_provider = false;
    public bool definition_provider = false;

    // To features...
    public bool references_provider = false;
    public bool rename_provider = false;
}

public class Iide.LspClient : Object {
    private int next_id = 1;
```

```
private Map<int, LspPromise> pending_requests = new HashMap<int, LspPromise> ();
private Map<int, Json.Object?> responses = new HashMap<int, Json.Object?> ();

//      LSP
private GLib.Subprocess process;

//           (              )
private GLib.DataInputStream input_stream;

//           (              )
private bool is_writing = false;
private Deque<string> write_queue = new LinkedList<string> ();
private OutputStream output_stream;

//                      UI
public signal void diagnostics_received (string uri, Gee.ArrayList<IdeLspDiagnostic> dia

//
public signal void log_message (int type, string message);

//
public ServerCapabilities capabilities { get; private set; }

//    ,       ,
public signal void initialized_with_capabilities (ServerCapabilities caps);

public bool is_initialized { get; private set; default = false; }

public LspClient () {
    this.capabilities = new ServerCapabilities ();
}

public async Json.Object? send_request (string method, Json.Object params) throws Error
    int id = next_id++;

    // 1.
    var root = new Json.Object ();
    root.set_string_member ("jsonrpc", "2.0");
    root.set_int_member ("id", id);
    root.set_string_member ("method", method);
    root.set_object_member ("params", params);

    // 2.              (Promise)
    //        send_request.callback,      Vala
    //                      yield
    pending_requests.set (id, new LspPromise (send_request.callback));
```

```
    // 3.       (      send_message_async         )
    yield this.send_message_async (root);

    // 4.
    yield;

    // 5.
    var response = responses.get (id);
    responses.unset (id);

    return response;
}

private async void send_message_async (Json.Object node) throws Error {
    // 1.
    var generator = new Json.Generator ();
    var root_node = new Json.Node (Json.NodeType.OBJECT);
    root_node.set_object (node);
    generator.set_root (root_node);

    string body = generator.to_data (null);
    string message = "Content-Length: %d\r\n\r\n%s".printf ((int) body.length, body);

    // 2.
    write_queue.add (message);

    // 3.          -         ,
    if (is_writing)return;

    is_writing = true;

    try {
        while (!write_queue.is_empty) {
            string current_msg = write_queue.poll_head ();
            yield output_stream.write_all_async (current_msg.data, Priority.DEFAULT, nul

            yield output_stream.flush_async (Priority.DEFAULT, null);
        }
    } finally {
        is_writing = false;
    }
}

private async void run_read_loop () {
    try {
```

```
        while (true) {
            // 1.           Content-Length
            string? line = yield input_stream.read_line_async (Priority.DEFAULT, null);

            if (line == null)break; //

            if (line.has_prefix ("Content-Length: ")) {
                int length = int.parse (line.substring (16).strip ());

                // 2.                                (\r\n\r\n)
                while (line != "" && line != null) {
                    line = yield input_stream.read_line_async (Priority.DEFAULT, null);

                    if (line != null)line = line.strip ();
                }

                // 3.         JSON
                uint8[] buffer = new uint8[length + 1];
                size_t bytes_read;
                yield input_stream.read_all_async (buffer[0 : length], Priority.DEFAULT,

                buffer[length] = '\0'; //

                // 4.
                this.handle_payload ((string) buffer);
            }
        }
    } catch (Error e) {
        if (!(e is IOError.CANCELLED)) {
            debug ("LSP Read Loop Error: %s", e.message);
        }
    }
}

private void handle_payload (string payload) {
    try {
        var parser = new Json.Parser ();
        parser.load_from_data (payload);
        var root = parser.get_root ().get_object ();

        //                (   'id')
        if (root.has_member ("id")) {
            this.handle_response (root);
        }
        //                        (   'method',    'id')
        else if (root.has_member ("method")) {
```

```
            this.handle_incoming_notification (root);
        }
    } catch (Error e) {
        warning ("Failed to parse LSP payload: %s", e.message);
    }
}

private void handle_response (Json.Object response) {
    int id = (int) response.get_int_member ("id");

    if (pending_requests.has_key (id)) {
        var promise = pending_requests.get (id);
        pending_requests.unset (id);

        //
        responses.set (id, response);

        //
        //       Idle.add,            MainContext (UI    )
        Idle.add ((owned) promise.callback);
    }
}

private void handle_incoming_notification (Json.Object root) {
    string method = root.get_string_member ("method");

    //                        (params)
    Json.Object? params = null;
    if (root.has_member ("params")) {
        params = root.get_object_member ("params");
    }

    switch (method) {
    case "textDocument/publishDiagnostics" :
        if (params != null) {
            string uri = params.get_string_member ("uri");
            var json_array = params.get_array_member ("diagnostics");

            //     JSON-                   IdeLspDiagnostic
            var diag_list = this.parse_diagnostics (json_array);

            //                   UI
            Idle.add (() => {
                this.diagnostics_received (uri, diag_list);
                return Source.REMOVE;
            });
```

```
        }
        break;

    case "window/logMessage" :
        if (params != null)handle_log_message (params);
        break;

    case "window/showMessage":
        //                                  GNOME
        if (params != null)debug ("LSP Show Message: %s", params.get_string_member ("mes
        break;

    default:
        debug ("Unhandled LSP notification: %s", method);
        break;
    }
}

private void handle_log_message (Json.Object params) {
    int type = (int) params.get_int_member ("type");
    string message = params.get_string_member ("message");

    Idle.add (() => {
        this.log_message (type, message);
        return Source.REMOVE;
    });
}

public async bool start_server_async (string command, string[] args, string? workspace_r
    try {
        // 1.
        string[] argv = { command };
        foreach (var arg in args)argv += arg;

        // 2.
        var launcher = new SubprocessLauncher (SubprocessFlags.STDOUT_PIPE | SubprocessF
        this.process = launcher.spawnv (argv);

        // 3.
        this.output_stream = this.process.get_stdin_pipe ();
        this.input_stream = new DataInputStream (this.process.get_stdout_pipe ());

        // 4.             (                  )
        this.run_read_loop.begin ();

        // 5.    INITIALIZE
```

```
        var init_params = build_init_params (workspace_root, initialization_options);
        var response = yield this.send_request ("initialize", init_params.get_object ()

        if (response != null && response.has_member ("result")) {
            var result = response.get_object_member ("result");
            //
            this.parse_capabilities (result);
        }

        //       Idle.add,
        is_initialized = true;
        Idle.add (() => {
            this.initialized_with_capabilities (this.capabilities);
            return Source.REMOVE; //
        });

        // 6.    INITIALIZED (              )
        yield this.send_notification_async ("initialized", new Json.Object ());

        debug ("LSP Server started and initialized for: %s", workspace_root ?? "unknown'
        return true;
    } catch (Error e) {
        warning ("Failed to start LSP server: %s", e.message);
        return false;
    }
}

private void parse_capabilities (Json.Object result) {
    if (!result.has_member ("capabilities"))return;
    var caps = result.get_object_member ("capabilities");

    //
    if (caps.has_member ("textDocumentSync")) {
        var sync = caps.get_member ("textDocumentSync");
        if (sync.get_node_type () == Json.NodeType.OBJECT) {
            var sync_obj = sync.get_object ();
            if (sync_obj.has_member ("change"))
                this.capabilities.sync_kind = (TextDocumentSyncKind) sync_obj.get_int_me
        } else if (sync.get_node_type () == Json.NodeType.VALUE) {
            this.capabilities.sync_kind = (TextDocumentSyncKind) sync.get_int ();
        }
    }

    //
    if (caps.has_member ("completionProvider")) {
        var comp = caps.get_object_member ("completionProvider");
```

```
        if (comp.has_member ("triggerCharacters")) {
            var triggers = comp.get_array_member ("triggerCharacters");
            this.capabilities.completion_triggers.clear ();
            foreach (var node in triggers.get_elements ()) {
                this.capabilities.completion_triggers.add (node.get_string ());
            }
        }
    }

    // Hover & Definition
    this.capabilities.hover_provider = caps.has_member ("hoverProvider") && caps.get_boo
    this.capabilities.definition_provider = caps.has_member ("definitionProvider") && ca
}

private Json.Node build_init_params (string? workspace_root, Json.Node? initialization_o
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("processId");
    builder.add_null_value ();
    builder.set_member_name ("clientInfo");
    builder.begin_object ();
    builder.set_member_name ("name");
    builder.add_string_value ("iide");
    builder.set_member_name ("version");
    builder.add_string_value ("0.1.0");
    builder.end_object ();
    builder.set_member_name ("rootUri");
    if (workspace_root != null) {
        builder.add_string_value (workspace_root);
    } else {
        builder.add_null_value ();
    }
    builder.set_member_name ("workspaceFolders");
    builder.begin_array ();
    if (workspace_root != null) {
        builder.begin_object ();
        builder.set_member_name ("uri");
        builder.add_string_value (workspace_root);
        builder.set_member_name ("name");
        builder.add_string_value ("workspace");
        builder.end_object ();
    }
    builder.end_array ();
    builder.set_member_name ("capabilities");
    builder.begin_object ();
    builder.set_member_name ("textDocument");
```

```
    builder.begin_object ();
    builder.set_member_name ("syncKind");
    builder.add_int_value (1);
    builder.end_object ();
    builder.end_object ();
    builder.set_member_name ("workspace");
    builder.begin_object ();
    builder.set_member_name ("workspaceFolders");
    builder.add_boolean_value (true);
    builder.end_object ();
    builder.end_object ();
    if (initialization_options != null) {
        builder.set_member_name ("initializationOptions");
        builder.add_value (initialization_options.copy ());
    }
    builder.end_object ();

    return builder.get_root ();
}

public async void send_notification_async (string method, Json.Object params) throws Err
    var root = new Json.Object ();
    root.set_string_member ("jsonrpc", "2.0");
    root.set_string_member ("method", method);
    root.set_object_member ("params", params);

    //
    yield this.send_message_async (root);
}

private Gee.ArrayList<IdeLspDiagnostic> parse_diagnostics (Json.Array diagnostics_array)
    var result = new Gee.ArrayList<IdeLspDiagnostic> ();

    foreach (var diag_node in diagnostics_array.get_elements ()) {
        var diag_obj = diag_node.get_object ();
        var d = new IdeLspDiagnostic ();

        //
        d.message = diag_obj.get_string_member ("message");
        if (diag_obj.has_member ("severity")) {
            d.severity = (int) diag_obj.get_int_member ("severity");
        }

        //       (Range   LSP      start   end     )
        var range = diag_obj.get_object_member ("range");
        var start = range.get_object_member ("start");
```

```
        var end = range.get_object_member ("end");

        d.start_line = (int) start.get_int_member ("line");
        d.start_column = (int) start.get_int_member ("character"); // character -> start
        d.end_line = (int) end.get_int_member ("line");
        d.end_column = (int) end.get_int_member ("character"); // character -> end_colum

        result.add (d);
    }

    return result;
}

public async void text_document_did_open (string uri, string language_id, int version, s
    var params = new Json.Object ();

    var doc = new Json.Object ();
    doc.set_string_member ("uri", uri);
    doc.set_string_member ("languageId", language_id);
    doc.set_int_member ("version", version);
    doc.set_string_member ("text", content);

    params.set_object_member ("textDocument", doc);

    //           (notification),           ID
    yield this.send_notification_async ("textDocument/didOpen", params);

    debug ("LSP: Sent didOpen for %s", uri);
}

public async void text_document_did_change (string uri, int version, string content) thr
    var params = new Json.Object ();

    // 1.
    var doc = new Json.Object ();
    doc.set_string_member ("uri", uri);
    doc.set_int_member ("version", version);
    params.set_object_member ("textDocument", doc);

    // 2.            (      range)
    var change = new Json.Object ();
    change.set_string_member ("text", content);

    var changes = new Json.Array ();
    changes.add_object_element (change);
    params.set_array_member ("contentChanges", changes);
```

```
    yield this.send_notification_async ("textDocument/didChange", params);
}

public async void send_did_change (string uri, int version, Gee.ArrayList<PendingChange>
    var params = new Json.Object ();

    // 1.
    var doc = new Json.Object ();
    doc.set_string_member ("uri", uri);
    doc.set_int_member ("version", version);
    params.set_object_member ("textDocument", doc);

    // 2.
    var json_changes = new Json.Array ();
    foreach (var c in changes) {
        var obj = new Json.Object ();

        //            (range)
        var range = new Json.Object ();

        var start = new Json.Object ();
        start.set_int_member ("line", c.start_line);
        start.set_int_member ("character", c.start_char);

        var end = new Json.Object ();
        end.set_int_member ("line", c.end_line);
        end.set_int_member ("character", c.end_char);

        range.set_object_member ("start", start);
        range.set_object_member ("end", end);

        obj.set_object_member ("range", range);
        obj.set_string_member ("text", c.text);

        json_changes.add_object_element (obj);
    }
    params.set_array_member ("contentChanges", json_changes);

    yield this.send_notification_async ("textDocument/didChange", params);
}

public async void text_document_did_close (string uri) throws Error {
    var params = new Json.Object ();
    var doc = new Json.Object ();
    doc.set_string_member ("uri", uri);
```

```
    params.set_object_member ("textDocument", doc);

    yield this.send_notification_async ("textDocument/didClose", params);
}

private IdeLspCompletionResult parse_completion_result (Json.Node node) {
    var res = new IdeLspCompletionResult ();
    res.items = new Gee.ArrayList<IdeLspCompletionItem> ();

    Json.Array? items_array = null;

    //     1:              CompletionList { isIncomplete, items }
    if (node.get_node_type () == Json.NodeType.OBJECT) {
        var obj = node.get_object ();
        if (obj.has_member ("isIncomplete")) {
            res.is_incomplete = obj.get_boolean_member ("isIncomplete");
        }
        if (obj.has_member ("items")) {
            items_array = obj.get_array_member ("items");
        }
    }
    //     2:                  CompletionItem[]
    else if (node.get_node_type () == Json.NodeType.ARRAY) {
        items_array = node.get_array ();
    }

    if (items_array != null) {
        foreach (var item_node in items_array.get_elements ()) {
            var item_obj = item_node.get_object ();
            var item = new IdeLspCompletionItem ();

            item.label = item_obj.get_string_member ("label");

            //
            if (item_obj.has_member ("insertText"))
                item.insert_text = item_obj.get_string_member ("insertText");
            else
                item.insert_text = item.label;

            if (item_obj.has_member ("detail"))
                item.detail = item_obj.get_string_member ("detail");

            if (item_obj.has_member ("kind"))
                item.kind = (int) item_obj.get_int_member ("kind");

            //              (                    MarkupContent)
```

```
            if (item_obj.has_member ("documentation")) {
                var doc_node = item_obj.get_member ("documentation");
                if (doc_node.get_node_type () == Json.NodeType.OBJECT)
                    item.documentation = doc_node.get_object ().get_string_member ("valu
                else
                    item.documentation = doc_node.get_string ();
            }

            res.items.add (item);
        }
    }

    return res;
}

public async IdeLspCompletionResult ? request_completion (string uri, int line, int char
    var params = new Json.Object ();

    var doc = new Json.Object ();
    doc.set_string_member ("uri", uri);
    params.set_object_member ("textDocument", doc);

    var pos = new Json.Object ();
    pos.set_int_member ("line", line);
    pos.set_int_member ("character", character);
    params.set_object_member ("position", pos);

    var context = new Json.Object ();
    context.set_int_member ("triggerKind", (int) trigger_kind);
    if (trigger_char != null) {
        context.set_string_member ("triggerCharacter", trigger_char);
    }
    params.set_object_member ("context", context);

    var response = yield this.send_request ("textDocument/completion", params);

    if (response == null || !response.has_member ("result"))return null;

    var result_node = response.get_member ("result");
    if (result_node.get_node_type () == Json.NodeType.NULL)return null;

    return parse_completion_result (result_node);
}

private string ? parse_hover_result (Json.Node node) {
    if (node.get_node_type () != Json.NodeType.OBJECT)
```

```
        return null;

    var result = node.get_object ();

    //     1:            'contents'
    if (!result.has_member ("contents"))
        return null;

    var contents = result.get_member ("contents");

    //           (MarkupContent)
    if (contents.get_node_type () == Json.NodeType.OBJECT) {
        var obj = contents.get_object ();
        if (obj.has_member ("value")) {
            return obj.get_string_member ("value");
        }
    }
    //
    else {
        return contents.get_string ();
    }

    return null;
}

public async string ? request_hover (string uri, int line, int character) throws Error {
    var params = new Json.Object ();

    var doc = new Json.Object ();
    doc.set_string_member ("uri", uri);
    params.set_object_member ("textDocument", doc);

    var pos = new Json.Object ();
    pos.set_int_member ("line", line);
    pos.set_int_member ("character", character);
    params.set_object_member ("position", pos);

    var response = yield this.send_request ("textDocument/hover", params);

    if (response == null || !response.has_member ("result"))return null;

    return parse_hover_result (response.get_member ("result"));
}

public async Gee.ArrayList<IdeLspLocation>? request_definition (string uri, int line, in
    var params = new Json.Object ();
```

```
    var doc = new Json.Object ();
    doc.set_string_member ("uri", uri);
    params.set_object_member ("textDocument", doc);

    var pos = new Json.Object ();
    pos.set_int_member ("line", line);
    pos.set_int_member ("character", character);
    params.set_object_member ("position", pos);

    var response = yield this.send_request ("textDocument/definition", params);

    if (response == null || !response.has_member ("result"))return null;

    var result_node = response.get_member ("result");
    if (result_node.get_node_type () == Json.NodeType.NULL)return null;

    return parse_definition_result (result_node);
}

private Gee.ArrayList<IdeLspLocation> parse_definition_result (Json.Node node) {
    var locations = new Gee.ArrayList<IdeLspLocation> ();

    //     1:           Location {}
    if (node.get_node_type () == Json.NodeType.OBJECT) {
        var loc = parse_single_location (node.get_object ());
        if (loc != null)locations.add (loc);
    }
    //     2:          Location[]
    else if (node.get_node_type () == Json.NodeType.ARRAY) {
        var array = node.get_array ();
        foreach (var element in array.get_elements ()) {
            var loc = parse_single_location (element.get_object ());
            if (loc != null)locations.add (loc);
        }
    }

    return locations;
}

private IdeLspLocation ? parse_single_location (Json.Object obj) {
    var loc = new IdeLspLocation ();
    loc.uri = obj.get_string_member ("uri");

    var range = obj.get_object_member ("range");
    var start = range.get_object_member ("start");
```

```
        var end = range.get_object_member ("end");

        loc.start_line = (int) start.get_int_member ("line");
        loc.start_column = (int) start.get_int_member ("character");
        loc.end_line = (int) end.get_int_member ("line");
        loc.end_column = (int) end.get_int_member ("character");

        return loc;
    }
}
```

File: src/Services/IconProvider.vala

```
public class Iide.IconProvider {
    private static IconProvider? instance;
    private IconProvider() {}

    public static IconProvider get_instance() {
        if (instance == null) {
            instance = new IconProvider();
        }
        return instance;
    }

    public static string ? get_mime_type_icon_name(string mime_type) {
        switch (mime_type) {
        case "text/x-vala" :
            return "text-x-vala";
        case "text/x-meson":
            return "text-x-meson";
        case "text/markdown":
            return "text-markdown";
        case "application/json":
            return "text-x-javascript";
        case "application/x-gtk-builder":
            return "text-xml";
        }
        return "text-x-generic";
    }
}
```

File: src/Services/LoggerService.vala

```
/*
 * loggerservice.vala
 *
```

```
 * Copyright 2026 kai
 *
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 * along with this program.  If not, see <https://www.gnu.org/licenses/>.
 *
 * SPDX-License-Identifier: GPL-3.0-or-later
 */

using GLib;

public enum Iide.LogLevel {
    DEBUG,
    INFO,
    WARNING,
    ERROR,
    CRITICAL;

    public string to_string () {
        switch (this) {
        case DEBUG:    return "DEBUG";
        case INFO:     return "INFO";
        case WARNING:  return "WARN";
        case ERROR:    return "ERROR";
        case CRITICAL: return "CRITICAL";
        default:       return "UNKNOWN";
        }
    }

    public string get_color () {
        switch (this) {
        case DEBUG:    return "#888888";
        case INFO:     return "#4a9eff";
        case WARNING:  return "#f5a623";
        case ERROR:    return "#e74c3c";
        case CRITICAL: return "#ff4757";
        default:       return "#ffffff";
```

```
        }
    }
}

public class Iide.LogEntry : Object {
    public int64 timestamp { get; set; }
    public Iide.LogLevel level { get; set; }
    public string domain { get; set; }
    public string message { get; set; }
    public string? details { get; set; }

    public string get_timestamp_string () {
        var time = new DateTime.from_unix_utc (timestamp / 1000000);
        var local = time.to_timezone (new TimeZone.local ());
        var msec = (int) ((timestamp / 1000) % 1000);
        return "%s.%03d".printf (local.format ("%H:%M:%S"), msec);
    }
}

public class Iide.LoggerService : Object {
    private static LoggerService? _instance;
    private Gee.ArrayList<Iide.LogEntry> entries;
    private int max_entries = 1000;
    private bool enable_logging = true;
    private File log_file;

    public signal void log_added (Iide.LogEntry entry);
    public signal void log_cleared ();

    public static LoggerService get_instance () {
        if (_instance == null) {
            _instance = new LoggerService ();
        }
        return _instance;
    }

    private LoggerService () {
        entries = new Gee.ArrayList<Iide.LogEntry> ();
        log_file = File.new_for_path (
                                        Path.build_filename (Environment.get_user_data_dir ()
        );
        try {
            var dir = log_file.get_parent ();
            if (dir != null && !dir.query_exists (null)) {
                dir.make_directory_with_parents (null);
            }
```

```
    } catch (Error e) {
        stderr.printf ("Failed to create log directory: %s\n", e.message);
    }
}

public void log (LogLevel level, string domain, string message, string? details = null)
    if (!enable_logging)return;

    var entry = new Iide.LogEntry () {
        timestamp = (int64) get_real_time (),
        level = level,
        domain = domain,
        message = message,
        details = details
    };

    entries.add (entry);

    if (entries.size > max_entries) {
        entries.remove_at (0);
    }

    log_to_file (entry);
    log_added (entry);
}

public void debug (string domain, string message, string? details = null) {
    log (LogLevel.DEBUG, domain, message, details);
}

public void info (string domain, string message, string? details = null) {
    log (LogLevel.INFO, domain, message, details);
}

public void warning (string domain, string message, string? details = null) {
    log (LogLevel.WARNING, domain, message, details);
}

public void error (string domain, string message, string? details = null) {
    log (LogLevel.ERROR, domain, message, details);
}

public void critical (string domain, string message, string? details = null) {
    log (LogLevel.CRITICAL, domain, message, details);
}
```

```
private void log_to_file (LogEntry entry) {
    try {
        var time = new DateTime.from_unix_utc (entry.timestamp / 1000000);
        var local = time.to_timezone (new TimeZone.local ());
        var timestamp = local.format ("%Y-%m-%d %H:%M:%S");
        var details_str = entry.details != null ? " | " + entry.details : "";
        var line = "[%s] [%s] [%s] %s%s\n".printf (timestamp, entry.level.to_string (),

        if (log_file.query_exists (null)) {
            log_file.append_to (FileCreateFlags.NONE, null).write (line.data);
        } else {
            log_file.create (FileCreateFlags.REPLACE_DESTINATION, null).write (line.data
        }
    } catch (Error e) {
        stderr.printf ("Failed to write to log file: %s\n", e.message);
    }
}

public Gee.ArrayList<Iide.LogEntry> get_entries () {
    return entries;
}

public Gee.ArrayList<Iide.LogEntry> get_entries_by_level (LogLevel level) {
    var result = new Gee.ArrayList<Iide.LogEntry> ();
    foreach (var entry in entries) {
        if (entry.level == level) {
            result.add (entry);
        }
    }
    return result;
}

public void clear () {
    entries.clear ();
    log_cleared ();
}

public void set_max_entries (int max) {
    max_entries = max;
    while (entries.size > max_entries) {
        entries.remove_at (0);
    }
}

public void set_logging_enabled (bool enabled) {
    enable_logging = enabled;
```

```
    }
}
```

File: src/Services/PanelLayoutHelper.vala

```
/*
 * panellayouthelper.vala
 *
 * Copyright 2026 kai
 *
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 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation, either version 3 of the License, or
 * (at your option) any later version.
 *
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 * but WITHOUT ANY WARRANTY; without even the implied warranty of
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 * GNU General Public License for more details.
 *
 * You should have received a copy of the GNU General Public License
 * along with this program.  If not, see <https://www.gnu.org/licenses/>.
 *
 * SPDX-License-Identifier: GPL-3.0-or-later
 */

using Adw;

public class Iide.PanelLayoutHelper : Object {

    public class WidgetInfo {
        public string title { get; set; }
        public string icon_name { get; set; }
        public string id { get; set; }
        public string kind { get; set; }
        public int area { get; set; }
        public uint column { get; set; }
        public uint row { get; set; }
        public uint depth { get; set; }
    }

    public class DocumentInfo {
        public string uri { get; set; }
        public uint column { get; set; }
        public uint row { get; set; }
    }
```

```
public static string serialize_dock (Panel.Dock dock) {
    var builder = new Json.Builder ();
    builder.begin_object ();

    builder.set_member_name ("reveal_start");
    builder.add_boolean_value (dock.reveal_start);
    builder.set_member_name ("reveal_end");
    builder.add_boolean_value (dock.reveal_end);
    builder.set_member_name ("reveal_bottom");
    builder.add_boolean_value (dock.reveal_bottom);
    builder.set_member_name ("start_width");
    builder.add_int_value (dock.start_width);
    builder.set_member_name ("end_width");
    builder.add_int_value (dock.end_width);
    builder.set_member_name ("bottom_height");
    builder.add_int_value (dock.bottom_height);

    builder.set_member_name ("widgets");
    builder.begin_array ();

    dock.foreach_frame ((frame) => {
        var pages = frame.get_pages ();
        var position = frame.get_position ();

        for (uint i = 0; i < pages.get_n_items (); i++) {
            var item = pages.get_item (i) as Adw.TabPage;
            if (item == null) {
                continue;
            }
            var child = item.get_child ();
            if (child == null) {
                continue;
            }
            var widget = child as Panel.Widget;
            if (widget == null) {
                continue;
            }

            builder.begin_object ();
            builder.set_member_name ("title");
            builder.add_string_value (widget.get_title () ?? "");
            builder.set_member_name ("icon_name");
            builder.add_string_value (widget.get_icon_name () ?? "");
            builder.set_member_name ("id");
            builder.add_string_value (widget.get_id () ?? "");
```

```
            builder.set_member_name ("kind");
            builder.add_string_value (widget.get_kind () ?? "");

            if (position != null) {
                builder.set_member_name ("area");
                builder.add_int_value ((int) position.get_area ());
                builder.set_member_name ("column");
                builder.add_int_value ((int) position.get_column ());
                builder.set_member_name ("row");
                builder.add_int_value ((int) position.get_row ());
                builder.set_member_name ("depth");
                builder.add_int_value ((int) position.get_depth ());
            }

            builder.end_object ();
        }
    });

    builder.end_array ();
    builder.end_object ();

    var generator = new Json.Generator ();
    generator.root = builder.get_root ();
    return generator.to_data (null);
}

public static Gee.ArrayList<WidgetInfo> parse_widgets (string data) {
    var result = new Gee.ArrayList<WidgetInfo> ();

    if (data == null || data == "") {
        return result;
    }

    try {
        var parser = new Json.Parser ();
        parser.load_from_data (data);
        var root = parser.get_root ().get_object ();

        if (root.has_member ("widgets")) {
            var widgets_array = root.get_array_member ("widgets");
            foreach (var node in widgets_array.get_elements ()) {
                var obj = node.get_object ();
                var info = new WidgetInfo ();
                info.title = obj.get_string_member ("title");
                info.icon_name = obj.has_member ("icon_name") ? obj.get_string_member ("
                info.id = obj.has_member ("id") ? obj.get_string_member ("id") : "";
```

```
                info.kind = obj.has_member ("kind") ? obj.get_string_member ("kind") : '
                info.area = obj.has_member ("area") ? (int) obj.get_int_member ("area")
                info.column = obj.has_member ("column") ? (uint) obj.get_int_member ("co
                info.row = obj.has_member ("row") ? (uint) obj.get_int_member ("row") :
                info.depth = obj.has_member ("depth") ? (uint) obj.get_int_member ("dept
                result.add (info);
            }
        }
    } catch (Error e) {
        warning ("Failed to parse widgets: %s", e.message);
    }

    return result;
}

public static void deserialize_dock (string data, Panel.Dock dock) {
    if (data == null || data == "") {
        return;
    }

    try {
        var parser = new Json.Parser ();
        parser.load_from_data (data);
        var root = parser.get_root ().get_object ();

        if (root.has_member ("reveal_start")) {
            dock.reveal_start = root.get_boolean_member ("reveal_start");
        }
        if (root.has_member ("reveal_end")) {
            dock.reveal_end = root.get_boolean_member ("reveal_end");
        }
        if (root.has_member ("reveal_bottom")) {
            dock.reveal_bottom = root.get_boolean_member ("reveal_bottom");
        }
        if (root.has_member ("start_width")) {
            dock.start_width = (int) root.get_int_member ("start_width");
        }
        if (root.has_member ("end_width")) {
            dock.end_width = (int) root.get_int_member ("end_width");
        }
        if (root.has_member ("bottom_height")) {
            dock.bottom_height = (int) root.get_int_member ("bottom_height");
        }
    } catch (Error e) {
        warning ("Failed to parse dock layout: %s", e.message);
    }
```

```
}

public static string serialize_grid (Panel.Grid grid) {
    var builder = new Json.Builder ();
    builder.begin_object ();

    builder.set_member_name ("n_columns");
    builder.add_int_value ((int) grid.get_n_columns ());

    builder.set_member_name ("documents");
    builder.begin_array ();

    grid.foreach_frame ((frame) => {
        var pages = frame.get_pages ();
        var position = frame.get_position ();

        for (uint i = 0; i < pages.get_n_items (); i++) {
            var item = pages.get_item (i) as Adw.TabPage;
            if (item == null) {
                continue;
            }
            var child = item.get_child ();
            if (child == null) {
                continue;
            }
            var widget = child as Panel.Widget;
            if (widget == null) {
                continue;
            }
            var text_view = widget as Iide.TextView;
            if (text_view == null) {
                continue;
            }

            builder.begin_object ();
            builder.set_member_name ("uri");
            builder.add_string_value (text_view.uri);
            if (position != null) {
                builder.set_member_name ("column");
                builder.add_int_value ((int) position.get_column ());
                builder.set_member_name ("row");
                builder.add_int_value ((int) position.get_row ());
            }
            builder.end_object ();
        }
    });
```

```
        builder.end_array ();
        builder.end_object ();

        var generator = new Json.Generator ();
        generator.root = builder.get_root ();
        return generator.to_data (null);
    }

    public static Gee.ArrayList<DocumentInfo> parse_grid_documents (string data) {
        var result = new Gee.ArrayList<DocumentInfo> ();

        if (data == null || data == "") {
            return result;
        }

        try {
            var parser = new Json.Parser ();
            parser.load_from_data (data);
            var root = parser.get_root ().get_object ();

            if (root.has_member ("documents")) {
                var docs_array = root.get_array_member ("documents");
                foreach (var node in docs_array.get_elements ()) {
                    var obj = node.get_object ();
                    var info = new DocumentInfo ();
                    info.uri = obj.get_string_member ("uri");
                    info.column = obj.has_member ("column") ? (uint) obj.get_int_member ("co
                    info.row = obj.has_member ("row") ? (uint) obj.get_int_member ("row") :
                    result.add (info);
                }
            }
        } catch (Error e) {
            warning ("Failed to parse grid documents: %s", e.message);
        }

        return result;
    }
}
```

File: src/Widgets/FuzzyFinderDialog.vala

```
public class Iide.FuzzyFinderDialog : Adw.Window {
    private Gtk.Entry search_entry;
    private Gtk.ListView list_view;
    private Gtk.SingleSelection selection;
```

```
private Gtk.StringList string_list;
private Gee.List<FileEntry> all_files;
private Gee.List<FileEntry> filtered_files;
private Iide.ProjectManager project_manager;
private Iide.DocumentManager document_manager;
private Window? parent_window;

private const int MAX_RESULTS = 50;

private class FileEntry : Object {
    public string path { get; construct; }
    public string name { get; construct; }
    public string relative_path { get; construct; }
    public string display_name { get; construct; }

    public FileEntry (string path, string name, string relative_path) {
        Object (path: path, name: name, relative_path: relative_path, display_name: "%s
    }
}

public FuzzyFinderDialog (Window parent_window, Iide.DocumentManager document_manager) {
    Object (
            title: _("Quick Open"),
            modal: true,
            destroy_with_parent: true,
            default_width: 600,
            default_height: 400
    );

    this.parent_window = parent_window;
    this.document_manager = document_manager;
    this.project_manager = Iide.ProjectManager.get_instance ();
    this.all_files = new Gee.ArrayList<FileEntry> ();
    this.filtered_files = new Gee.ArrayList<FileEntry> ();

    setup_ui ();
    load_project_files ();
}

private bool on_key_pressed (Gtk.EventControllerKey controller, uint keyval, uint keyco
    if (keyval == Gdk.Key.Escape) {
        this.close ();
        return true;
    } else if (keyval == Gdk.Key.Return || keyval == Gdk.Key.KP_Enter) {
        open_selected ();
        return true;
```

```
    } else if (keyval == Gdk.Key.Up || keyval == Gdk.Key.KP_Up) {
        if (selection.selected > 0) {
            selection.selected -= 1;
            list_view.scroll_to (selection.selected, Gtk.ListScrollFlags.NONE, null);
        }
        return true;
    } else if (keyval == Gdk.Key.Down || keyval == Gdk.Key.KP_Down) {
        if (selection.selected < (int) string_list.get_n_items () - 1) {
            selection.selected += 1;
            list_view.scroll_to (selection.selected, Gtk.ListScrollFlags.NONE, null);
        }
        return true;
    }
    return false;
}

private void setup_ui () {
    var vbox = new Gtk.Box (Gtk.Orientation.VERTICAL, 8);
    vbox.margin_top = 12;
    vbox.margin_bottom = 12;
    vbox.margin_start = 12;
    vbox.margin_end = 12;

    search_entry = new Gtk.Entry () {
        placeholder_text = _("Search files..."),
        hexpand = true
    };
    vbox.append (search_entry);

    string_list = new Gtk.StringList (new string[0]);
    selection = new Gtk.SingleSelection (string_list);
    list_view = new Gtk.ListView (selection, null);
    list_view.hexpand = true;
    list_view.vexpand = true;

    var factory = new Gtk.SignalListItemFactory ();
    factory.setup.connect ((item) => {
        var list_item = item as Gtk.ListItem;
        var item_box = new Gtk.Box (Gtk.Orientation.VERTICAL, 2);
        item_box.margin_start = 6;
        item_box.margin_end = 6;
        item_box.margin_top = 6;
        item_box.margin_bottom = 6;

        var name_label = new Gtk.Label (null);
        name_label.xalign = 0;
```

```
    name_label.add_css_class ("title-5");
    name_label.hexpand = true;

    var path_label = new Gtk.Label (null);
    path_label.xalign = 0;
    path_label.add_css_class ("dim-label");
    path_label.hexpand = true;

    item_box.append (name_label);
    item_box.append (path_label);
    list_item.set_child (item_box);
});
factory.bind.connect ((item) => {
    var list_item = item as Gtk.ListItem;
    var item_box = list_item.get_child () as Gtk.Box;
    var name_label = item_box.get_first_child () as Gtk.Label;
    var path_label = name_label.get_next_sibling () as Gtk.Label;

    var index = list_item.get_position ();
    if (index >= 0 && index < filtered_files.size) {
        var entry = filtered_files[(int) index];
        name_label.set_label (entry.name);
        path_label.set_label (entry.relative_path);
    }
});
list_view.factory = factory;

var scrolled = new Gtk.ScrolledWindow ();
scrolled.child = list_view;
scrolled.hexpand = true;
scrolled.vexpand = true;
vbox.append (scrolled);

set_content (vbox);

search_entry.changed.connect (on_search_changed);

var key_controller = new Gtk.EventControllerKey ();
key_controller.key_pressed.connect (on_key_pressed);
search_entry.add_controller (key_controller);

list_view.activate.connect (() => {
    open_selected ();
});

search_entry.activate.connect (() => {
```

```
        open_selected ();
    });
}

private void load_project_files () {
    var project_root = project_manager.get_current_project_root ();
    if (project_root == null) {
        return;
    }

    var root_path = project_root.get_path ();
    if (root_path == null) {
        return;
    }

    try {
        scan_directory (project_root, root_path);
    } catch (Error e) {
        warning ("Error scanning directory: %s", e.message);
    }

    all_files.sort ((a, b) => a.name.collate (b.name));
    on_search_changed ();
}

private void scan_directory (GLib.File dir, string base_path) throws Error {
    var enumerator = dir.enumerate_children (
                                            "standard::name,standard::type",
                                            FileQueryInfoFlags.NONE,
                                            null
    );

    FileInfo? info;
    while ((info = enumerator.next_file (null)) != null) {
        var name = info.get_name ();
        if (name.has_prefix (".")) {
            continue;
        }

        var file_type = info.get_file_type ();
        if (file_type == FileType.DIRECTORY) {
            if (name == "node_modules" || name == "target" || name == "build" || name ==
                continue;
            }
            scan_directory (dir.get_child (name), base_path);
        } else if (file_type == FileType.REGULAR) {
```

```
            var path = dir.get_child (name).get_path ();
            if (path != null) {
                var relative = path.substring (base_path.length + 1);
                all_files.add (new FileEntry (path, name, relative));
            }
        }
    }
}

private void on_search_changed () {
    var query = search_entry.get_text ().down ();
    filtered_files.clear ();

    if (query == "") {
        foreach (var f in all_files) {
            if (filtered_files.size >= MAX_RESULTS)break;
            filtered_files.add (f);
        }
    } else {
        foreach (var f in all_files) {
            if (fuzzy_match (f.name.down (), query)) {
                filtered_files.add (f);
                if (filtered_files.size >= MAX_RESULTS)break;
            }
        }
    }

    update_results ();
}

private bool fuzzy_match (string text, string query) {
    int qi = 0;
    for (int i = 0; i < text.length && qi < query.length; i++) {
        if (text[i] == query[qi]) {
            qi++;
        }
    }
    return qi == query.length;
}

private void update_results () {
    var strings = new string[filtered_files.size];
    for (int i = 0; i < filtered_files.size; i++) {
        strings[i] = filtered_files[i].display_name;
    }
```

```
        string_list.splice (0, string_list.get_n_items (), strings);

        if (filtered_files.size > 0) {
            selection.selected = 0;
        }
    }

    private void open_selected () {
        var index = (int) selection.selected;
        if (index >= 0 && index < filtered_files.size) {
            var entry = filtered_files[index];
            var file = GLib.File.new_for_path (entry.path);
            document_manager.open_document (file, null);
            this.close ();
        }
    }

    public override void show () {
        base.show ();
        search_entry.grab_focus ();
    }
}
```

File: src/Widgets/PreferencesDialog.vala

```
/*
 * preferencesdialog.vala
 *
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 *
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 *
 * SPDX-License-Identifier: GPL-3.0-or-later
 */
```

```
public class Iide.PreferencesDialog : Adw.PreferencesWindow {
    private Iide.SettingsService settings;
    private Adw.StyleManager style_manager;

    private Adw.ComboRow color_scheme_row;
    private Adw.SwitchRow show_minimap_row;
    private Adw.SwitchRow show_line_numbers_row;
    private Adw.SwitchRow highlight_current_line_row;
    private Adw.SwitchRow auto_indent_row;

    public PreferencesDialog () {
        Object (modal: true, destroy_with_parent: true);
        settings = Iide.SettingsService.get_instance ();
        style_manager = Adw.StyleManager.get_default ();

        title = _("Preferences");
        set_default_size (500, 450);

        build_ui ();
    }

    private void build_ui () {
        var appearance_page = new Adw.PreferencesPage () {
            title = _("Appearance"),
            icon_name = "preferences-desktop-appearance-symbolic"
        };

        var appearance_group = new Adw.PreferencesGroup () {
            title = _("Appearance")
        };

        color_scheme_row = new Adw.ComboRow () {
            title = _("Color Scheme"),
            model = new Gtk.StringList ({ "System", "Light", "Dark" })
        };
        var scheme_index = settings.color_scheme;
        color_scheme_row.selected = (uint) scheme_index;
        color_scheme_row.notify["selected"].connect (() => {
            var scheme = (ColorScheme) color_scheme_row.selected;
            settings.color_scheme = scheme;
            style_manager.color_scheme = scheme.to_adw_color_scheme ();
        });
        appearance_group.add (color_scheme_row);
        appearance_page.add (appearance_group);

        var editor_page = new Adw.PreferencesPage () {
```

```
    title = _("Editor"),
    icon_name = "accessories-text-editor-symbolic"
};

var editor_group = new Adw.PreferencesGroup () {
    title = _("Editor")
};

show_minimap_row = new Adw.SwitchRow () {
    title = _("Show Minimap"),
    subtitle = _("Display a minimap showing the document overview")
};
show_minimap_row.active = settings.show_minimap;
show_minimap_row.notify["active"].connect (() => {
    settings.show_minimap = show_minimap_row.active;
});
editor_group.add (show_minimap_row);

show_line_numbers_row = new Adw.SwitchRow () {
    title = _("Show Line Numbers"),
    subtitle = _("Display line numbers on the editor gutter")
};
show_line_numbers_row.active = settings.show_line_numbers;
show_line_numbers_row.notify["active"].connect (() => {
    settings.show_line_numbers = show_line_numbers_row.active;
});
editor_group.add (show_line_numbers_row);

highlight_current_line_row = new Adw.SwitchRow () {
    title = _("Highlight Current Line"),
    subtitle = _("Highlight the line where the cursor is positioned")
};
highlight_current_line_row.active = settings.highlight_current_line;
highlight_current_line_row.notify["active"].connect (() => {
    settings.highlight_current_line = highlight_current_line_row.active;
});
editor_group.add (highlight_current_line_row);

auto_indent_row = new Adw.SwitchRow () {
    title = _("Auto Indent"),
    subtitle = _("Automatically indent new lines based on the previous line")
};
auto_indent_row.active = settings.auto_indent;
auto_indent_row.notify["active"].connect (() => {
    settings.auto_indent = auto_indent_row.active;
});
```

```
editor_group.add (auto_indent_row);

var sizes = FontSizeHelper.get_available_sizes ();
var size_strings = new string[sizes.length];
for (int i = 0; i < sizes.length; i++) {
    size_strings[i] = "%d px".printf (sizes[i]);
}
var font_size_model = new Gtk.StringList (size_strings);
var current_level = settings.editor_font_size;
if (current_level < FontSizeHelper.MIN_ZOOM_LEVEL || current_level > FontSizeHelper.
    current_level = FontSizeHelper.DEFAULT_ZOOM_LEVEL;
}

var font_size_row = new Adw.ComboRow () {
    title = _("Font Size"),
    model = font_size_model,
    selected = (uint) (current_level - 1)
};
font_size_row.notify["selected"].connect (() => {
    settings.editor_font_size = (int) font_size_row.selected + 1;
});
editor_group.add (font_size_row);
editor_page.add (editor_group);

var projects_page = new Adw.PreferencesPage () {
    title = _("Projects"),
    icon_name = "folder-open-symbolic"
};

var projects_group = new Adw.PreferencesGroup () {
    title = _("Projects")
};

var recent_projects_row = new Adw.ActionRow () {
    title = _("Recent Projects")
};
var recent_projects = settings.recent_projects;
if (recent_projects.length > 0) {
    var subtitle = string.joinv ("\n", recent_projects);
    recent_projects_row.subtitle = subtitle;
} else {
    recent_projects_row.subtitle = _("No recent projects");
}
var clear_button = new Gtk.Button () {
    icon_name = "edit-clear-symbolic",
    tooltip_text = _("Clear Recent Projects")
```

```
};
clear_button.clicked.connect (() => {
    settings.clear_recent_projects ();
    recent_projects_row.subtitle = _("No recent projects");
});
recent_projects_row.add_suffix (clear_button);
recent_projects_row.activatable_widget = clear_button;
projects_group.add (recent_projects_row);
projects_page.add (projects_group);

var shortcuts_page = new Adw.PreferencesPage () {
    title = _("Shortcuts"),
    icon_name = "input-keyboard-symbolic"
};

var shortcuts_group = new Adw.PreferencesGroup () {
    title = _("Keyboard Shortcuts")
};

var action_manager = Iide.ActionManager.get_instance ();
var actions = action_manager.get_all_actions ();

var list_box = new Gtk.ListBox () {
    selection_mode = Gtk.SelectionMode.NONE,
    show_separators = true
};

var sorted_actions = new Gee.ArrayList<Iide.Action> ();
foreach (var action in actions) {
    sorted_actions.add (action);
}
sorted_actions.sort ((a, b) => {
    var cat_cmp = (a.category ?? "").collate (b.category ?? "");
    if (cat_cmp != 0)return cat_cmp;
    return a.name.collate (b.name);
});

string? current_category = null;
foreach (var action in sorted_actions) {
    if (action.category != current_category) {
        current_category = action.category;
        var header_row = new Gtk.ListBoxRow () {
            selectable = false,
            activatable = false
        };
        var header_label = new Gtk.Label (action.category ?? _("Other")) {
```

```
                    halign = Gtk.Align.START,
                    margin_top = 8,
                    margin_bottom = 4
                };
                header_label.add_css_class ("title");
                header_label.add_css_class ("dim-label");
                header_row.child = header_label;
                list_box.append (header_row);
            }
            var row = new ShortcutRow (action);
            list_box.append (row);
        }

        var scrolled = new Gtk.ScrolledWindow () {
            child = list_box,
            hexpand = true,
            vexpand = true
        };
        scrolled.set_size_request (-1, 300);
        shortcuts_group.add (scrolled);
        shortcuts_page.add (shortcuts_group);

        add (appearance_page);
        add (editor_page);
        add (projects_page);
        add (shortcuts_page);
    }
}

private class ShortcutRow : Gtk.ListBoxRow {
    private Iide.Action action;
    private Gtk.Label shortcut_label;

    public ShortcutRow (Iide.Action action) {
        this.action = action;

        var box = new Gtk.Box (Gtk.Orientation.HORIZONTAL, 12) {
            margin_start = 12,
            margin_end = 12,
            margin_top = 8,
            margin_bottom = 8
        };

        var info_box = new Gtk.Box (Gtk.Orientation.VERTICAL, 2);
        var name_label = new Gtk.Label (action.name) {
            halign = Gtk.Align.START,
```

```
        hexpand = true
    };
    name_label.add_css_class ("title");
    var desc_label = new Gtk.Label (action.description ?? "") {
        halign = Gtk.Align.START,
        hexpand = true
    };
    desc_label.add_css_class ("caption");
    desc_label.add_css_class ("dim-label");
    info_box.append (name_label);
    info_box.append (desc_label);
    box.append (info_box);

    shortcut_label = new Gtk.Label (action.shortcut != null ? action.shortcut : _("None'
        halign = Gtk.Align.END,
        margin_start = 12
    };
    shortcut_label.add_css_class ("caption");
    shortcut_label.add_css_class ("dim-label");
    box.append (shortcut_label);

    var clear_button = new Gtk.Button () {
        icon_name = "edit-clear-symbolic",
        tooltip_text = _("Clear shortcut"),
        valign = Gtk.Align.CENTER
    };
    clear_button.clicked.connect (on_clear_clicked);
    box.append (clear_button);

    var capture_button = new Gtk.Button () {
        label = _("Change"),
        valign = Gtk.Align.CENTER
    };
    capture_button.clicked.connect (on_capture_clicked);
    box.append (capture_button);

    child = box;

    action.shortcut_changed.connect ((new_shortcut) => {
        shortcut_label.set_label (new_shortcut != null ? new_shortcut : _("None"));
    });
}

private void on_clear_clicked () {
    Iide.ActionManager.get_instance ().set_shortcut (action.id, null);
}
```

```
    private void on_capture_clicked () {
        var window = this.get_ancestor (typeof (Gtk.Window)) as Gtk.Window;
        var dialog = new ShortcutCaptureWindow (action, window);
        dialog.show ();
    }
}

private class ShortcutCaptureWindow : Gtk.Window {
    private Iide.Action action;
    private Gtk.Label label;
    private uint keyval = 0;
    private Gdk.ModifierType modifiers;

    public ShortcutCaptureWindow (Iide.Action action, Gtk.Window? parent) {
        this.action = action;
        title = _("Set Shortcut for %s").printf (action.name);
        modal = true;
        if (parent != null) {
            set_transient_for (parent);
        }

        var content_box = new Gtk.Box (Gtk.Orientation.VERTICAL, 12) {
            margin_top = 20,
            margin_bottom = 20,
            margin_start = 20,
            margin_end = 20
        };

        label = new Gtk.Label (_("Press a key combination...")) {
            halign = Gtk.Align.CENTER
        };
        content_box.append (label);

        var current = action.shortcut;
        if (current != null && current != "") {
            var hint = new Gtk.Label (_("Current: %s").printf (current)) {
                halign = Gtk.Align.CENTER
            };
            hint.add_css_class ("caption");
            hint.add_css_class ("dim-label");
            content_box.append (hint);
        }

        var event_controller = new Gtk.EventControllerKey ();
        event_controller.key_pressed.connect (on_key_pressed);
```

```
        content_box.add_controller (event_controller);

        var cancel_button = new Gtk.Button () {
            label = _("Cancel")
        };
        cancel_button.clicked.connect (() => destroy ());

        var clear_button = new Gtk.Button () {
            label = _("Clear")
        };
        clear_button.clicked.connect (() => {
            Iide.ActionManager.get_instance ().set_shortcut (action.id, null);
            destroy ();
        });

        var save_button = new Gtk.Button () {
            label = _("Save")
        };
        save_button.add_css_class ("suggested-action");
        save_button.clicked.connect (on_save);

        var button_box = new Gtk.Box (Gtk.Orientation.HORIZONTAL, 6);
        button_box.append (cancel_button);
        button_box.append (clear_button);
        button_box.append (save_button);
        button_box.halign = Gtk.Align.END;
        content_box.append (button_box);

        child = content_box;
        set_size_request (350, 120);
    }

    private bool on_key_pressed (uint keyval, uint keycode, Gdk.ModifierType state) {
        var modifiers = state & (Gdk.ModifierType.SHIFT_MASK | Gdk.ModifierType.CONTROL_MASK

        if (keyval == Gdk.Key.Escape) {
            destroy ();
            return true;
        }

        if (keyval == Gdk.Key.BackSpace && modifiers == 0) {
            Iide.ActionManager.get_instance ().set_shortcut (action.id, null);
            destroy ();
            return true;
        }
```

```
        if (modifiers == 0 || (modifiers & Gdk.ModifierType.CONTROL_MASK) != 0 ||
            (modifiers & Gdk.ModifierType.SHIFT_MASK) != 0 ||
            (modifiers & Gdk.ModifierType.ALT_MASK) != 0 ||
            (modifiers & Gdk.ModifierType.SUPER_MASK) != 0) {

            if (modifiers != 0) {
                this.keyval = keyval;
                this.modifiers = modifiers;
                label.set_label (format_shortcut (keyval, modifiers));
            }
            return true;
        }
        return false;
    }

    private string format_shortcut (uint keyval, Gdk.ModifierType modifiers) {
        var accel_string = "";

        if ((modifiers & Gdk.ModifierType.CONTROL_MASK) != 0)accel_string += "<primary>";
        if ((modifiers & Gdk.ModifierType.ALT_MASK) != 0)accel_string += "<Alt>";
        if ((modifiers & Gdk.ModifierType.SHIFT_MASK) != 0)accel_string += "<Shift>";
        if ((modifiers & Gdk.ModifierType.SUPER_MASK) != 0)accel_string += "<Super>";

        var key_name = Gdk.keyval_name (keyval);
        if (key_name != null) {
            accel_string += key_name;
        }

        return accel_string;
    }

    private void on_save () {
        if (keyval != 0) {
            var shortcut = format_shortcut (keyval, modifiers);
            Iide.ActionManager.get_instance ().set_shortcut (action.id, shortcut);
        }
        destroy ();
    }
}
```

File: src/Widgets/SearchInFilesDialog.vala

```
public class Iide.SearchInFilesDialog : Adw.Window {
    private Gtk.Entry search_entry;
    private Gtk.ListView results_view;
    private Gtk.SingleSelection selection;
```

```
private Iide.ProjectManager project_manager;
private Iide.DocumentManager document_manager;
private Window? parent_window;
private string project_root_path;
private string current_query;

private const int MAX_RESULTS = 200;

private interface ListItem : Object {
    public abstract string get_display_text ();
    public abstract bool is_header ();
}

private class SearchResult : Object, ListItem {
    public string file_path { get; construct; }
    public string file_name { get; construct; }
    public string relative_path { get; construct; }
    public int line_number { get; construct; }
    public string line_content { get; construct; }
    public int match_start { get; construct; }
    public int match_end { get; construct; }

    public SearchResult (string file_path, string file_name, string relative_path, int l
        Object (
                file_path: file_path,
                file_name: file_name,
                relative_path: relative_path,
                line_number: line_number,
                line_content: line_content,
                match_start: match_start,
                match_end: match_end
        );
    }

    public string get_display_text () {
        return "%d: %s".printf (line_number + 1, line_content);
    }

    public bool is_header () {
        return false;
    }
}

private class ResultGroup : Object, ListItem {
    public string file_path { get; construct; }
    public string file_name { get; construct; }
```

```
    public string relative_path { get; construct; }
    public int result_count { get; construct; }

    public ResultGroup (string file_path, string file_name, string relative_path, int re
        Object (
                file_path: file_path,
                file_name: file_name,
                relative_path: relative_path,
                result_count: result_count
        );
    }

    public string get_display_text () {
        return relative_path;
    }

    public bool is_header () {
        return true;
    }
}

private Gee.List<ListItem> all_items;
private Gee.List<ListItem> filtered_items;
private Gtk.StringList string_list;

public SearchInFilesDialog (Window parent_window, Iide.DocumentManager document_manager)
    Object (
            title: _("Search in Files"),
            modal: true,
            destroy_with_parent: true,
            default_width: 800,
            default_height: 500
    );

    this.parent_window = parent_window;
    this.document_manager = document_manager;
    this.project_manager = Iide.ProjectManager.get_instance ();
    this.all_items = new Gee.ArrayList<ListItem> ();
    this.filtered_items = new Gee.ArrayList<ListItem> ();

    var project_root = project_manager.get_current_project_root ();
    if (project_root != null) {
        this.project_root_path = project_root.get_path () ?? "";
    } else {
        this.project_root_path = "";
    }
```

```
    setup_ui ();
}

private void setup_ui () {
    var vbox = new Gtk.Box (Gtk.Orientation.VERTICAL, 8);
    vbox.margin_top = 12;
    vbox.margin_bottom = 12;
    vbox.margin_start = 12;
    vbox.margin_end = 12;

    search_entry = new Gtk.Entry () {
        placeholder_text = _("Search text..."),
        hexpand = true
    };
    vbox.append (search_entry);

    var list_model = new Gtk.StringList (new string[0]);
    string_list = list_model;
    selection = new Gtk.SingleSelection (list_model);
    results_view = new Gtk.ListView (selection, null);
    results_view.hexpand = true;
    results_view.vexpand = true;
    results_view.show_separators = true;

    var factory = new Gtk.SignalListItemFactory ();
    factory.setup.connect ((item) => {
        var list_item = item as Gtk.ListItem;
        list_item.set_child (new Gtk.Label (null));
    });
    factory.bind.connect ((item) => {
        var list_item = item as Gtk.ListItem;
        var label = list_item.get_child () as Gtk.Label;
        var index = list_item.get_position ();

        if (index >= 0 && index < filtered_items.size) {
            var list_item_obj = filtered_items[(int) index];
            if (list_item_obj.is_header ()) {
                var group = list_item_obj as ResultGroup;
                label.set_label ("%s (%d)".printf (group.relative_path, group.result_cou
                label.add_css_class ("title-5");
                label.add_css_class ("dim-label");
                label.xalign = 0;
            } else {
                label.set_label (list_item_obj.get_display_text ());
                label.add_css_class ("monospace");
```

```
                label.add_css_class ("body");
                label.xalign = 0;
            }
        }
    });
    results_view.factory = factory;

    var scrolled = new Gtk.ScrolledWindow ();
    scrolled.child = results_view;
    scrolled.hexpand = true;
    scrolled.vexpand = true;
    vbox.append (scrolled);

    set_content (vbox);

    search_entry.changed.connect (on_search_changed);

    var key_controller = new Gtk.EventControllerKey ();
    key_controller.key_pressed.connect (on_key_pressed);
    search_entry.add_controller (key_controller);

    results_view.activate.connect (() => {
        open_selected ();
    });

    search_entry.activate.connect (() => {
        open_selected ();
    });

    search_entry.focus_on_click = false;

    this.set_default_widget (search_entry);
}

private bool on_key_pressed (Gtk.EventControllerKey controller, uint keyval, uint keyco
    if (keyval == Gdk.Key.Escape) {
        this.close ();
        return true;
    } else if (keyval == Gdk.Key.Return || keyval == Gdk.Key.KP_Enter) {
        open_selected ();
        return true;
    } else if (keyval == Gdk.Key.Up || keyval == Gdk.Key.KP_Up) {
        navigate_up ();
        return true;
    } else if (keyval == Gdk.Key.Down || keyval == Gdk.Key.KP_Down) {
        navigate_down ();
```

```
        return true;
    }
    return false;
}

private void navigate_up () {
    var current = (int) selection.selected;
    if (current <= 0)return;

    for (int i = current - 1; i >= 0; i--) {
        if (filtered_items[i] is SearchResult) {
            selection.selected = i;
            results_view.scroll_to (i, Gtk.ListScrollFlags.NONE, null);
            return;
        }
    }
}

private void navigate_down () {
    var current = (int) selection.selected;
    var max = (int) string_list.get_n_items () - 1;
    if (current >= max)return;

    for (int i = current + 1; i < filtered_items.size; i++) {
        if (filtered_items[i] is SearchResult) {
            selection.selected = i;
            results_view.scroll_to (i, Gtk.ListScrollFlags.NONE, null);
            return;
        }
    }
}

private void on_search_changed () {
    current_query = search_entry.get_text ();

    if (current_query == "" || current_query.length < 2) {
        filtered_items = new Gee.ArrayList<ListItem> ();
        update_results ();
        return;
    }

    var query_lower = current_query.down ();
    var results_by_file = new Gee.HashMap<string, Gee.List<SearchResult>> ();

    foreach (var item in all_items) {
        if (item is ResultGroup) {
```

```
            continue;
        }
        var result = item as SearchResult;
        if (result.line_content.down ().contains (query_lower)) {
            if (!results_by_file.has_key (result.file_path)) {
                results_by_file[result.file_path] = new Gee.ArrayList<SearchResult> ();
            }
            results_by_file[result.file_path].add (result);
        }
    }

    filtered_items = new Gee.ArrayList<ListItem> ();
    foreach (var entry in results_by_file) {
        if (filtered_items.size >= MAX_RESULTS)break;

        var group = new ResultGroup (
                                    entry.key,
                                    entry.value[0].file_name,
                                    entry.value[0].relative_path,
                                    entry.value.size
        );
        filtered_items.add (group);

        foreach (var result in entry.value) {
            if (filtered_items.size >= MAX_RESULTS)break;
            filtered_items.add (result);
        }
    }

    update_results ();
}

private void update_results () {
    var strings = new string[filtered_items.size];
    for (int i = 0; i < filtered_items.size; i++) {
        strings[i] = filtered_items[i].get_display_text ();
    }

    string_list.splice (0, string_list.get_n_items (), strings);

    if (filtered_items.size > 0) {
        select_first_result ();
    }
}

private void select_first_result () {
```

```
    for (int i = 0; i < filtered_items.size; i++) {
        if (filtered_items[i] is SearchResult) {
            selection.selected = i;
            return;
        }
    }
}

private void open_selected () {
    var index = (int) selection.selected;
    if (index >= 0 && index < filtered_items.size) {
        var item = filtered_items[index];
        if (item is SearchResult) {
            var result = item as SearchResult;
            var file = GLib.File.new_for_path (result.file_path);

            var query_lower = current_query.down ();
            var line_lower = result.line_content.down ();
            var pos = line_lower.index_of (query_lower);
            var start_col = pos >= 0 ? pos : 0;
            var end_col = pos >= 0 ? pos + (int) current_query.length : start_col;

            document_manager.open_document_with_selection (file, result.line_number, sta
            this.close ();
        }
    }
}

public void start_search (string query) {
    search_entry.set_text (query);
}

private void scan_files_for_search () {
    if (project_root_path == "") {
        return;
    }

    try {
        scan_directory (GLib.File.new_for_path (project_root_path));
    } catch (Error e) {
        warning ("Error scanning for search: %s", e.message);
    }

    on_search_changed ();
}
```

```
private void scan_directory (GLib.File dir) throws Error {
    var enumerator = dir.enumerate_children (
                                            "standard::name,standard::type",
                                            FileQueryInfoFlags.NONE,
                                            null
    );

    FileInfo? info;
    while ((info = enumerator.next_file (null)) != null) {
        var name = info.get_name ();
        if (name.has_prefix (".")) {
            continue;
        }

        var file_type = info.get_file_type ();
        if (file_type == FileType.DIRECTORY) {
            if (name == "node_modules" || name == "target" || name == "build" ||
                name == "__pycache__" || name == ".git") {
                continue;
            }
            scan_directory (dir.get_child (name));
        } else if (file_type == FileType.REGULAR) {
            var path = dir.get_child (name).get_path ();
            if (path != null) {
                search_file (path);
            }
        }
    }
}

private void search_file (string file_path) {
    try {
        var file = GLib.File.new_for_path (file_path);
        var dis = new DataInputStream (file.read ());
        string line;
        int line_num = 0;

        string relative_path = file_path;
        if (file_path.has_prefix (project_root_path)) {
            relative_path = file_path.substring (project_root_path.length);
            if (relative_path.has_prefix ("/")) {
                relative_path = relative_path.substring (1);
            }
        }

        while ((line = dis.read_line ()) != null) {
```

```
                all_items.add (new SearchResult (
                                                    file_path,
                                                    file.get_basename (),
                                                    relative_path,
                                                    line_num,
                                                    line.strip (),
                                                    0, 0
                ));
                line_num++;
            }
            dis.close ();
        } catch (Error e) {
        }
    }

    public override void show () {
        base.show ();
        if (all_items.size == 0) {
            scan_files_for_search ();
        }
        search_entry.grab_focus ();
    }
}
```

File: src/iide.gresource.xml

```
<?xml version="1.0" encoding="UTF-8" ?>
<gresources>
  <gresource prefix="/org/github/kai66673/iide">
    <file preprocess="xml-stripblanks">window.ui</file>
    <file preprocess="xml-stripblanks">shortcuts-dialog.ui</file>
    <file>style.css</file>
  </gresource>
  <gresource prefix="/org/github/kai66673/iide/icons">
     <file
            compressed="true"
            alias="scalable/mimetypes/text-x-python-symbolic.svg"
        >icons/mimetypes/text-x-python-symbolic.svg</file>
    <file
            compressed="true"
            alias="scalable/mimetypes/text-x-vala-symbolic.svg"
        >icons/mimetypes/text-x-vala-symbolic.svg</file>
    <file
            compressed="true"
            alias="scalable/mimetypes/text-x-meson-symbolic.svg"
        >icons/mimetypes/text-x-meson-symbolic.svg</file>
```

```
    <file
            compressed="true"
            alias="scalable/mimetypes/text-markdown-symbolic.svg"
        >icons/mimetypes/text-markdown-symbolic.svg</file>
    <file
            compressed="true"
            alias="scalable/mimetypes/text-x-javascript-symbolic.svg"
        >icons/mimetypes/text-x-javascript-symbolic.svg</file>
    <file
            compressed="true"
            alias="scalable/mimetypes/text-xml-symbolic.svg"
        >icons/mimetypes/text-xml-symbolic.svg</file>
  </gresource>
</gresources>
```

File: .gitmodules

```
[submodule "vendor/tree-sitter"]
    path = vendor/tree-sitter
    url = https://github.com/albfan/tree-sitter
[submodule "parsers/c"]
    path = parsers/c
    url = https://github.com/tree-sitter/tree-sitter-c
[submodule "parsers/bash"]
    path = parsers/bash
    url = https://github.com/tree-sitter/tree-sitter-bash
[submodule "parsers/json"]
    path = parsers/json
    url = https://github.com/tree-sitter/tree-sitter-json
[submodule "parsers/python"]
    path = parsers/python
    url = https://github.com/tree-sitter/tree-sitter-python
[submodule "parsers/php"]
    path = parsers/php
    url = https://github.com/tree-sitter/tree-sitter-php
[submodule "parsers/cpp"]
    path = parsers/cpp
    url = https://github.com/tree-sitter/tree-sitter-cpp
[submodule "parsers/go"]
    path = parsers/go
    url = https://github.com/tree-sitter/tree-sitter-go
[submodule "parsers/html"]
    path = parsers/html
    url = https://github.com/tree-sitter/tree-sitter-html
[submodule "parsers/javascript"]
    path = parsers/javascript
```

```
    url = https://github.com/tree-sitter/tree-sitter-javascript
[submodule "parsers/ruby"]
    path = parsers/ruby
    url = https://github.com/tree-sitter/tree-sitter-ruby
[submodule "parsers/rust"]
    path = parsers/rust
    url = https://github.com/tree-sitter/tree-sitter-rust
[submodule "parsers/typescript"]
    path = parsers/typescript
    url = https://github.com/tree-sitter/tree-sitter-typescript
[submodule "parsers/yaml"]
    path = parsers/yaml
    url = https://github.com/tree-sitter-grammars/tree-sitter-yaml
[submodule "parsers/xml"]
    path = parsers/xml
    url = https://github.com/tree-sitter-grammars/tree-sitter-xml
[submodule "parsers/vala"]
    path = parsers/vala
    url = https://github.com/vala-lang/tree-sitter-vala
```

File: src/Widgets/TextView/GutterMarkRenderer.vala

```
using Gtk;
using GtkSource;
using GLib;

public class Iide.GutterMarkRenderer : GutterRenderer {
    private int current_icon_size = 16;

    public void set_icons_size (int size) {
        if (current_icon_size == size) {
            return;
        }
        current_icon_size = size;
        queue_resize ();

        var gutter = (Gutter) get_parent ();
        if (gutter != null) {
            gutter.queue_allocate ();
        }
    }

    public override void measure (Gtk.Orientation orientation, int for_size, out int minimum
        minimum_baseline = natural_baseline = -1;
        if (orientation == Gtk.Orientation.HORIZONTAL) {
            minimum = natural = current_icon_size;
```

```
    } else {
        minimum = natural = 0;
    }
}

public override void snapshot_line (Gtk.Snapshot snapshot, GutterLines lines, uint line)
    var buffer = get_buffer ();
    if (buffer == null) {
        return;
    }

    Gtk.TextIter iter;
    lines.get_iter_at_line (out iter, line);
    var marks = iter.get_marks ();

    string? icon_name_to_draw = null;

    foreach (var text_mark in marks) {
        var lsp_mark = text_mark as LspDiagnosticsMark;
        if (lsp_mark != null) {
            icon_name_to_draw = lsp_mark.get_icon_name ();
        }
    }

    if (icon_name_to_draw == null) {
        return;
    }

    var display = Gdk.Display.get_default ();
    if (display == null) {
        return;
    }

    var theme = Gtk.IconTheme.get_for_display (display);
    var gicon = new GLib.ThemedIcon (icon_name_to_draw);
    var paintable = theme.lookup_by_gicon (gicon, current_icon_size, 1, Gtk.TextDirectio
    if (paintable == null) {
        return;
    }

    int y, height;
    lines.get_line_yrange (line, GutterRendererAlignmentMode.CELL, out y, out height);

    var snapshot_size = (float) current_icon_size;
    var x = (float) xpad;
    var y_pos = (float) y + ((float) height - snapshot_size) / 2.0f;
```

```
        var point = Graphene.Point () { x = x, y = y_pos };
        snapshot.translate (point);
        paintable.snapshot (snapshot, snapshot_size, snapshot_size);
        snapshot.translate (Graphene.Point () { x = -x, y = -y_pos });
    }
}
```

File: src/Widgets/ToolViews/ProjectView.vala

```
using Gtk;

public class Iide.FileTreeView : Box {
    private ColumnView column_view;
    private SingleSelection selection;
    private GLib.File? _root_directory = null;
    private CustomSorter sorter;

    public signal void file_activated (FileItem item);

    //              null
    public GLib.File? root_directory {
        get { return _root_directory; }
        set {
            _root_directory = value;

            if (_root_directory == null) {
                //         null,                View
                column_view.model = null;
                selection = null;
                return;
            }

            //
            var root_store = create_file_model (_root_directory);
            var sorted_root = new SortListModel (root_store, sorter);

            //          ,
            var tree_model = new TreeListModel (sorted_root, false, false, (item) => {
                var file_item = item as FileItem;
                if (file_item != null && file_item.is_directory) {
                    var child_store = create_file_model (file_item.file);
                    return new SortListModel (child_store, sorter);
                }
                return null;
            });
```

```
        selection = new SingleSelection (tree_model);
        column_view.model = selection;
    }
}

public FileTreeView (GLib.File? root_dir = null) {
    Object (orientation : Orientation.VERTICAL);

    //          View
    column_view = new ColumnView (null);
    //
    column_view.get_first_child ().set_visible (false);

    //                  ,
    if (root_dir != null) {
        root_directory = root_dir;
    }

    setup_view ();
}

public void set_root_file (GLib.File? root_dir) {
    root_directory = root_dir;
}

private void setup_view () {
    var column = new ColumnViewColumn (null, create_factory ());
    column.expand = true;
    sorter = new CustomSorter ((a, b) => {
        var fi1 = a as FileItem;
        var fi2 = b as FileItem;
        if (fi1 == null || fi2 == null)return 0;

        //              :
        if (fi1.is_directory != fi2.is_directory) {
            return fi1.is_directory ? -1 : 1;
        }
        //
        return strcmp (fi1.name.down (), fi2.name.down ());
    });
    sorter.set_sort_func ((a, b) => {
        var fi1 = a as FileItem;
        var fi2 = b as FileItem;
        if (fi1 == null || fi2 == null)return 0;
```

```
        //              :
        if (fi1.is_directory != fi2.is_directory) {
            return fi1.is_directory ? -1 : 1;
        }
        //
        return strcmp (fi1.name.down (), fi2.name.down ());
    });
    column.sorter = sorter;
    column_view.append_column (column);

    var scroll = new ScrolledWindow ();
    scroll.vexpand = true;
    scroll.set_child (column_view);
    this.append (scroll);

    column_view.activate.connect ((pos) => {
        var item = selection.get_item (pos) as TreeListRow;
        if (item != null) {
            var file_item = item.get_item () as FileItem;
            if (file_item != null) {
                file_activated (file_item);
            }
        }
    });
}

private GLib.ListStore create_file_model (GLib.File dir) {
    var store = new GLib.ListStore (typeof (FileItem));
    try {
        //
        var enumerator = dir.enumerate_children ("standard::display-name,standard::type,
        GLib.FileInfo info;
        while ((info = enumerator.next_file (null)) != null) {
            store.append (new FileItem (dir.get_child (info.get_name ()), info));
        }
    } catch (Error e) {
        //
        debug ("Cannot read directory: %s", e.message);
    }
    return store;
}

private ListItemFactory create_factory () {
    var factory = new SignalListItemFactory ();

    factory.setup.connect ((list_item) => {
```

```
            var item = (Gtk.ListItem) list_item;
            var expander = new TreeExpander ();
            var box = new Box (Orientation.HORIZONTAL, 6);
            var icon = new Image ();
            var label = new Label ("");

            box.append (icon);
            box.append (label);
            expander.set_child (box);
            item.set_child (expander);
        });

        factory.bind.connect ((list_item) => {
            var item = (Gtk.ListItem) list_item;
            var tree_row = item.get_item () as TreeListRow;
            if (tree_row == null)return;

            var file_item = tree_row.get_item () as FileItem;
            var expander = item.get_child () as TreeExpander;
            var box = expander.get_child () as Box;
            var icon = box.get_first_child () as Image;
            var label = icon.get_next_sibling () as Label;

            expander.list_row = tree_row;
            if (file_item != null) {
                label.label = file_item.name;
                icon.icon_name = file_item.is_directory ? "folder-symbolic" : IconProvider.g

                if (!file_item.is_directory) {
                    var click = new Gtk.GestureClick ();
                    click.released.connect (() => {
                        file_activated (file_item);
                    });
                    box.add_controller (click);
                }
            }
        });

        return factory;
    }
}

//
public class Iide.FileItem : Object {
    public GLib.File file { get; construct; }
    public string name { get; construct; }
```

```
    public bool is_directory { get; construct; }

    public FileItem (GLib.File file, GLib.FileInfo info) {
        Object (
                file : file,
                name : info.get_display_name (),
                is_directory: info.get_file_type () == GLib.FileType.DIRECTORY
        );
    }
}
```

File: tests/test_lsp_service.vala

```
/*
 * test_lsp_service.vala
 *
 * Copyright 2026 kai
 *
 * This program is free software: you can redistribute it and/or modify
 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation, either version 3 of the License, or
 * (at your option) any later version.
 *
 * This program is distributed in the hope that it will be useful,
 * but WITHOUT ANY WARRANTY; without even the implied warranty of
 * MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
 * GNU General Public License for more details.
 *
 * You should have received a copy of the GNU General Public License
 * along with this program.  If not, see <https://www.gnu.org/licenses/>.
 *
 * SPDX-License-Identifier: GPL-3.0-or-later
 */

using GLib;

public static int main (string[] args) {
    Test.init (ref args);

    Test.add_func ("/lsp/json_builder", () => {
        test_json_builder ();
    });

    Test.add_func ("/lsp/json_reader", () => {
        test_json_reader ();
    });
```

```
    Test.add_func ("/lsp/content_length", () => {
        test_content_length_parsing ();
    });

    Test.add_func ("/lsp/clangd_server", () => {
        test_clangd_server_start ();
    });

    Test.add_func ("/lsp/did_change_valid_json", () => {
        test_did_change_valid_json ();
    });

    Test.add_func ("/lsp/did_change_content_length", () => {
        test_did_change_content_length ();
    });

    Test.add_func ("/lsp/clangd_diagnostics_parse", () => {
        test_clangd_diagnostics_parse ();
    });

    Test.add_func ("/lsp/did_change_full_document", () => {
        test_did_change_full_document ();
    });

    Test.add_func ("/lsp/didopen_diagnostics_didchange_diagnostics", () => {
        test_didopen_diagnostics_didchange_diagnostics ();
    });

    Test.add_func ("/lsp/integration_didopen_didchange_diagnostics", () => {
        test_integration_didopen_didchange_diagnostics ();
    });

    return Test.run ();
}

private void test_json_builder () {
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("method");
    builder.add_string_value ("test");
    builder.set_member_name ("params");
    builder.begin_object ();
    builder.set_member_name ("value");
    builder.add_int_value (42);
    builder.end_object ();
```

```
    builder.end_object ();

    var generator = new Json.Generator ();
    generator.root = builder.get_root ();
    string json = generator.to_data (null);

    assert (json != null);
    assert (json.contains ("\"method\":\"test\""));
    assert (json.contains ("\"value\":42"));
}

private void test_json_reader () {
    string json_str = """
        {
            "method": "test",
            "value": 42,
            "name": "hello"
        }
    """;

    var parser = new Json.Parser ();
    try {
        parser.load_from_data (json_str);
    } catch (Error e) {
        assert_not_reached ();
    }
    var reader = new Json.Reader (parser.get_root ());

    reader.read_member ("method");
    string method = reader.get_string_value ();
    reader.end_member ();
    assert (method == "test");

    reader.read_member ("value");
    int value = (int) reader.get_int_value ();
    reader.end_member ();
    assert (value == 42);

    reader.read_member ("name");
    string name = reader.get_string_value ();
    reader.end_member ();
    assert (name == "hello");
}

private void test_content_length_parsing () {
    string header = "Content-Length: 123\r\n\r\n";
```

```
    int length = -1;

    string[] lines = header.split ("\r\n");
    foreach (var line in lines) {
        if (line.has_prefix ("Content-Length: ")) {
            string len_str = line.substring (15);
            length = int.parse (len_str);
        }
    }

    assert (length == 123);
}

private void test_clangd_server_start () {
    var main_loop = new MainLoop ();
    bool server_initialized = false;
    bool error_received = false;

    var client = new Iide.IdeLspClient ();
    client.initialized.connect (() => {
        server_initialized = true;
        main_loop.quit ();
    });
    client.error_occurred.connect ((msg) => {
        error_received = true;
        print ("LSP Error: %s\n", msg);
        main_loop.quit ();
    });

    string? clangd_path = Environment.find_program_in_path ("clangd");
    assert (clangd_path != null);
    if (clangd_path == null) {
        print ("clangd not found, skipping test\n");
        return;
    }

    string workspace_uri = "file://" + Environment.get_current_dir ();
    client.start_server.begin (clangd_path, {}, workspace_uri);

    Timeout.add (10000, () => {
        main_loop.quit ();
        return Source.REMOVE;
    });

    main_loop.run ();
```

```
    assert (server_initialized == true);
}

private void test_did_change_valid_json () {
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("jsonrpc");
    builder.add_string_value ("2.0");
    builder.set_member_name ("method");
    builder.add_string_value ("textDocument/didChange");
    builder.set_member_name ("params");
    builder.begin_object ();
    builder.set_member_name ("textDocument");
    builder.begin_object ();
    builder.set_member_name ("uri");
    builder.add_string_value ("file:///test.c");
    builder.set_member_name ("version");
    builder.add_int_value (2);
    builder.end_object ();
    builder.set_member_name ("contentChanges");
    builder.begin_array ();
    builder.begin_object ();
    builder.set_member_name ("text");
    builder.add_string_value ("int main() { return 0; }");
    builder.end_object ();
    builder.end_array ();
    builder.end_object ();
    builder.end_object ();

    var generator = new Json.Generator ();
    generator.root = builder.get_root ();
    string json = generator.to_data (null);

    assert (json.contains ("\"method\":\"textDocument/didChange\""));
    assert (json.contains ("\"version\":2"));
    assert (json.contains ("file:///test.c"));
}

private void test_did_change_content_length () {
    string content = """
        {
            "jsonrpc": "2.0",
            "method": "textDocument/didChange",
            "params": {
                "textDocument": {
                    "uri": "file:///test.c",
```

```
                    "version": 2
                },
                "contentChanges": [
                    {
                        "text": "int main() { return 0; }"
                    }
                ]
            }
        }
    """;

    int body_length = content.length;
    string header = "Content-Length: %d\r\n\r\n".printf (body_length);

    int parsed_length = -1;
    string[] lines = header.split ("\r\n");
    foreach (var line in lines) {
        if (line.has_prefix ("Content-Length: ")) {
            string len_str = line.substring (15);
            parsed_length = int.parse (len_str);
        }
    }

    assert (parsed_length == body_length);
    assert (parsed_length > 0);
}

private void test_clangd_diagnostics_parse () {
    var diagnostic_json = """
        {
            "jsonrpc": "2.0",
            "method": "textDocument/publishDiagnostics",
            "params": {
                "uri": "file:///test.c",
                "version": 1,
                "diagnostics": [
                    {
                        "severity": 1,
                        "range": {
                            "start": {"line": 0, "character": 5},
                            "end": {"line": 0, "character": 9}
                        },
                        "message": "expected ';' after return statement"
                    }
                ]
            }
```

```
        }
    """;

    var parser = new Json.Parser ();
    try {
        parser.load_from_data (diagnostic_json);
    } catch (Error e) {
        assert_not_reached ();
    }

    var root = parser.get_root ();
    var reader = new Json.Reader (root);

    reader.read_member ("method");
    string method = reader.get_string_value ();
    reader.end_member ();
    assert (method == "textDocument/publishDiagnostics");

    reader.read_member ("params");
    assert (reader.is_object ());

    assert (reader.read_member ("uri"));
    string uri = reader.get_string_value ();
    reader.end_member ();
    assert (uri == "file:///test.c");

    assert (reader.read_member ("diagnostics"));
    assert (reader.is_array ());
    int count = 0;
    while (reader.read_element (count)) {
        count++;
        reader.end_element ();
    }
    assert (count == 1);
    reader.end_member ();

    reader.end_member ();
    reader.end_member ();
}

private void test_did_change_full_document () {
    string full_content = """
#include <stdio.h>

int main() {
    printf("Hello, World!\\n");
```

```
    return 0;
}
""";

    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("textDocument");
    builder.begin_object ();
    builder.set_member_name ("uri");
    builder.add_string_value ("file:///test.c");
    builder.set_member_name ("version");
    builder.add_int_value (1);
    builder.end_object ();
    builder.set_member_name ("contentChanges");
    builder.begin_array ();
    builder.begin_object ();
    builder.set_member_name ("text");
    builder.add_string_value (full_content);
    builder.end_object ();
    builder.end_array ();
    builder.end_object ();

    var generator = new Json.Generator ();
    generator.root = builder.get_root ();
    string json = generator.to_data (null);

    assert (json.contains ("#include <stdio.h>"));
    assert (json.contains ("printf"));
    assert (json.contains ("\"version\":1"));
}

private void test_didopen_diagnostics_didchange_diagnostics () {
    var test_uri = "file:///test.c";
    var initial_content = """
#include <stdio.h>

int main() {
    printf("Hello\n");
    return 0;
}
""";

    var updated_content = """
#include <stdio.h>

int main() {
```

```
    printf("Hello\n");
    return 0
}
""";

    var did_open_json = build_did_open_message (test_uri, "c", 1, initial_content);
    assert (did_open_json.contains ("\"method\":\"textDocument/didOpen\"")); 
    assert (did_open_json.contains ("\"uri\":\"file:///test.c\"")); 
    assert (did_open_json.contains ("\"languageId\":\"c\"")); 
    assert (did_open_json.contains ("\"version\":1"));

    var diagnostics_json_1 = build_publish_diagnostics_message (test_uri, 1, "expected ';' a
    assert (diagnostics_json_1.contains ("\"method\":\"textDocument/publishDiagnostics\"")); 
    assert (diagnostics_json_1.contains ("\"uri\":\"file:///test.c\"")); 
    assert (diagnostics_json_1.contains ("\"severity\":1"));

    var did_change_json = build_did_change_message (test_uri, 2, updated_content);
    assert (did_change_json.contains ("\"method\":\"textDocument/didChange\"")); 
    assert (did_change_json.contains ("\"version\":2")); 
    assert (did_change_json.contains ("printf")); 
    assert (did_change_json.contains ("return 0"));

    var diagnostics_json_2 = build_publish_diagnostics_message (test_uri, 2, "expected ';' a
    assert (diagnostics_json_2.contains ("\"method\":\"textDocument/publishDiagnostics\"")); 
    assert (diagnostics_json_2.contains ("\"version\":2"));

    print ("didOpen -> publishDiagnostics -> didChange -> publishDiagnostics chain: OK\n");
}

private void test_integration_didopen_didchange_diagnostics () {
    string? clangd_path = Environment.find_program_in_path ("clangd");
    if (clangd_path == null) {
        print ("clangd not found, skipping integration test\n");
        return;
    }

    var main_loop = new MainLoop ();
    int diagnostics_count = 0;
    string? last_uri = null;
    bool first_diag_received = false;

    var test_uri = "file:///test.c";
    var initial_content = "int main() { return 0; }";
    var updated_content = "int main() { return 0 }";

    var client = new Iide.IdeLspClient ();
```

```
    client.initialized.connect (() => {
        print ("Client initialized, sending didOpen...\n");
        client.text_document_did_open.begin (test_uri, "c", 1, initial_content);
    });

    client.diagnostics_received.connect ((uri, diagnostics) => {
        last_uri = uri;
        diagnostics_count++;
        print ("Diagnostics #%d received for %s\n", diagnostics_count, uri);

        if (diagnostics_count == 1) {
            first_diag_received = true;
            print ("First diagnostics (after didOpen) received, sending didChange...\n");
            client.text_document_did_change.begin (test_uri, 2, updated_content, null, null)
        } else if (diagnostics_count >= 2) {
            print ("Second diagnostics (after didChange) received\n");
            main_loop.quit ();
        }
    });

    client.error_occurred.connect ((msg) => {
        print ("LSP Error: %s\n", msg);
    });

    string workspace_uri = "file://" + Environment.get_current_dir ();
    client.start_server.begin (clangd_path, {}, workspace_uri);

    Timeout.add (20000, () => {
        print ("Timeout. Diagnostics received: %d\n", diagnostics_count);
        main_loop.quit ();
        return Source.REMOVE;
    });

    main_loop.run ();

    assert (last_uri == test_uri);
    assert (first_diag_received);
    print ("Integration test: passed (didOpen -> publishDiagnostics received)\n");
}

private string build_did_open_message (string uri, string language_id, int version, string c
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("jsonrpc");
    builder.add_string_value ("2.0");
```

```
    builder.set_member_name ("method");
    builder.add_string_value ("textDocument/didOpen");
    builder.set_member_name ("params");
    builder.begin_object ();
    builder.set_member_name ("textDocument");
    builder.begin_object ();
    builder.set_member_name ("uri");
    builder.add_string_value (uri);
    builder.set_member_name ("languageId");
    builder.add_string_value (language_id);
    builder.set_member_name ("version");
    builder.add_int_value (version);
    builder.set_member_name ("text");
    builder.add_string_value (content);
    builder.end_object ();
    builder.end_object ();
    builder.end_object ();

    var generator = new Json.Generator ();
    generator.root = builder.get_root ();
    return generator.to_data (null);
}

private string build_publish_diagnostics_message (string uri, int version, string message, i
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("jsonrpc");
    builder.add_string_value ("2.0");
    builder.set_member_name ("method");
    builder.add_string_value ("textDocument/publishDiagnostics");
    builder.set_member_name ("params");
    builder.begin_object ();
    builder.set_member_name ("uri");
    builder.add_string_value (uri);
    builder.set_member_name ("version");
    builder.add_int_value (version);
    builder.set_member_name ("diagnostics");
    builder.begin_array ();
    builder.begin_object ();
    builder.set_member_name ("severity");
    builder.add_int_value (1);
    builder.set_member_name ("message");
    builder.add_string_value (message);
    builder.set_member_name ("range");
    builder.begin_object ();
    builder.set_member_name ("start");
```

```
    builder.begin_object ();
    builder.set_member_name ("line");
    builder.add_int_value (start_line);
    builder.set_member_name ("character");
    builder.add_int_value (start_col);
    builder.end_object ();
    builder.set_member_name ("end");
    builder.begin_object ();
    builder.set_member_name ("line");
    builder.add_int_value (end_line);
    builder.set_member_name ("character");
    builder.add_int_value (end_col);
    builder.end_object ();
    builder.end_object ();
    builder.end_object ();
    builder.end_array ();
    builder.end_object ();
    builder.end_object ();

    var generator = new Json.Generator ();
    generator.root = builder.get_root ();
    return generator.to_data (null);
}

private string build_did_change_message (string uri, int version, string content) {
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("jsonrpc");
    builder.add_string_value ("2.0");
    builder.set_member_name ("method");
    builder.add_string_value ("textDocument/didChange");
    builder.set_member_name ("params");
    builder.begin_object ();
    builder.set_member_name ("textDocument");
    builder.begin_object ();
    builder.set_member_name ("uri");
    builder.add_string_value (uri);
    builder.set_member_name ("version");
    builder.add_int_value (version);
    builder.end_object ();
    builder.set_member_name ("contentChanges");
    builder.begin_array ();
    builder.begin_object ();
    builder.set_member_name ("text");
    builder.add_string_value (content);
    builder.end_object ();
```

```
    builder.end_array ();
    builder.end_object ();
    builder.end_object ();

    var generator = new Json.Generator ();
    generator.root = builder.get_root ();
    return generator.to_data (null);
}
```

File: vapi/libtreesitter.vapi

```
[CCode (cheader_filename = "tree_sitter/api.h")]
namespace TreeSitter {

    [CCode (cname = "TREE_SITTER_LANGUAGE_VERSION")]
    public const int LANGUAGE_VERSION;

    [CCode (cname = "TREE_SITTER_MIN_COMPATIBLE_LANGUAGE_VERSION")]
    public const int MIN_COMPATIBLE_LANGUAGE_VERSION;

    [CCode (cname = "TSInputEncoding", cprefix = "TSInputEncoding", has_type_id = false)]
    public enum InputEncoding {
        UTF8,
        UTF16
    }

    [CCode (cname = "TSSymbolType", cprefix = "TSSymbolType", has_type_id = false)]
    public enum SymbolType {
        Regular,
        Anonymous,
        Auxiliary
    }

    [CCode (cname = "TSLogType", cprefix = "TSLogType", has_type_id = false)]
    public enum LogType {
        Parse,
        Lex
    }

    [SimpleType]
    [CCode (cname = "TSSymbol", has_type_id = false)]
    public struct Symbol : uint16 {
    }

    [CCode (cname = "TSLanguage")]
    [Compact]
```

```
    public class Language {
        uint32 version;
        uint32 symbol_count;
        uint32 alias_count;
        uint32 token_count;
        uint32 external_token_count;
        // const char **symbol_names;
        // const TSSymbolMetadata *symbol_metadata;
        // const uint16 *parse_table;
        // const TSParseActionEntry *parse_actions;
        // const TSLexMode *lex_modes;
        // const TSSymbol *alias_sequences;
        uint16 max_alias_sequence_length;
        // bool (*lex_fn)(TSLexer *, TSStateId);
        // bool (*keyword_lex_fn)(TSLexer *, TSStateId);
        // Symbol keyword_capture_token;
/*
   struct {
    const bool *states;
    const TSSymbol *symbol_map;
    void *(*create)();
    void (*destroy)(void *);
    bool (*scan)(void *, TSLexer *, const bool *symbol_whitelist);
    unsigned (*serialize)(void *, char *);
    void (*deserialize)(void *, const char *, unsigned);
   } external_scanner;
 */
        [CCode (cname = "ts_language_symbol_count")]
        public uint32 get_symbol_count ();

        [CCode (cname = "ts_language_symbol_name")]
        public unowned string get_symbol_name (Symbol symbol);

        [CCode (cname = "ts_language_symbol_for_name")]
        public Symbol symbol_for_name (string name);

        [CCode (cname = "ts_language_symbol_type")]
        public SymbolType get_symbol_type (Symbol symbol);

        [CCode (cname = "ts_language_abi_version")]
        public uint32 get_version ();
    }

    [CCode (cname = "TSParser", free_function = "ts_parser_delete")]
    [Compact]
    public class Parser {
```

```
    [CCode (cname = "ts_parser_new")]
    public Parser ();

    [CCode (cname = "ts_parser_language")]
    public unowned Language get_language ();

    [CCode (cname = "ts_parser_set_language")]
    public bool set_language (Language language);

    [CCode (cname = "ts_parser_parse_string", array_length_type = "uint32_t")]
    public Tree parse_string ([CCode (transfer = "none")] Tree? tree, uint8[] source);
}

[CCode (cname = "TSTree", free_function = "ts_tree_delete")]
[Compact]
public class Tree {
    [CCode (cname = "ts_tree_root_node")]
    public Node root_node ();

    [CCode (cname = "ts_tree_edit")]
    public void edit (InputEdit edit);

    [CCode (cname = "ts_tree_get_changed_ranges")]
    public Range[] get_changed_ranges (Tree new_tree);
}

[SimpleType]
[CCode (cname = "TSPoint", has_type_id = false)]
public struct Point {
    uint32 row;
    uint32 column;
}

[CCode (cname = "TSRange", has_type_id = false)]
public struct Range {
    Point start_point;
    Point end_point;
    uint32 start_byte;
    uint32 end_byte;
}

[CCode (cname = "TSInput", has_type_id = false)]
public struct Input {
    void* payload;
    // const char *(*read)(void *payload, uint32 byte_index, Point position, uint32 *byt
    InputEncoding encoding;
```

```
}

[CCode (cname = "TSLogger", has_type_id = false)]
public struct Logger {
    void* payload;
    // void (*log)(void *payload, TSLogType, const char *);
}

[CCode (cname = "TSInputEdit", has_type_id = false)]
public struct InputEdit {
    uint32 start_byte;
    uint32 old_end_byte;
    uint32 new_end_byte;
    Point start_point;
    Point old_end_point;
    Point new_end_point;
}

[SimpleType]
[CCode (cname = "TSNode", has_type_id = false)]
public struct Node {

    [CCode (cname = "ts_node_type")]
    public unowned string type ();

    [CCode (cname = "ts_node_string")]
    public string to_str ();

    [CCode (cname = "ts_node_start_point")]
    public Point start_point ();

    [CCode (cname = "ts_node_end_point")]
    public Point end_point ();

    [CCode (cname = "ts_node_start_byte")]
    public uint32 start_byte ();

    [CCode (cname = "ts_node_end_byte")]
    public uint32 end_byte ();

    [CCode (cname = "ts_node_child_count")]
    public uint32 child_count ();

    [CCode (cname = "ts_node_child")]
    public unowned Node child (uint32 index);
}
```

```
    [CCode (cname = "TSTreeCursor", free_function = "ts_tree_cursor_delete", has_type_id = f
    [Compact]
    public class TreeCursor {
        [CCode (cname = "ts_tree_cursor_goto_next_sibling")]
        public bool goto_next_sibling ();

        [CCode (cname = "ts_tree_cursor_goto_first_child")]
        public bool goto_first_child ();

        [CCode (cname = "ts_tree_cursor_current_node")]
        public Node current_node ();

        [CCode (cname = "ts_tree_cursor_goto_parent")]
        public bool goto_parent ();
    }
}

/*

   TSLogger ts_parser_logger(const TSParser *);
   void ts_parser_set_logger(TSParser *, TSLogger);
   void ts_parser_print_dot_graphs(TSParser *, FILE *);
   void ts_parser_halt_on_error(TSParser *, bool);
   TSTree *ts_parser_parse(TSParser *, const TSTree *, TSInput);
   bool ts_parser_enabled(const TSParser *);
   void ts_parser_set_enabled(TSParser *, bool);
   size_t ts_parser_operation_limit(const TSParser *);
   void ts_parser_set_operation_limit(TSParser *, size_t);
   void ts_parser_reset(TSParser *);
   void ts_parser_set_included_ranges(TSParser *, const TSRange *, uint32_t);
   const TSRange *ts_parser_included_ranges(const TSParser *, uint32_t *);

   TSTree *ts_tree_copy(const TSTree *);
   void ts_tree_edit(TSTree *, const TSInputEdit *);
   TSRange *ts_tree_get_changed_ranges(const TSTree *, const TSTree *, uint32_t *);
   void ts_tree_print_dot_graph(const TSTree *, FILE *);
   const TSLanguage *ts_tree_language(const TSTree *);

   TSSymbol ts_node_symbol(TSNode);
   char *ts_node_string(TSNode);
   bool ts_node_eq(TSNode, TSNode);
   bool ts_node_is_null(TSNode);
   bool ts_node_is_named(TSNode);
   bool ts_node_is_missing(TSNode);
   bool ts_node_has_changes(TSNode);
```

```
  bool ts_node_has_error(TSNode);
  TSNode ts_node_parent(TSNode);
  TSNode ts_node_child(TSNode, uint32_t);
  TSNode ts_node_named_child(TSNode, uint32_t);
  uint32_t ts_node_child_count(TSNode);
  uint32_t ts_node_named_child_count(TSNode);
  TSNode ts_node_next_sibling(TSNode);
  TSNode ts_node_next_named_sibling(TSNode);
  TSNode ts_node_prev_sibling(TSNode);
  TSNode ts_node_prev_named_sibling(TSNode);
  TSNode ts_node_first_child_for_byte(TSNode, uint32_t);
  TSNode ts_node_first_named_child_for_byte(TSNode, uint32_t);
  TSNode ts_node_descendant_for_byte_range(TSNode, uint32_t, uint32_t);
  TSNode ts_node_named_descendant_for_byte_range(TSNode, uint32_t, uint32_t);
  TSNode ts_node_descendant_for_point_range(TSNode, TSPoint, TSPoint);
  TSNode ts_node_named_descendant_for_point_range(TSNode, TSPoint, TSPoint);
  void ts_node_edit(TSNode *, const TSInputEdit *);

  void ts_tree_cursor_delete(TSTreeCursor *);
  void ts_tree_cursor_reset(TSTreeCursor *, TSNode);
  int64_t ts_tree_cursor_goto_first_child_for_byte(TSTreeCursor *, uint32_t);

 */
```

File: src/Services/TreeSitter/TreeSitterManager.vala

```
// [CCode (cname = "tree_sitter_rust")]
// extern unowned TreeSitter.Language ? get_language_rust ();
// [CCode (cname = "tree_sitter_c")]
// extern unowned TreeSitter.Language ? get_language_c ();
// [CCode (cname = "tree_sitter_javascript")]
// extern unowned TreeSitter.Language ? get_language_javascript ();
// [CCode (cname = "tree_sitter_python")]
// extern unowned TreeSitter.Language ? get_language_python ();
// [CCode (cname = "tree_sitter_ruby")]
// extern unowned TreeSitter.Language ? get_language_ruby ();
[CCode(cname = "tree_sitter_cpp")]
extern unowned TreeSitter.Language ? get_language_cpp();
[CCode(cname = "tree_sitter_vala")]
extern unowned TreeSitter.Language ? get_language_vala();
// [CCode (cname = "tree_sitter_go")]
// extern unowned TreeSitter.Language ? get_language_go ();
// [CCode (cname = "tree_sitter_bash")]
// extern unowned TreeSitter.Language ? get_language_bash ();
// [CCode (cname = "tree_sitter_json")]
// extern unowned TreeSitter.Language ? get_language_json ();
```

```
// [CCode (cname = "tree_sitter_php")]
// extern unowned TreeSitter.Language ? get_language_php ();
// [CCode (cname = "tree_sitter_html")]
// extern unowned TreeSitter.Language ? get_language_html ();
// [CCode (cname = "tree_sitter_xml")]
// extern unowned TreeSitter.Language ? get_language_xml ();
// [CCode (cname = "tree_sitter_typescript")]
// extern unowned TreeSitter.Language ? get_language_typescript ();
// [CCode (cname = "tree_sitter_yaml")]
// extern unowned TreeSitter.Language ? get_language_yaml ();



class Iide.TreeSitterManager : GLib.Object {
    public BaseTreeSitterHighlighter ? get_ts_highlighter(GtkSource.View view) {
        var language_name = ((GtkSource.Buffer) view.buffer).language.name.down();
        message("LANG Detected: " + language_name);
        switch (language_name) {
        case "python" :
            return new PythonTreeSitterHighlighter(view);
        case "c++":
            return new CppTreeSitterHighlighter(view);
        case "vala":
            return new ValaTreeSitterHighlighter(view);
        }
        return null;
    }

    // public unowned TreeSitter.Language? get_ts_language (GtkSource.Buffer buffer) {
    // var language_name = buffer.language.name.down ();
    // unowned TreeSitter.Language? language = null;
    // switch (language_name) {
    // case "bash" :
    // language = get_language_bash ();
    // break;
    // case "c" :
    // language = get_language_c ();
    // break;
    // case "cpp" :
    // language = get_language_cpp ();
    // break;
    // case "go" :
    // language = get_language_go ();
    // break;
    // case "javascript" :
    // language = get_language_javascript ();
```

```
    // break;
    // case "json" :
    // language = get_language_json ();
    // break;
    // case "html" :
    // language = get_language_html ();
    // break;
    // case "php" :
    // language = get_language_php ();
    // break;
    // case "python" :
    // language = get_language_python ();
    // break;
    // case "ruby" :
    // language = get_language_ruby ();
    // break;
    // case "rust" :
    // language = get_language_rust ();
    // break;
    // case "typescript" :
    // language = get_language_typescript ();
    // break;
    // case "xml" :
    // language = get_language_xml ();
    // break;
    // case "yaml" :
    // language = get_language_yaml ();
    // break;
    // }
    // return language;
    // }
}
```

File: data/org.github.kai66673.iide.gschema.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<schemalist gettext-domain="iide">
    <schema id="org.github.kai66673.iide" path="/org/github/kai66673/iide/">
        <key name="color-scheme" type="s">
            <default>"system"</default>
            <summary>Color scheme</summary>
            <description>Color scheme: light, dark, or system</description>
        </key>

        <key name="editor-font-size" type="d">
            <default>6</default>
```

```
    <summary>Editor zoom level</summary>
    <description>Editor zoom level (1-15)</description>
</key>

<key name="show-minimap" type="b">
    <default>true</default>
    <summary>Show minimap</summary>
    <description>Whether to show the minimap in text editors</description>
</key>

<key name="show-line-numbers" type="b">
    <default>true</default>
    <summary>Show line numbers</summary>
    <description>Whether to show line numbers in text editors</description>
</key>

<key name="highlight-current-line" type="b">
    <default>true</default>
    <summary>Highlight current line</summary>
    <description>Whether to highlight the current line in text editors</description>
</key>

<key name="auto-indent" type="b">
    <default>true</default>
    <summary>Auto indent</summary>
    <description>Whether to enable auto indentation</description>
</key>

<key name="panel-start-width" type="d">
    <default>250</default>
    <summary>Start panel width</summary>
    <description>Width of the start panel (file tree)</description>
</key>

<key name="panel-end-width" type="d">
    <default>250</default>
    <summary>End panel width</summary>
    <description>Width of the end panel</description>
</key>

<key name="panel-bottom-height" type="d">
    <default>200</default>
    <summary>Bottom panel height</summary>
    <description>Height of the bottom panel (terminal)</description>
</key>
```

```
<key name="panel-bottom-width" type="d">
    <default>500</default>
    <summary>Bottom panel width</summary>
    <description>Width of the bottom panel split between terminal and log</descripti
</key>

<key name="reveal-start-panel" type="b">
    <default>true</default>
    <summary>Reveal start panel</summary>
    <description>Whether the start panel is visible</description>
</key>

<key name="reveal-end-panel" type="b">
    <default>false</default>
    <summary>Reveal end panel</summary>
    <description>Whether the end panel is visible</description>
</key>

<key name="reveal-bottom-panel" type="b">
    <default>false</default>
    <summary>Reveal bottom panel</summary>
    <description>Whether the bottom panel is visible</description>
</key>

<key name="recent-projects" type="as">
    <default>[]</default>
    <summary>Recent projects</summary>
    <description>List of recently opened project paths</description>
</key>

<key name="max-recent-projects" type="d">
    <default>10</default>
    <summary>Maximum recent projects</summary>
    <description>Maximum number of recent projects to remember</description>
</key>

<key name="last-open-directory" type="s">
    <default>""</default>
    <summary>Last open directory</summary>
    <description>Last directory used for opening files or projects</description>
</key>

<key name="current-project-path" type="s">
    <default>""</default>
    <summary>Current project path</summary>
    <description>Path to the currently open project</description>
```

```
        </key>

        <key name="open-documents" type="as">
            <default>[]</default>
            <summary>Open documents</summary>
            <description>List of currently open document URIs</description>
        </key>

        <key name="panel-layout" type="s">
            <default>""</default>
            <summary>Panel layout</summary>
            <description>JSON string describing panel dock layout</description>
        </key>

        <key name="grid-layout" type="s">
            <default>""</default>
            <summary>Grid layout</summary>
            <description>JSON string describing grid layout</description>
        </key>

        <key name="window-width" type="d">
            <default>1200</default>
            <summary>Window width</summary>
            <description>Width of the main window</description>
        </key>

        <key name="window-height" type="d">
            <default>800</default>
            <summary>Window height</summary>
            <description>Height of the main window</description>
        </key>

        <key name="window-maximized" type="b">
            <default>false</default>
            <summary>Window maximized</summary>
            <description>Whether the window is maximized</description>
        </key>

        <key name="shortcuts" type="s">
            <default>""</default>
            <summary>Custom shortcuts</summary>
            <description>JSON string mapping action IDs to keyboard shortcuts</description>
        </key>
    </schema>
</schemalist>
```

File: src/Services/ProjectManager.vala

```
using Gtk;
using GLib;
using Gee;

public class Iide.LanguageConfig : GLib.Object {
    public string language_id { get; construct set; }
    public string[] server_command { get; construct set; }
    public string[] file_patterns { get; construct set; }

    public LanguageConfig (string language_id, string[] server_command, string[] file_patter
        Object (
            language_id: language_id,
            server_command: server_command,
            file_patterns: file_patterns
        );
    }
}

public class Iide.ProjectManager : Object {
    private static ProjectManager? _instance;
    private GLib.File? current_project_root;
    private string? current_project_name;
    private Iide.SettingsService settings;
    private Gee.HashMap<string, LanguageConfig> language_configs;

    public signal void project_opened (GLib.File project_root);
    public signal void project_closed ();

    public static unowned ProjectManager get_instance () {
        if (_instance == null) {
            _instance = new ProjectManager ();
        }
        return _instance;
    }

    public ProjectManager () {
        current_project_root = null;
        current_project_name = null;
        settings = Iide.SettingsService.get_instance ();
        language_configs = new Gee.HashMap<string, LanguageConfig> ();
        init_default_language_configs ();
    }

    private void init_default_language_configs () {
```

```
    language_configs.set ("c", new LanguageConfig ("c", { "clangd" }, { "*.c", "*.h" }))
    language_configs.set ("cpp", new LanguageConfig ("cpp", { "clangd" }, { "*.cpp", "*.
    language_configs.set ("python", new LanguageConfig ("python", { "basedpyright-langse
    language_configs.set ("rust", new LanguageConfig ("rust", { "rust-analyzer" }, { "*.
    language_configs.set ("go", new LanguageConfig ("go", { "gopls" }, { "*.go" }));
    language_configs.set ("typescript", new LanguageConfig ("typescript", { "typescript-
    language_configs.set ("javascript", new LanguageConfig ("javascript", { "typescript-
    language_configs.set ("json", new LanguageConfig ("json", { "vscode-json-languageser
    language_configs.set ("html", new LanguageConfig ("html", { "vscode-html-language-se
    language_configs.set ("css", new LanguageConfig ("css", { "vscode-css-language-serve
}

private string? last_loaded_config_path;

public Gee.Collection<LanguageConfig> get_language_configs () {
    return language_configs.values;
}

public LanguageConfig? get_language_config (string language_id) {
    return language_configs.get (language_id);
}

public void load_lsp_config (GLib.File project_root) {
    string? config_path = project_root.get_path ();
    if (config_path == null) {
        config_path = project_root.get_uri ();
    }

    if (config_path != null && config_path == last_loaded_config_path) {
        return;
    }

    init_default_language_configs ();

    var config_dir = project_root.get_child (".iide");
    var lsp_file = config_dir.get_child ("lsp.json");

    if (!lsp_file.query_exists (null)) {
        message ("No .iide/lsp.json found at %s", lsp_file.get_path ());
        last_loaded_config_path = config_path;
        return;
    }

    try {
        var parser = new Json.Parser ();
        parser.load_from_file (lsp_file.get_path ());
```

```
        var root = parser.get_root ();
        var obj = (Json.Object) root;

        if (obj.has_member ("languages")) {
            var languages = obj.get_array_member ("languages");
            int lang_count = (int) languages.get_length ();
            for (int j = 0; j < lang_count; j++) {
                var node = languages.get_element (j);
                var lang_obj = node.get_object ();
                if (lang_obj == null) continue;

                string? lang_id = null;
                string[]? server_cmd = null;
                string[]? patterns = null;

                if (lang_obj.has_member ("id")) {
                    lang_id = lang_obj.get_string_member ("id");
                }
                if (lang_obj.has_member ("server")) {
                    var server_arr = lang_obj.get_array_member ("server");
                    int server_count = (int) server_arr.get_length ();
                    server_cmd = new string[server_count];
                    for (int i = 0; i < server_count; i++) {
                        server_cmd[i] = server_arr.get_string_element (i);
                    }
                }
                if (lang_obj.has_member ("patterns")) {
                    var patterns_arr = lang_obj.get_array_member ("patterns");
                    int patterns_count = (int) patterns_arr.get_length ();
                    patterns = new string[patterns_count];
                    for (int i = 0; i < patterns_count; i++) {
                        patterns[i] = patterns_arr.get_string_element (i);
                    }
                }

                if (lang_id != null) {
                    merge_language_config (lang_id, server_cmd, patterns);
                }
            }
        }

        message ("Loaded LSP config from %s", lsp_file.get_path ());
        last_loaded_config_path = config_path;
    } catch (Error e) {
        warning ("Failed to load LSP config: %s", e.message);
    }
```

```
}

private void merge_language_config (string lang_id, string[]? server_cmd, string[]? patt
    var config = language_configs.get (lang_id);
    if (config != null) {
        if (server_cmd != null && server_cmd.length > 0) {
            config.server_command = server_cmd;
        }
        if (patterns != null && patterns.length > 0) {
            config.file_patterns = patterns;
        }
        message ("Merged LSP config for '%s'", lang_id);
        return;
    }

    if (server_cmd != null && patterns != null) {
        language_configs.set (lang_id, new LanguageConfig (lang_id, server_cmd, patterns
        message ("Added new LSP config for '%s'", lang_id);
    }
}

public async void open_project_async (GLib.File project_root) {
    if (!project_root.query_exists (null)) {
        stderr.printf ("Project directory does not exist: %s\n", project_root.get_path (
        return;
    }

    if (current_project_root != null) {
        close_project ();
    }

    init_default_language_configs ();

    current_project_root = project_root;
    current_project_name = project_root.get_basename ();

    settings.current_project_path = project_root.get_path ();
    settings.add_recent_project (project_root.get_path ());
    settings.last_open_directory = project_root.get_parent ().get_path ();

    load_lsp_config (project_root);

    project_opened (project_root);
}

public void close_project () {
```

```
    if (current_project_root != null) {
        current_project_root = null;
        current_project_name = null;
        settings.current_project_path = "";
        project_closed ();
    }
}

public string? get_workspace_root_path () {
    if (current_project_root != null) {
        return current_project_root.get_path ();
    }
    return null;
}

public void open_project_by_path (string path) {
    if (path != null && path != "") {
        var file = GLib.File.new_for_path (path);
        if (file.query_exists (null)) {
            open_project_async.begin (file);
        }
    }
}

public GLib.File? get_current_project_root () {
    return current_project_root;
}

public string? get_current_project_name () {
    return current_project_name;
}

public bool has_open_project () {
    return current_project_root != null;
}

public async void open_project_dialog (Window parent_window) {
    var dialog = new FileDialog () {
        title = _("Open Project"),
        modal = true
    };

    var last_dir = settings.last_open_directory;
    if (last_dir != null && last_dir != "") {
        dialog.initial_folder = GLib.File.new_for_path (last_dir);
    }
```

```
        try {
            var file = yield dialog.select_folder (parent_window, null);
            if (file != null) {
                yield open_project_async (file);
            }
        } catch {
            // User dismissed dialog or other error - silently ignore
        }
    }
}
```

File: src/Services/SettingsService.vala

```
/*
 * settingsservice.vala
 *
 * Copyright 2026 kai
 *
 * This program is free software: you can redistribute it and/or modify
 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation, either version 3 of the License, or
 * (at your option) any later version.
 *
 * This program is distributed in the hope that it will be useful,
 * but WITHOUT ANY WARRANTY; without even the implied warranty of
 * MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
 * GNU General Public License for more details.
 *
 * You should have received a copy of the GNU General Public License
 * along with this program.  If not, see <https://www.gnu.org/licenses/>.
 *
 * SPDX-License-Identifier: GPL-3.0-or-later
 */

public enum ColorScheme {
    SYSTEM,
    LIGHT,
    DARK;

    public static ColorScheme from_string (string value) {
        switch (value) {
            case "light":
                return LIGHT;
            case "dark":
                return DARK;
```

```
            default:
                return SYSTEM;
        }
    }

    public string to_string () {
        switch (this) {
            case LIGHT:
                return "light";
            case DARK:
                return "dark";
            default:
                return "system";
        }
    }

    public Adw.ColorScheme to_adw_color_scheme () {
        switch (this) {
            case LIGHT:
                return Adw.ColorScheme.FORCE_LIGHT;
            case DARK:
                return Adw.ColorScheme.FORCE_DARK;
            default:
                return Adw.ColorScheme.DEFAULT;
        }
    }
}

public class FontSizeHelper : Object {
    public const int DEFAULT_ZOOM_LEVEL = 6;
    public const int MIN_ZOOM_LEVEL = 1;
    public const int MAX_ZOOM_LEVEL = 15;
    private const int[] FONT_SIZES = { 4, 6, 8, 10, 12, 14, 16, 18, 20, 24, 28, 32, 36, 40,

    public static int get_size_for_zoom_level (int zoom_level) {
        if (zoom_level >= MIN_ZOOM_LEVEL && zoom_level <= MAX_ZOOM_LEVEL) {
            return FONT_SIZES[zoom_level - 1];
        }
        return FONT_SIZES[DEFAULT_ZOOM_LEVEL - 1];
    }

    public static int[] get_available_sizes () {
        return FONT_SIZES;
    }
}
```

```
public class Iide.SettingsService : Object {
    private static SettingsService? _instance;
    private Settings settings;

    public signal void editor_setting_changed (string key);

    public static SettingsService get_instance () {
        if (_instance == null) {
            _instance = new SettingsService ();
        }
        return _instance;
    }

    private SettingsService () {
        settings = new Settings ("org.github.kai66673.iide");
    }

    public ColorScheme color_scheme {
        get {
            return ColorScheme.from_string (settings.get_string ("color-scheme"));
        }
        set {
            settings.set_string ("color-scheme", value.to_string ());
            editor_setting_changed ("color-scheme");
        }
    }

    public int editor_font_size {
        get {
            return (int) settings.get_double ("editor-font-size");
        }
        set {
            settings.set_double ("editor-font-size", (double) value);
            editor_setting_changed ("editor-font-size");
        }
    }

    public bool show_minimap {
        get {
            return settings.get_boolean ("show-minimap");
        }
        set {
            settings.set_boolean ("show-minimap", value);
            editor_setting_changed ("show-minimap");
        }
    }
```

```
public bool show_line_numbers {
    get {
        return settings.get_boolean ("show-line-numbers");
    }
    set {
        settings.set_boolean ("show-line-numbers", value);
        editor_setting_changed ("show-line-numbers");
    }
}

public bool highlight_current_line {
    get {
        return settings.get_boolean ("highlight-current-line");
    }
    set {
        settings.set_boolean ("highlight-current-line", value);
        editor_setting_changed ("highlight-current-line");
    }
}

public bool auto_indent {
    get {
        return settings.get_boolean ("auto-indent");
    }
    set {
        settings.set_boolean ("auto-indent", value);
        editor_setting_changed ("auto-indent");
    }
}

public int panel_start_width {
    get {
        return (int) settings.get_double ("panel-start-width");
    }
    set {
        settings.set_double ("panel-start-width", (double) value);
    }
}

public int panel_end_width {
    get {
        return (int) settings.get_double ("panel-end-width");
    }
    set {
        settings.set_double ("panel-end-width", (double) value);
```

```
    }
}

public int panel_bottom_height {
    get {
        return (int) settings.get_double ("panel-bottom-height");
    }
    set {
        settings.set_double ("panel-bottom-height", (double) value);
    }
}

public int panel_bottom_width {
    get {
        return (int) settings.get_double ("panel-bottom-width");
    }
    set {
        settings.set_double ("panel-bottom-width", (double) value);
    }
}

public bool reveal_start_panel {
    get {
        return settings.get_boolean ("reveal-start-panel");
    }
    set {
        settings.set_boolean ("reveal-start-panel", value);
    }
}

public bool reveal_end_panel {
    get {
        return settings.get_boolean ("reveal-end-panel");
    }
    set {
        settings.set_boolean ("reveal-end-panel", value);
    }
}

public bool reveal_bottom_panel {
    get {
        return settings.get_boolean ("reveal-bottom-panel");
    }
    set {
        settings.set_boolean ("reveal-bottom-panel", value);
    }
```

```
}

public string[] recent_projects {
    owned get {
        return settings.get_strv ("recent-projects");
    }
}

public int max_recent_projects {
    get {
        return (int) settings.get_double ("max-recent-projects");
    }
    set {
        settings.set_double ("max-recent-projects", (double) value);
    }
}

public string last_open_directory {
    owned get {
        return settings.get_string ("last-open-directory");
    }
    set {
        settings.set_string ("last-open-directory", value);
    }
}

public string current_project_path {
    owned get {
        return settings.get_string ("current-project-path");
    }
    set {
        settings.set_string ("current-project-path", value);
    }
}

public Gee.ArrayList<string> open_documents {
    owned get {
        var arr = settings.get_strv ("open-documents");
        var list = new Gee.ArrayList<string> ();
        foreach (var s in arr) {
            list.add (s);
        }
        return (owned) list;
    }
    set {
        var arr = new string[value.size];
```

```
        int i = 0;
        foreach (var s in value) {
            arr[i++] = s;
        }
        settings.set_strv ("open-documents", arr);
    }
}

public string panel_layout {
    owned get {
        return settings.get_string ("panel-layout");
    }
    set {
        settings.set_string ("panel-layout", value);
    }
}

public string grid_layout {
    owned get {
        return settings.get_string ("grid-layout");
    }
    set {
        settings.set_string ("grid-layout", value);
        Settings.sync ();
    }
}

public int window_width {
    get {
        return (int) settings.get_double ("window-width");
    }
    set {
        settings.set_double ("window-width", (double) value);
    }
}

public int window_height {
    get {
        return (int) settings.get_double ("window-height");
    }
    set {
        settings.set_double ("window-height", (double) value);
    }
}

public bool window_maximized {
```

```
    get {
        return settings.get_boolean ("window-maximized");
    }
    set {
        settings.set_boolean ("window-maximized", value);
    }
}

public void add_recent_project (string path) {
    var projects = new List<string> ();
    projects.prepend (path);

    foreach (var p in recent_projects) {
        if (p != path) {
            bool found = false;
            foreach (var existing in projects) {
                if (existing == path) {
                    found = true;
                    break;
                }
            }
            if (!found) {
                projects.append (p);
            }
        }
    }

    var arr = new string[projects.length ()];
    int i = 0;
    foreach (var p in projects) {
        arr[i++] = p;
    }
    settings.set_strv ("recent-projects", arr);
}

public void remove_recent_project (string path) {
    var projects = new List<string> ();
    foreach (var p in recent_projects) {
        if (p != path) {
            projects.append (p);
        }
    }

    var arr = new string[projects.length ()];
    int i = 0;
    foreach (var p in projects) {
```

```
            arr[i++] = p;
        }
        settings.set_strv ("recent-projects", arr);
    }

    public void clear_recent_projects () {
        settings.set_strv ("recent-projects", {});
    }

    public signal void setting_changed (string key);
}
```

File: src/style.css

```
textview {
 ----
.log-view text {
 ----
.text-view.zoom-1 {
 ----
.lsp_error_line {
 ----
.lsp_warning_line {
 ----
.lsp_info_line {
 ----
.text-view.zoom-2 {
 ----
.text-view.zoom-3 {
 ----
.text-view.zoom-4 {
 ----
.text-view.zoom-5 {
 ----
.text-view.zoom-6 {
 ----
.text-view.zoom-7 {
 ----
.text-view.zoom-8 {
 ----
.text-view.zoom-9 {
 ----
.text-view.zoom-10 {
 ----
.text-view.zoom-11 {
 ----
```

```
.text-view.zoom-12 {
----
.text-view.zoom-13 {
----
.text-view.zoom-14 {
----
.text-view.zoom-15 {
----
.textview-map {
```

File: meson.build

```
project(
    'iide',
    ['c', 'vala'],
    version: '0.1.0',
    meson_version: '>= 1.0.0',
    default_options: [
        'warning_level=2',
        'werror=false',
    ],
)

add_project_arguments('-DGETTEXT_PACKAGE="' + meson.project_name() + '"', language: 'c')

tree_sitter_sources = [
    'vendor/tree-sitter/lib/src/lib.c',
    'parsers/ts_helper.c',
    'parsers/bash/src/parser.c',
    'parsers/bash/src/scanner.c',
    'parsers/json/src/parser.c',
    'parsers/cpp/src/parser.c',
    'parsers/cpp/src/scanner.c',
    'parsers/go/src/parser.c',
    'parsers/html/src/parser.c',
    'parsers/html/src/scanner.c',
    'parsers/javascript/src/parser.c',
    'parsers/javascript/src/scanner.c',
    'parsers/python/src/parser.c',
    'parsers/python/src/scanner.c',
    'parsers/ruby/src/parser.c',
    'parsers/ruby/src/scanner.c',
    'parsers/rust/src/parser.c',
    'parsers/rust/src/scanner.c',
    'parsers/yaml/src/parser.c',
    'parsers/yaml/src/scanner.c',
```

```
    'parsers/vala/src/parser.c',
]

tree_sitter_lib = static_library(
    'tree-sitter-runtime',
    tree_sitter_sources,
    include_directories: [
        include_directories('vendor/tree-sitter/lib/src'),
        include_directories('vendor/tree-sitter/lib/include'),
        include_directories('parsers/vala/src'),
    ],
)

tree_sitter_vapi = meson.get_compiler('vala').find_library('libtreesitter', dirs: join_paths

tree_sitter_xml_sources = [
    'parsers/xml/xml/src/parser.c',
    'parsers/xml/xml/src/scanner.c',
]

tree_sitter_xml = static_library(
    'tree-sitter-xml',
    tree_sitter_xml_sources,
    include_directories: [
        include_directories('parsers/xml/xml/src'),
    ],
)

tree_sitter_c_sources = [
    'parsers/c/src/parser.c',
]

tree_sitter_c = static_library(
    'tree-sitter-c',
    tree_sitter_c_sources,
    include_directories: [
        include_directories('parsers/c/src'),
    ],
)

tree_sitter_typescript_sources = [
    'parsers/typescript/typescript/src/parser.c',
    'parsers/typescript/typescript/src/scanner.c',
]

tree_sitter_typescript = static_library(
```

```
    'tree-sitter-typescript',
    tree_sitter_typescript_sources,
    include_directories: [
        include_directories('parsers/typescript/typescript/src'),
    ],
)

tree_sitter_php_sources = [
    'parsers/php/php/src/parser.c',
    'parsers/php/php/src/scanner.c',
]

tree_sitter_php = static_library(
    'tree-sitter-php',
    tree_sitter_php_sources,
    include_directories: [
        include_directories('parsers/php/php/src'),
    ],
)

tree_sitter_dep = declare_dependency(
    link_with: [
        tree_sitter_lib,
        tree_sitter_xml,
        tree_sitter_c,
        tree_sitter_typescript,
        tree_sitter_php,
    ],
    include_directories: include_directories(join_paths('vendor', 'tree-sitter', 'lib', 'inc
    dependencies: tree_sitter_vapi,
)

i18n = import('i18n')
gnome = import('gnome')
valac = meson.get_compiler('vala')

srcdir = meson.project_source_root() / 'src'

config_h = configuration_data()
config_h.set_quoted('PACKAGE_VERSION', meson.project_version())
config_h.set_quoted('GETTEXT_PACKAGE', 'iide')
config_h.set_quoted('LOCALEDIR', get_option('prefix') / get_option('localedir'))
configure_file(output: 'config.h', configuration: config_h)

config_dep = valac.find_library('config', dirs: srcdir)
config_inc = include_directories('.')
```

```
subdir('data')
subdir('src')
subdir('po')

test_lsp_service_sources = [
    'tests/test_lsp_service.vala',
    'src/Services/LSP/IdeLspClient.vala',
]

test_lsp_service_deps = [
    dependency('glib-2.0'),
    dependency('gtk4'),
    dependency('gio-2.0'),
    dependency('json-glib-1.0'),
    dependency('gee-0.8'),
    dependency('jsonrpc-glib-1.0'),
]

test_lsp_service = executable(
    'test_lsp_service',
    test_lsp_service_sources,
    dependencies: test_lsp_service_deps,
    install: false,
)

test(
    'lsp_service',
    test_lsp_service,
    protocol: 'exitcode',
)

gnome.post_install(
    glib_compile_schemas: true,
    gtk_update_icon_cache: true,
    update_desktop_database: true,
)

local_settings_conf = configuration_data()
local_settings_conf.set('PREFIX', get_option('prefix'))
local_settings_sh = configure_file(
    input: 'scripts/install-local-settings.sh.in',
    output: 'install-local-settings.sh',
    configuration: local_settings_conf,
)
meson.add_install_script(local_settings_sh)
```

File: src/Widgets/TextView/FontZoomer.vala

```
using Gtk;
using GtkSource;
using GLib;

public class FontZoomer : Object {
    private View src_view;
    private int zoom_level;
    private Iide.SettingsService settings;
    public signal void zoom_changed(int level);

    public FontZoomer(View src_view) {
        this.src_view = src_view;
        this.settings = Iide.SettingsService.get_instance();

        if (!this.src_view.has_css_class("text-view")) {
            this.src_view.add_css_class("text-view");
        }

        this.zoom_level = settings.editor_font_size;
        if (this.zoom_level < FontSizeHelper.MIN_ZOOM_LEVEL || this.zoom_level > FontSizeHel
            this.zoom_level = FontSizeHelper.DEFAULT_ZOOM_LEVEL;
        }
        this.src_view.add_css_class("zoom-" + zoom_level.to_string());

        var scroll_controller = new EventControllerScroll(EventControllerScrollFlags.VERTICA
        scroll_controller.scroll.connect((dx, dy) => {
            var event = scroll_controller.get_current_event();
            if (event == null)return false;

            var modifiers = event.get_modifier_state();
            if ((modifiers & Gdk.ModifierType.CONTROL_MASK) != 0) {
                if (dy < 0) {
                    zoom_in();
                } else if (dy > 0) {
                    zoom_out();
                }
                return true;
            }
            return false;
        });

        this.src_view.add_controller(scroll_controller);
    }
```

```
    public void zoom_in() {
        if (zoom_level < FontSizeHelper.MAX_ZOOM_LEVEL) {
            src_view.remove_css_class("zoom-" + zoom_level.to_string());
            zoom_level++;
            src_view.add_css_class("zoom-" + zoom_level.to_string());
            settings.editor_font_size = zoom_level;
            zoom_changed(zoom_level);
        }
    }

    public void zoom_out() {
        if (zoom_level > FontSizeHelper.MIN_ZOOM_LEVEL) {
            src_view.remove_css_class("zoom-" + zoom_level.to_string());
            zoom_level--;
            src_view.add_css_class("zoom-" + zoom_level.to_string());
            settings.editor_font_size = zoom_level;
            zoom_changed(zoom_level);
        }
    }

    public void zoom_reset() {
        src_view.remove_css_class("zoom-" + zoom_level.to_string());
        zoom_level = FontSizeHelper.DEFAULT_ZOOM_LEVEL;
        src_view.add_css_class("zoom-" + zoom_level.to_string());
        settings.editor_font_size = zoom_level;
        zoom_changed(zoom_level);
    }

    public void set_zoom_level(int level) {
        if (level < FontSizeHelper.MIN_ZOOM_LEVEL || level > FontSizeHelper.MAX_ZOOM_LEVEL)
            return;
        }
        if (zoom_level == level) {
            return;
        }
        src_view.remove_css_class("zoom-" + zoom_level.to_string());
        zoom_level = level;
        src_view.add_css_class("zoom-" + zoom_level.to_string());
        zoom_changed(zoom_level);
    }

    public int get_zoom_level() {
        return zoom_level;
    }
}
```

File: src/Widgets/TextView/SourceView.vala

```
public class Iide.WordRange {
    public int line;
    public int start_column;
    public int end_column;

    public WordRange (int line, int start_column, int end_column) {
        this.line = line;
        this.start_column = start_column;
        this.end_column = end_column;
    }

    public bool is_equal (WordRange? other) {
        if (other == null) {
            return false;
        }
        return line == other.line && start_column == other.start_column && end_column == oth
    }
}

public class Iide.LspTooltipWidget : Gtk.Box {
    private Gtk.Label label;
    private Gtk.Spinner spinner;

    public LspTooltipWidget () {
        Object (
                orientation: Gtk.Orientation.HORIZONTAL,
                spacing: 6,
                margin_top: 8,
                margin_bottom: 8,
                margin_start: 8,
                margin_end: 8
        );

        spinner = new Gtk.Spinner ();
        label = new Gtk.Label ("    ...");
        label.use_markup = true;
        label.wrap = true;
        label.max_width_chars = 60;

        append (spinner);
        append (label);
        spinner.start ();
    }
```

```
    public void update_text (string? text, bool show_spinner) {
        if (show_spinner) {
            spinner.start ();
            spinner.show ();
        } else {
            spinner.stop ();
            spinner.hide ();
        }

        if (text != null && text != "") {
            label.set_markup (text);
        } else {
            label.set_text ("        ");
        }
    }
}

public class Iide.SourceView : GtkSource.View {
    public Window window;
    public string uri { get; private set; }
    private Iide.TreeSitterManager ts_manager;
    public BaseTreeSitterHighlighter? ts_highlighter;
    public GutterMarkRenderer mark_renderer;
    public string icon_name = "text-x-generic";
    private Gtk.TextIter? pending_scroll_iter = null;

    private WordRange? last_hover_range = null;
    private LspTooltipWidget tooltip_widget;
    private string tooltip_separator = "                       ";

    // Incremental didChange...
    private Gee.ArrayList<PendingChange> pending_queue = new Gee.ArrayList<PendingChange> ()
    private uint debounce_id = 0;
    private uint debounce_full_id = 0;
    private int document_version = 0;

    private ulong insert_handler_id = 0;
    private ulong delete_handler_id = 0;
    private int lsp_sync_kind = 0; // 0 -           , 1 - incremental, 2 - full sync

    // private LoggerService logger = LoggerService.get_instance ();

    public SourceView (Window window, string uri, GtkSource.Buffer buffer) {
        Object (buffer : buffer);
        this.window = window;
        this.uri = uri;
```

```
this.ts_manager = new TreeSitterManager ();
this.ts_highlighter = null;

this.tooltip_widget = new LspTooltipWidget ();

// LSP-complete
var completion = get_completion ();
completion.select_on_show = true;
completion.show_icons = true;
completion.page_size = 18;
completion.remember_info_visibility = false;
var provider = new LspCompletionProvider (this);
completion.add_provider (provider);

// Build extra menu for context menu
var zoom_section = new GLib.Menu ();
zoom_section.append (_("Zoom In"), "app.zoom_in");
zoom_section.append (_("Zoom Out"), "app.zoom_out");
zoom_section.append (_("Reset Zoom"), "app.zoom_reset");

var view_section = new GLib.Menu ();
view_section.append (_("Minimap"), "app.toggle_minimap");

var extra_menu = new GLib.Menu ();
extra_menu.append_section (null, zoom_section);
extra_menu.append_section (null, view_section);
this.extra_menu = extra_menu;

var settings = Iide.SettingsService.get_instance ();

show_line_numbers = settings.show_line_numbers;
highlight_current_line = settings.highlight_current_line;
auto_indent = settings.auto_indent;
indent_on_tab = true;

// Marks gutter
set_show_line_marks (false);
var left_gutter = get_gutter (Gtk.TextWindowType.LEFT);
left_gutter.visible = true;

mark_renderer = new GutterMarkRenderer ();
mark_renderer.set_icons_size (FontSizeHelper.get_size_for_zoom_level (settings.edito
left_gutter.insert (mark_renderer, 0);

LspDiagnosticsMark.set_mark_attributes (this);
```

```
// Connect to settings changes to apply to all open documents
settings.editor_setting_changed.connect ((key) => {
    switch (key) {
        case "show-line-numbers" :
            show_line_numbers = settings.show_line_numbers;
            break;
        case "highlight-current-line" :
            highlight_current_line = settings.highlight_current_line;
            break;
        case "auto-indent":
            auto_indent = settings.auto_indent;
            break;
    }
});

// SpaceDrawer
var space_drawer = get_space_drawer ();

// 2.
space_drawer.set_enable_matrix (true);
space_drawer.set_types_for_locations (
                                      GtkSource.SpaceLocationFlags.ALL,
                                      GtkSource.SpaceTypeFlags.NONE
);
space_drawer.set_types_for_locations (
                                      GtkSource.SpaceLocationFlags.LEADING | GtkSour
                                      GtkSource.SpaceTypeFlags.SPACE | GtkSource.Spa
);

// Tree-sitter
detect_language ();
ts_highlighter = ts_manager.get_ts_highlighter (this);
if (ts_highlighter != null) {
    ((GtkSource.Buffer) (buffer)).highlight_syntax = false;
}

// LSP-tooltips
has_tooltip = true;
query_tooltip.connect (on_query_tooltip);

// Control-click controller (goto definition)
var click_gest = new Gtk.GestureClick ();
click_gest.set_button (1);
click_gest.pressed.connect (on_click_pressed);
add_controller (click_gest);
```

```
    // setup_buffer_signals ();
    // setup_buffer_signals_test ();
    this.connect_incremental_signals ();

    buffer.set_modified (false);
}

//
public void setup_lsp_sync (LspClient client) {
    this.apply_sync_strategy (client.capabilities, client);
}

public void setup_no_lsp_sync () {
    lsp_sync_kind = 0;
    if (insert_handler_id > 0)buffer.disconnect (insert_handler_id);
    if (delete_handler_id > 0)buffer.disconnect (delete_handler_id);

    //                  (        )
    this.pending_queue.clear ();
}

private void connect_incremental_signals () {
    insert_handler_id = buffer.insert_text.connect ((ref location, text, len) => {
        var change = new PendingChange (text, location);
        this.add_change (change);
    });

    delete_handler_id = buffer.delete_range.connect ((start, end) => {
        var change = new PendingChange ("", start, end);
        this.add_change (change);
    });
}

private void apply_sync_strategy (ServerCapabilities caps, LspClient client) {
    //                      (Full Sync)
    if (caps.sync_kind != TextDocumentSyncKind.INCREMENTAL) {
        //       'before'
        if (insert_handler_id > 0)buffer.disconnect (insert_handler_id);
        if (delete_handler_id > 0)buffer.disconnect (delete_handler_id);

        //                 (            Full Sync)
        this.pending_queue.clear ();

        //         Full Sync (                  )
        buffer.changed.connect_after (() => {
            this.start_debounce_full_timer ();
```

```
        });

        //                          (              )
        client.text_document_did_change.begin (this.uri, this.document_version, this.buf
        lsp_sync_kind = 2;
    } else {
        lsp_sync_kind = 1;
        reset_timer ();
    }
}

private void start_debounce_full_timer () {
    // 1.               ,
    if (debounce_full_id > 0) {
        Source.remove (debounce_full_id);
    }

    // 2.               (    , 500       )
    debounce_full_id = Timeout.add (500, () => {
        this.flush_changes_full_async.begin (); //
        debounce_full_id = 0;
        return Source.REMOVE;
    });
}

private async void flush_changes_full_async () {
    this.document_version++;

    //                (         ,          Main Loop       )
    string current_text = this.buffer.text;

    var client = IdeLspService.get_instance ().get_client_for_uri (this.uri);
    if (client != null) {
        try {
            //
            yield client.text_document_did_change (this.uri, this.document_version, curr

            debug ("LSP: Full sync sent for %s (v%d)", this.uri, this.document_version);
        } catch (Error e) {
            warning ("LSP: Full sync error: %s", e.message);
        }
    }
}

private void add_change (PendingChange nc) {
    if (!pending_queue.is_empty) {
```

```
        // var last = pending_queue.get (pending_queue.size - 1);

        //
        //    :        char_count()
        // if (nc.text != "" && last.text != "" && nc.start_offset == last.end_offset) {

        // last.text += nc.text;

        ////
        // last.end_offset = nc.end_offset;
        // last.end_line = nc.end_line;
        // last.end_char = nc.end_char;

        // reset_timer ();
        // return;
        // }

        // TODO: merge mergeable delete_range changes...
        //           (Backspace)
        //          -                 ,
        // if (nc.text == "" && last.text == "" && nc.end_offset == last.start_offset) {
        // last.start_offset = nc.start_offset;
        // last.start_line = nc.start_line;
        // last.start_char = nc.start_char;
        // reset_timer ();
        // return;
        // }
    }

    pending_queue.add (nc);
    reset_timer ();
}

private void reset_timer () {
    if (debounce_id > 0)Source.remove (debounce_id);
    debounce_id = Timeout.add (400, () => {
        flush_changes ();
        return Source.REMOVE;
    });
}

public void flush_changes () {
    if (lsp_sync_kind != 1)return;
    if (pending_queue.is_empty)return;

    var changes = pending_queue;
```

```
    pending_queue = new Gee.ArrayList<PendingChange> ();
    if (debounce_id > 0) {
        Source.remove (debounce_id);
        debounce_id = 0;
    }

    this.document_version++;

    //
    var lsp_service = IdeLspService.get_instance ();
    lsp_service.send_did_change.begin (this.uri, this.document_version, changes);
}

private async void flush_changes_async () {
    if (pending_queue.is_empty)return;

    var changes = pending_queue;
    pending_queue = new Gee.ArrayList<PendingChange> ();
    if (debounce_id > 0) {
        Source.remove (debounce_id);
        debounce_id = 0;
    }

    this.document_version++;

    //
    var lsp_service = IdeLspService.get_instance ();
    yield lsp_service.send_did_change (this.uri, this.document_version, changes);
}

public async void sync_changes_async () {
    switch (lsp_sync_kind) {
    case 1:
        yield flush_changes_async ();

        break;
    case 2:
        yield flush_changes_full_async ();

        break;
    }
}

public GtkSource.Language? language {
    set {
        ((GtkSource.Buffer) buffer).language = value;
```

```
    }
    get {
        return ((GtkSource.Buffer) buffer).language;
    }
}

public void detect_language () {
    var manager = GtkSource.LanguageManager.get_default ();
    var file = GLib.File.new_for_uri (uri);

    string mime_type = mime_type_for_file (file);
    message ("MIME: _ " + mime_type);

    icon_name = IconProvider.get_mime_type_icon_name (mime_type);
    language = manager.guess_language (file.get_path (), mime_type);

    // Fake file type detection
    // "Not all files are equal"
    if (file.get_basename () == "CMakeLists.txt") {
        language = manager.get_language ("cmake");
        icon_name = "text-x-cmake"; // Specific icon for CMake
    }
}

public WordRange ? get_word_range_under_cursor (Gtk.TextIter cursor_iter) {
    // 2.    :                          ,
    unichar c = cursor_iter.get_char ();
    if (c.isspace () || c == '\0') {
        return null;
    }

    // 3.
    Gtk.TextIter end_iter = cursor_iter;

    // 4.
    //                   , backward_word_start     false               .
    //          ,                   .
    if (!cursor_iter.starts_word ()) {
        cursor_iter.backward_word_start ();
    }

    // 5.
    if (!end_iter.ends_word ()) {
        end_iter.forward_word_end ();
    }
```

```
    return new WordRange (cursor_iter.get_line (), cursor_iter.get_line_offset (), end_i
}

private bool on_lsp_diagnostics_tooltip (GLib.SList<weak GtkSource.Mark> marks, Gtk.Tool
    var sb = new StringBuilder ();

    foreach (var mark in marks) {
        var lsp_mark = mark as LspDiagnosticsMark;
        if (lsp_mark == null) {
            continue;
        }

        string icon;
        string header_color;

        switch (lsp_mark.severity) {
        case 1 :
            icon = " ";
            header_color = "#F44336"; //
            break;
        case 2 :
            icon = " ";
            header_color = "#FF9800"; //
            break;
        case 3:
        case 4:
            icon = " ";
            header_color = "#2196F3"; //
            break;
        default:
            icon = " ";
            header_color = "#F44336"; //
            break;
        }

        if (sb.len > 0) {
            sb.append ("\n" + tooltip_separator + "\n");
        }

        //
        sb.append_printf ("%s <span font_weight='bold' foreground='%s'>%s</span>\n",
                          icon, header_color, lsp_mark.category.up ());
        sb.append_printf ("<span>%s</span>", GLib.Markup.escape_text (lsp_mark.diagnosti
    }

    if (sb.len > 0) {
```

```
        tooltip_widget.update_text (sb.str, false);
        tooltip.set_custom (tooltip_widget);
        return true;
    }

    return false;
}

private bool on_lsp_hover_tooltip (Gtk.TextIter iter, Gtk.Tooltip tooltip) {
    WordRange? word_range = get_word_range_under_cursor (iter);
    if (word_range == null) {
        last_hover_range = null;
        return false;
    }

    tooltip.set_custom (tooltip_widget);

    //      ,             100
    if (word_range.is_equal (last_hover_range)) {
        return true;
    }
    tooltip_widget.update_text ("Loading...", true);
    last_hover_range = word_range;

    flush_changes ();
    fetch_lsp_hover_async.begin (word_range.line, word_range.start_column + 1);

    return true;
}

private bool on_query_tooltip (int x, int y, bool keyboard_mode, Gtk.Tooltip tooltip) {
    Gtk.TextIter iter;

    //
    int buffer_x, buffer_y;
    window_to_buffer_coords (Gtk.TextWindowType.WIDGET, x, y, out buffer_x, out buffer_y

    //
    if (!get_iter_at_location (out iter, buffer_x, buffer_y)) {
        return false;
    }

    //                  (        "error")
    var buffer = (GtkSource.Buffer) buffer;
    var marks = buffer.get_source_marks_at_line (iter.get_line (), null);
```

```
    if (marks.length () > 0) {
        return on_lsp_diagnostics_tooltip (marks, tooltip);
    }

    return on_lsp_hover_tooltip (iter, tooltip);
}

private async void fetch_lsp_hover_async (int line, int col) {
    var lsp_service = IdeLspService.get_instance ();
    string? markdown = yield lsp_service.request_hover (uri, line, col);

    tooltip_widget.update_text (markdown, false);
}

private void on_click_pressed (Gtk.GestureClick gesture, int n_press, double x, double y
    //
    var modifiers = gesture.get_current_event_state ();

    if ((modifiers & Gdk.ModifierType.CONTROL_MASK) != 0) {
        Gtk.TextIter iter;
        //   GTK4
        //                     (           )
        int buf_x, buf_y;
        window_to_buffer_coords (Gtk.TextWindowType.WIDGET, (int) x, (int) y, out buf_x,

        if (get_iter_at_location (out iter, buf_x, buf_y)) {
            get_buffer ().place_cursor (iter);

            //
            flush_changes ();
            handle_ctrl_click_async.begin (iter.get_line (), iter.get_line_offset ());
        }
    }
}

private async void handle_ctrl_click_async (int line, int col) {
    var lsp_service = IdeLspService.get_instance ();
    var locations = yield lsp_service.goto_definition (uri, line, col);

    if (locations == null || locations.size == 0) {
        LoggerService.get_instance ().warning ("LSP", "No locations found for goto defin
        return;
    }

    var loc = locations.get (0);
    window.get_document_manager ().open_document_with_selection (File.new_for_uri (loc.u
```

```
    }

    public void select_and_scroll (int line, int start_col, int end_col, bool is_new) {
        if (line >= buffer.get_line_count ()) {
            return;
        }

        Gtk.TextIter start_iter;
        buffer.get_iter_at_line_offset (out start_iter, line, start_col);

        Gtk.TextIter end_iter;
        buffer.get_iter_at_line_offset (out end_iter, line, end_col);

        buffer.select_range (start_iter, end_iter);
        scroll_to_iter (start_iter, 0.0, true, 0.5, 0.5);
        pending_scroll_iter = null;
        if (is_new) {
            pending_scroll_iter = start_iter;
        }
    }

    public override void size_allocate (int width, int height, int baseline) {
        base.size_allocate (width, height, baseline);
        if (pending_scroll_iter != null) {
            scroll_to_iter (pending_scroll_iter, 0.0, true, 0.5, 0.5);
            pending_scroll_iter = null;
        }
    }
}
```

File: src/Widgets/TextView/LspCompletionProvider.vala

```
/*
 * LspCompletionProvider.vala
 *
 * Copyright 2026 kai
 *
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 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation, either version 3 of the License, or
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 * but WITHOUT ANY WARRANTY; without even the implied warranty of
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```

```
 *
 * You should have received a copy of the GNU General Public License
 * along with this program.  If not, see <https://www.gnu.org/licenses/>.
 *
 * SPDX-License-Identifier: GPL-3.0-or-later
 */

using GLib;
using Gee;
using Gtk;
using GtkSource;

namespace Iide {

    public class LspCompletionProposal : GLib.Object, CompletionProposal {
        public IdeLspCompletionItem item { get; private set; }

        public LspCompletionProposal (IdeLspCompletionItem item) {
            this.item = item;
        }

        public string get_label () {
            return item.label;
        }

        public string ? get_markup () {
            return item.label;
        }

        public string get_text () {
            return item.insert_text;
        }

        public Icon ? get_icon () {
            return null;
        }

        public string ? get_info () {
            return item.documentation;
        }
    }

    public class LspCompletionProvider : GLib.Object, CompletionProvider {
        private weak SourceView source_view;
        private IdeLspService lsp_service;
        private GLib.ListStore base_store;
```

```
private Gtk.FilterListModel filter_model;
private string current_word = "";

private bool filter_proposals (Object item) {
    var proposal = item as LspCompletionProposal;
    if (proposal == null)return false;

    //          filter_text   LSP,         ,     label
    string haystack = proposal.get_label ();
    string needle = this.current_word;

    if (needle == "")return true;

    //                  GtkSource.Completion
    uint priority;
    return GtkSource.Completion.fuzzy_match (haystack, needle, out priority);
}

public LspCompletionProvider (SourceView view) {
    this.source_view = view;
    this.lsp_service = IdeLspService.get_instance ();

    // 1.
    base_store = new GLib.ListStore (typeof (LspCompletionProposal));

    // 2.         ,                 filter_proposals
    var custom_filter = new Gtk.CustomFilter (filter_proposals);

    // 3.         -
    filter_model = new Gtk.FilterListModel (base_store, custom_filter);
}

public virtual bool is_trigger (Gtk.TextIter iter, unichar ch) {
    // 1.
    var client = IdeLspService.get_instance ().get_client_for_uri (this.source_view.

    if (client == null || !client.is_initialized) {
        //               ,                 (         )
        return ch == '.';
    }

    // 2.      ,
    string s = ch.to_string ();
    if (client.capabilities.completion_triggers.contains (s)) {
        return true;
    }
```

```
    return false;
}

public virtual CompletionActivation get_activation (CompletionContext context) {
    //        :   Ctrl+Space
    return CompletionActivation.INTERACTIVE | CompletionActivation.USER_REQUESTED;
}

public virtual async GLib.ListModel populate_async (CompletionContext context, GLib.
    base_store.remove_all ();
    current_word = context.get_word ();

    if (source_view == null) {
        return filter_model;
    }

    yield source_view.sync_changes_async ();

    var buffer = source_view.buffer;
    var insert_mark = buffer.get_insert ();
    TextIter iter;
    buffer.get_iter_at_mark (out iter, insert_mark);

    int line = iter.get_line ();
    int character = iter.get_line_offset ();
    string uri = source_view.uri;

    string? trigger_char = null;
    var trigger_kind = CompletionTriggerKind.INVOKED; // Invoked

    TextIter prev = iter;
    if (prev.backward_char ()) {
        unichar ch = prev.get_char ();
        string s = ch.to_string ();

        var client = IdeLspService.get_instance ().get_client_for_uri (this.source_v
        if (client != null && client.capabilities.completion_triggers.contains (s))
            trigger_char = s;
            trigger_kind = CompletionTriggerKind.TRIGGER_CHARACTER;
        }
    }

    var result = yield lsp_service.request_completion (uri, line, character, trigger

    if (result == null || result.items.size == 0) {
```

```
        return filter_model;
    }

    foreach (var item in result.items) {
        var proposal = new LspCompletionProposal (item);
        base_store.append (proposal);
    }

    return filter_model;
}

private string get_icon_name_for_kind (int kind) {
    switch (kind) {
    case 2 : case 3: return "code-function-symbolic"; // Method, Function
    case 4: return "code-class-symbolic"; // Constructor
    case 5: return "code-variable-symbolic"; // Field
    case 6: return "code-variable-symbolic"; // Variable
    case 7: return "code-class-symbolic"; // Class
    case 8: return "code-class-symbolic"; // Interface
    case 9: return "code-context-menu-symbolic"; // Module
    case 10: return "code-variable-symbolic"; // Property
    case 11: return "code-variable-symbolic"; // Unit
    case 12: return "code-variable-symbolic"; // Value
    case 13: return "code-class-symbolic"; // Enum
    case 14: return "code-variable-symbolic"; // Keyword
    case 15: return "code-context-menu-symbolic"; // Snippet
    case 16: return "code-variable-symbolic"; // Color
    case 17: return "code-variable-symbolic"; // File
    case 18: return "code-variable-symbolic"; // Reference
    default: return "code-variable-symbolic";
    }
}

public virtual void display (CompletionContext context, CompletionProposal proposal,
    var p = (LspCompletionProposal) proposal;

    switch (cell.column) {
    case CompletionColumn.ICON:
        cell.set_icon_name (get_icon_name_for_kind (p.item.kind));
        break;
    case CompletionColumn.TYPED_TEXT:
        cell.text = p.get_label ();
        break;
    case CompletionColumn.DETAILS: {
        string? doc = p.item.documentation;
        if (doc != null && doc != "") {
```

```
            cell.set_markup (doc);
        }
        break;
    }
    case CompletionColumn.COMMENT: {
        string? doc = p.item.detail;
        if (doc != null && doc != "") {
            cell.set_markup (doc);
        }
        break;
    }
    default:
        break;
    }
}

// 2.            (   )
public virtual void activate (CompletionContext context, CompletionProposal proposal
    var view = context.get_view ();
    var buffer = view.get_buffer ();

    TextIter start, end;

    //                ,      GSV            "           "
    if (context.get_bounds (out start, out end)) {
        //             ,
        buffer.delete (ref start, ref end);
    } else {
        //           (       ),              ,
        buffer.get_iter_at_mark (out start, buffer.get_insert ());
        end = start;
        string? word = context.get_word ();
        if (word != null && word != "") {
            start.backward_chars (word.char_count ());
            buffer.delete (ref start, ref end);
        }
    }

    //
    var p = (LspCompletionProposal) proposal;
    buffer.insert (ref start, p.get_label (), -1);
}

//            -
public virtual void refilter (CompletionContext context, GLib.ListModel model) {
    current_word = context.get_word ();
```

```
            // 6.
            var filter = filter_model.get_filter () as Gtk.CustomFilter;
            filter.changed (Gtk.FilterChange.DIFFERENT);

            if (model.get_n_items () == 0) {
                context.get_completion ().hide ();
            }
        }

        public virtual string ? get_title () { return "Simple"; }
        public virtual int get_priority (CompletionContext context) { return 100; }
        public virtual bool is_running (CompletionContext context) { return true; }
    }
}
```

File: src/Services/LSP/IdeLspService.vala

```
using GLib;
using Gee;

public class Iide.IdeLspService : GLib.Object {
    private static IdeLspService? _instance;
    private Gee.HashMap<string, LspClient> clients;
    private Gee.HashMap<string, string> uri_to_client_key;
    private Gee.HashMap<string, int> document_versions;
    private Gee.HashMap<string, bool> client_starting;
    private Gee.ArrayList<PendingOpen> pending_opens;
    private Gee.HashMap<string, LanguageConfig> language_configs;

    private LoggerService logger = LoggerService.get_instance ();

    public signal void diagnostics_updated (string uri, ArrayList<IdeLspDiagnostic> diagnost

    public class PendingOpen {
        public string uri;
        public string language_id;
        public string content;
        public string? workspace_root;
        public SourceView view;

        public PendingOpen (string uri, string language_id, string content, string? workspac
            this.uri = uri;
            this.language_id = language_id;
            this.content = content;
            this.workspace_root = workspace_root;
```

```
        this.view = view;
    }
}

public class LanguageConfig {
    public string command;
    public string[] args;
    public string? workspace_root;

    public LanguageConfig (string command, string[] args, string? workspace_root = null)
        this.command = command;
        this.args = args;
        this.workspace_root = workspace_root;
    }
}

private void init_language_configs () {
    language_configs.set ("python", new LanguageConfig ("basedpyright-langserver", { "--
    language_configs.set ("python3", new LanguageConfig ("basedpyright-langserver", { "-
    language_configs.set ("cpp", new LanguageConfig ("clangd", new string[0]));
    language_configs.set ("c++", new LanguageConfig ("clangd", new string[0]));
    language_configs.set ("c", new LanguageConfig ("clangd", new string[0]));
    language_configs.set ("vala", new LanguageConfig ("vala-language-server", new string
    language_configs.set ("rust", new LanguageConfig ("rust-analyzer", new string[0]));
    language_configs.set ("go", new LanguageConfig ("gopls", new string[0]));
}

construct {
    clients = new Gee.HashMap<string, LspClient> ();
    uri_to_client_key = new Gee.HashMap<string, string> ();
    document_versions = new Gee.HashMap<string, int> ();
    client_starting = new Gee.HashMap<string, bool> ();
    pending_opens = new Gee.ArrayList<PendingOpen> ();
    language_configs = new Gee.HashMap<string, LanguageConfig> ();

    init_language_configs ();
}

public static unowned IdeLspService get_instance () {
    if (_instance == null) {
        _instance = new IdeLspService ();
    }
    return _instance;
}

public async void open_document (string uri, string language_id, string content, string?
```

```
var server_key = get_server_key_for_language (language_id);
if (server_key == null) {
    debug ("IdeLspService: No LSP server configured for language: %s", language_id);
    Idle.add (() => {
        view.setup_no_lsp_sync ();
        return Source.REMOVE;
    });
    return;
}

debug ("IdeLspService: Opening document %s (lang=%s)", uri, language_id);

if (clients.has_key (server_key)) {
    var client = clients.get (server_key);
    uri_to_client_key.set (uri, server_key);
    document_versions.set (uri, 1);
    yield client.text_document_did_open (uri, language_id, 1, content);

    Idle.add (() => {
        view.setup_lsp_sync (client);
        return Source.REMOVE;
    });

    return;
}

if (client_starting.get (server_key) == true) {
    pending_opens.add (new PendingOpen (uri, language_id, content, workspace_root, v
    return;
}

client_starting.set (server_key, true);

var config = language_configs.get (server_key);
if (config == null) {
    client_starting.set (server_key, false);
    return;
}

var client = new LspClient ();
client.diagnostics_received.connect ((uri, diagnostics) => {
    diagnostics_updated (uri, diagnostics);
});

bool started = yield client.start_server_async (config.command, config.args, workspa
```

```
    logger.info ("LSP", "Started server for %s: %b (%s -- %s)".printf (server_key, start

    if (started) {
        clients.set (server_key, client);
        uri_to_client_key.set (uri, server_key);
        document_versions.set (uri, 1);
        yield client.text_document_did_open (uri, language_id, 1, content);

        Idle.add (() => {
            view.setup_lsp_sync (client);
            return Source.REMOVE;
        });

        yield process_pending_opens ();
    } else {
        // TODO: restart logic...
        Idle.add (() => {
            view.setup_no_lsp_sync ();
            return Source.REMOVE;
        });

        warning ("IdeLspService: Failed to start LSP server for %s", server_key);
    }

    client_starting.set (server_key, false);
}

private async void process_pending_opens () {
    var opens_to_process = new ArrayList<PendingOpen> ();
    foreach (var open in pending_opens) {
        opens_to_process.add (open);
    }
    pending_opens.clear ();

    foreach (var open in opens_to_process) {
        yield open_document (open.uri, open.language_id, open.content, open.workspace_ro
    }
}

public async void change_document (string uri, string content, int? change_start = null,
    var server_key = uri_to_client_key.get (uri);

    logger.debug ("LSP", "Changed doc: " + uri + " / server_key: " + server_key);
    if (server_key == null || !clients.has_key (server_key)) {
        return;
    }
```

```
    var client = clients.get (server_key);
    var version = document_versions.get (uri);
    document_versions.set (uri, version + 1);

    yield client.text_document_did_change (uri, version + 1, content);
}

public async void send_did_change (string uri, int version, Gee.ArrayList<PendingChange>
    var server_key = uri_to_client_key.get (uri);
    if (server_key == null || !clients.has_key (server_key)) {
        return;
    }

    var client = clients.get (server_key);
    yield client.send_did_change (uri, version, changes);
}

public async void close_document (string uri) {
    var server_key = uri_to_client_key.get (uri);
    if (server_key == null || !clients.has_key (server_key)) {
        return;
    }

    var client = clients.get (server_key);
    yield client.text_document_did_close (uri);

    uri_to_client_key.unset (uri);
    document_versions.unset (uri);
}

private string ? get_server_key_for_language (string language_id) {
    if (language_configs.has_key (language_id)) {
        return language_id;
    }

    if (language_id.down ().contains ("python")) {
        return "python";
    }
    if (language_id.down ().contains ("c++") || language_id.down ().contains ("cpp")) {
        return "cpp";
    }
    if (language_id.down ().contains ("c")) {
        return "c";
    }
```

```
        return null;
    }

    public void set_language_config (string language_id, string command, string[] args, stri
        language_configs.set (language_id, new LanguageConfig (command, args, workspace_root
    }

    public LspClient ? get_client_for_uri (string uri) {
        var server_key = uri_to_client_key.get (uri);
        if (server_key == null) {
            return null;
        }
        return clients.get (server_key);
    }

    public async IdeLspCompletionResult ? request_completion (string uri,
                                                              int line,
                                                              int character,
                                                              string? trigger_character = nu
                                                              CompletionTriggerKind trigger_
        var client = get_client_for_uri (uri);
        if (client == null) {
            return null;
        }
        return yield client.request_completion (uri, line, character, trigger_character, tri
    }

    public async string ? request_hover (string uri, int line, int character) {
        var client = get_client_for_uri (uri);
        if (client == null) {
            logger.debug ("HOVER", "client is null!");
            return null;
        }
        return yield client.request_hover (uri, line, character);
    }

    public async Gee.ArrayList<IdeLspLocation>? goto_definition (string uri, int line, int c
        var client = get_client_for_uri (uri);
        if (client == null)return null;
        return yield client.request_definition (uri, line, character);
    }
}
```

File: src/Services/LSP/IdeLspClient.vala

```
using GLib;
```

```
using Json;


public enum Iide.CompletionTriggerKind {
    /**
     *                 (     , Ctrl+Space)
     *                ,               .
     */
    INVOKED = 1,

    /**
     *                            -
     * (     , '.', ':', '->').
     */
    TRIGGER_CHARACTER = 2,

    /**
     *                       (     ,
     *               'incomplete').
     */
    TRIGGER_FOR_INCOMPLETE_COMPLETIONS = 3
}

public class Iide.PendingChange : GLib.Object {
    public int start_offset;
    public int end_offset;
    public string text;

    //               LSP
    public int start_line;
    public int start_char;
    public int end_line;
    public int end_char;

    public PendingChange (string t, Gtk.TextIter s_iter, Gtk.TextIter? e_iter = null) {
        this.text = t;
        this.start_offset = s_iter.get_offset ();
        this.end_offset = e_iter != null? e_iter.get_offset () : this.start_offset;

        //
        this.calculate_lsp_pos (s_iter, out this.start_line, out this.start_char);
        if (e_iter != null) {
            this.calculate_lsp_pos (e_iter, out this.end_line, out this.end_char);
        } else {
            this.end_line = this.start_line;
            this.end_char = this.start_char;
```

```
        }
    }

    private void calculate_lsp_pos (Gtk.TextIter iter, out int lsp_line, out int lsp_char) {
        lsp_line = iter.get_line ();

        //
        Gtk.TextIter line_start = iter;
        line_start.set_line_offset (0);

        //
        string line_text = line_start.get_text (iter);

        //      UTF-16 code units
        int utf16_count = 0;
        int i = 0;
        unichar c;
        while (line_text.get_next_char (ref i, out c)) {
            if (c <= 0xFFFF) {
                utf16_count += 1;
            } else {
                utf16_count += 2;
            }
        }
        lsp_char = utf16_count;
    }
}

public class Iide.IdeLspDiagnostic : GLib.Object {
    public int severity { get; set; default = 1; }
    public string message { get; set; default = ""; }
    public int start_line { get; set; default = 0; }
    public int start_column { get; set; default = 0; }
    public int end_line { get; set; default = 0; }
    public int end_column { get; set; default = 0; }

    public string to_string () {
        return "Diagnostic: (%d:%d-%d:%d) %s".printf (start_line, start_column, end_line, en
    }
}

public class Iide.IdeLspCompletionItem : GLib.Object {
    public string label { get; set; default = ""; }
    public string? detail { get; set; }
    public string? documentation { get; set; }
    public int sort_text_priority { get; set; default = 0; }
```

```
    public int insert_text_priority { get; set; default = 0; }
    public string insert_text { get; set; default = ""; }
    public string? text_edit { get; set; }
    public int start_line { get; set; default = 0; }
    public int start_column { get; set; default = 0; }
    public int end_line { get; set; default = 0; }
    public int end_column { get; set; default = 0; }
    public IdeLspCompletionKind kind { get; set; default = IdeLspCompletionKind.TEXT; }
}

public enum Iide.IdeLspCompletionKind {
    TEXT = 1,
    METHOD = 2,
    FUNCTION = 3,
    CONSTRUCTOR = 4,
    FIELD = 5,
    VARIABLE = 6,
    CLASS = 7,
    INTERFACE = 8,
    MODULE = 9,
    PROPERTY = 10,
    UNIT = 11,
    VALUE = 12,
    ENUM = 13,
    KEYWORD = 14,
    SNIPPET = 15,
    COLOR = 16,
    FILE = 17,
    REFERENCE = 18,
    FOLDER = 19,
    ENUM_MEMBER = 20,
    CONSTANT = 21,
    STRUCT = 22,
    EVENT = 23,
    OPERATOR = 24,
    TYPE_PARAMETER = 25
}

public class Iide.IdeLspCompletionResult : GLib.Object {
    public Gee.ArrayList<IdeLspCompletionItem> items { get; set; }
    public bool is_incomplete { get; set; default = false; }
}

private class Iide.CallbackWrapper {
    public SourceFunc callback;
    public CallbackWrapper (owned SourceFunc cb) {
```

```
        this.callback = (owned) cb;
    }
}

public class Iide.IdeLspLocation : GLib.Object {
    public string uri { get; set; }
    public int start_line { get; set; }
    public int start_column { get; set; }
    public int end_line { get; set; }
    public int end_column { get; set; }
}

public class Iide.IdeLspClient : GLib.Object {
    public signal void initialized ();
    public signal void diagnostics_received (string uri, Gee.ArrayList<IdeLspDiagnostic> dia
    public signal void completion_received (string uri, IdeLspCompletionResult result);
    public signal void error_occurred (string message);

    private SubprocessLauncher? launcher;
    private Subprocess? process;
    private InputStream? stdout_stream;
    private OutputStream? stdin_stream;

    private uint next_request_id = 1;
    private Gee.HashMap<int, Cancellable?> pending_requests;
    private bool server_initialized = false;
    private string? workspace_root;

    private uint8[] read_buffer;
    private Thread<void>? reader_thread;

    private bool is_writing = false;
    private bool is_writing_queue = false;
    private Gee.ArrayList<string> message_queue;
    private Gee.HashMap<int, string> pending_responses;

    private Gee.HashMap<int, CallbackWrapper> pending_callbacks;
    private Gee.HashMap<int, Json.Object> response_data;


    public IdeLspClient () {
        pending_requests = new Gee.HashMap<int, Cancellable?> ();
        pending_responses = new Gee.HashMap<int, string> ();
        message_queue = new Gee.ArrayList<string> ();
        read_buffer = new uint8[65536];
```

```
    pending_callbacks = new Gee.HashMap<int, CallbackWrapper> ();
    response_data = new Gee.HashMap<int, Json.Object> ();
}

private string build_request_json (string method, Json.Node? params) {
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("jsonrpc");
    builder.add_string_value ("2.0");
    builder.set_member_name ("id");
    builder.add_int_value (next_request_id++);
    builder.set_member_name ("method");
    builder.add_string_value (method);
    builder.set_member_name ("params");
    if (params != null) {
        builder.add_value (params);
    } else {
        builder.begin_object ();
        builder.end_object ();
    }
    builder.end_object ();

    var generator = new Json.Generator ();
    generator.root = builder.get_root ();
    return generator.to_data (null);
}

private string build_notification_json (string method, Json.Node? params = null) {
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("jsonrpc");
    builder.add_string_value ("2.0");
    builder.set_member_name ("method");
    builder.add_string_value (method);
    builder.set_member_name ("params");
    if (params != null) {
        builder.add_value (params);
    } else {
        builder.begin_object ();
        builder.end_object ();
    }
    builder.end_object ();

    var generator = new Json.Generator ();
    generator.root = builder.get_root ();
    return generator.to_data (null);
```

```
}

public async bool start_server (string command, string[] args, string? workspace_root, 
    this.workspace_root = workspace_root;

    try {
        var argv = new Gee.ArrayList<string> ();
        argv.add (command);
        foreach (var arg in args) {
            argv.add (arg);
        }

        launcher = new SubprocessLauncher (SubprocessFlags.STDOUT_PIPE | SubprocessFlags
        process = launcher.spawnv (argv.to_array ());

        stdout_stream = process.get_stdout_pipe ();
        stdin_stream = process.get_stdin_pipe ();

        reader_thread = new Thread<void> ("lsp-reader", () => {
            read_loop ();
        });

        SourceFunc cont = start_server.callback;
        var init_params = build_init_params (workspace_root, initialization_options);

        new Thread<void> ("lsp-init", () => {
            var json = build_request_json ("initialize", init_params);
            send_message_sync (json);

            var initialized_json = build_notification_json ("initialized");
            send_message_sync (initialized_json);

            server_initialized = true;
            Idle.add ((owned) cont);
        });

        yield;

        initialized ();
        return true;
    } catch (Error e) {
        error_occurred ("Failed to start LSP server: %s".printf (e.message));
        return false;
    }
}
```

```
private void send_message_sync (string message) {
    if (stdin_stream == null) {
        return;
    }

    try {
        var full_message = "Content-Length: %d\r\n\r\n%s".printf (message.length, messag
        var data = full_message.data;
        stdin_stream.write (data);
        stdin_stream.flush ();
    } catch (Error e) {
        warning ("LSP sync write error: %s", e.message);
    }
}

private Json.Node build_init_params (string? workspace_root, Json.Node? initialization_o
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("processId");
    builder.add_null_value ();
    builder.set_member_name ("clientInfo");
    builder.begin_object ();
    builder.set_member_name ("name");
    builder.add_string_value ("iide");
    builder.set_member_name ("version");
    builder.add_string_value ("0.1.0");
    builder.end_object ();
    builder.set_member_name ("rootUri");
    if (workspace_root != null) {
        builder.add_string_value (workspace_root);
    } else {
        builder.add_null_value ();
    }
    builder.set_member_name ("workspaceFolders");
    builder.begin_array ();
    if (workspace_root != null) {
        builder.begin_object ();
        builder.set_member_name ("uri");
        builder.add_string_value (workspace_root);
        builder.set_member_name ("name");
        builder.add_string_value ("workspace");
        builder.end_object ();
    }
    builder.end_array ();
    builder.set_member_name ("capabilities");
    builder.begin_object ();
```

```
    builder.set_member_name ("textDocument");
    builder.begin_object ();
    builder.set_member_name ("syncKind");
    builder.add_int_value (1);
    builder.end_object ();
    builder.end_object ();
    builder.set_member_name ("workspace");
    builder.begin_object ();
    builder.set_member_name ("workspaceFolders");
    builder.add_boolean_value (true);
    builder.end_object ();
    builder.end_object ();
    if (initialization_options != null) {
        builder.set_member_name ("initializationOptions");
        builder.add_value (initialization_options.copy ());
    }
    builder.end_object ();

    return builder.get_root ();
}

private async void send_notification (string method, Json.Node? params) {
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("jsonrpc");
    builder.add_string_value ("2.0");
    builder.set_member_name ("method");
    builder.add_string_value (method);
    builder.set_member_name ("params");
    if (params != null) {
        builder.add_value (params);
    } else {
        builder.begin_object ();
        builder.end_object ();
    }
    builder.end_object ();

    var generator = new Json.Generator ();
    generator.root = builder.get_root ();
    string json = generator.to_data (null);

    yield send_message (json);
}

public async IdeLspCompletionResult ? request_completion (string uri,
                                                          int line,
```

```
                                                        int character,
                                                        string? trigger_character = nu
                                                        CompletionTriggerKind trigger_
int id = (int) next_request_id; //            ID

//                       LSP
var builder = new Json.Builder ();
builder.begin_object ();
builder.set_member_name ("textDocument");
builder.begin_object ();
builder.set_member_name ("uri");
builder.add_string_value (uri);
builder.set_member_name ("triggerKind");
builder.add_int_value ((int) trigger_kind);
if (trigger_character != null) {
    builder.set_member_name ("triggerCharacter");
    builder.add_string_value (trigger_character);
}
builder.end_object ();

builder.set_member_name ("position");
builder.begin_object ();
builder.set_member_name ("line");
builder.add_int_value (line);
builder.set_member_name ("character");
builder.add_int_value (character);
builder.end_object ();

builder.end_object ();

var json = build_request_json ("textDocument/completion", builder.get_root ());

//       callback
pending_callbacks.set (id, new CallbackWrapper (request_completion.callback));

//
send_message_sync (json);

//                        callback   handle_incoming_message
yield;

//                   ,
var response = response_data.get (id);
response_data.unset (id);

if (response == null || !response.has_member ("result"))return null;
```

```
    return parse_completion_result (response.get_member ("result"));
}

public async string ? request_hover (string uri, int line, int character) {
    int id = (int) next_request_id;

    //                      LSP
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("textDocument");
    builder.begin_object ();
    builder.set_member_name ("uri");
    builder.add_string_value (uri);
    builder.end_object ();

    builder.set_member_name ("position");
    builder.begin_object ();
    builder.set_member_name ("line");
    builder.add_int_value (line);
    builder.set_member_name ("character");
    builder.add_int_value (character);
    builder.end_object ();

    builder.end_object ();

    var json = build_request_json ("textDocument/hover", builder.get_root ());

    //       callback
    pending_callbacks.set (id, new CallbackWrapper (request_hover.callback));

    //
    send_message_sync (json);

    //                       callback   handle_incoming_message
    yield;

    //                  ,
    var response = response_data.get (id);
    response_data.unset (id);

    if (response == null || !response.has_member ("result"))return null;

    return parse_hover_result (response.get_member ("result"));
}
```

```
private string ? parse_hover_result (Json.Node node) {
    if (node.get_node_type () != Json.NodeType.OBJECT)
        return null;

    var result = node.get_object ();

    //     1:            'contents'
    if (!result.has_member ("contents"))
        return null;

    var contents = result.get_member ("contents");

    //           (MarkupContent)
    if (contents.get_node_type () == Json.NodeType.OBJECT) {
        var obj = contents.get_object ();
        if (obj.has_member ("value")) {
            return obj.get_string_member ("value");
        }
    }
    //
    else {
        return contents.get_string ();
    }

    return null;
}

private IdeLspCompletionResult ? parse_completion_result (Json.Node node) {
    var result = new IdeLspCompletionResult ();
    result.items = new Gee.ArrayList<IdeLspCompletionItem> ();

    Json.Array items_array = null;

    // 1. LSP                ,        CompletionList
    if (node.get_node_type () == Json.NodeType.ARRAY) {
        items_array = node.get_array ();
    } else if (node.get_node_type () == Json.NodeType.OBJECT) {
        var obj = node.get_object ();
        if (obj.has_member ("items")) {
            items_array = obj.get_array_member ("items");
        }
        if (obj.has_member ("isIncomplete")) {
            result.is_incomplete = obj.get_boolean_member ("isIncomplete");
        }
    }
```

```
    if (items_array == null)return result;

    // 2.
    items_array.foreach_element ((array, index, item_node) => {
        var item_obj = item_node.get_object ();
        var item = new IdeLspCompletionItem ();

        item.label = item_obj.get_string_member ("label");

        if (item_obj.has_member ("detail"))
            item.detail = item_obj.get_string_member ("detail");

        if (item_obj.has_member ("insertText"))
            item.insert_text = item_obj.get_string_member ("insertText");
        else
            item.insert_text = item.label;

        if (item_obj.has_member ("kind"))
            item.kind = (IdeLspCompletionKind) item_obj.get_int_member ("kind");

        //              (                   MarkupContent)
        if (item_obj.has_member ("documentation")) {
            var doc_node = item_obj.get_member ("documentation");
            if (doc_node.get_node_type () == Json.NodeType.OBJECT) {
                item.documentation = doc_node.get_object ().get_string_member ("value");
            } else {
                item.documentation = item_obj.get_string_member ("documentation");
            }
        }

        result.items.add (item);
    });

    return result;
}

private async void send_message (string message) {
    if (stdin_stream == null) {
        return;
    }

    if (is_writing) {
        message_queue.add (message);
        return;
    }
```

```
    is_writing = true;

    try {
        var full_message = "Content-Length: %d\r\n\r\n%s".printf (message.length, messag
        var data = full_message.data;
        size_t bytes_written = 0;
        yield stdin_stream.write_all_async (data, Priority.DEFAULT, null, out bytes_writ

        yield stdin_stream.flush_async (Priority.DEFAULT, null);
    } catch (Error e) {
        warning ("LSP Write error: %s", e.message);
        error_occurred ("Write error: %s".printf (e.message));
        is_writing = false;
        return;
    }

    while (message_queue.size > 0) {
        if (is_writing_queue) {
            break;
        }
        is_writing_queue = true;
        var queued = message_queue.get (0);
        message_queue.remove_at (0);
        try {
            var full_message = "Content-Length: %d\r\n\r\n%s".printf (queued.length, que
            var data = full_message.data;
            size_t bytes_written = 0;
            yield stdin_stream.write_all_async (data, Priority.DEFAULT, null, out bytes_

            yield stdin_stream.flush_async (Priority.DEFAULT, null);
        } catch (Error e) {
            warning ("LSP queued write error: %s", e.message);
            is_writing_queue = false;
            break;
        }
        is_writing_queue = false;
    }
    is_writing = false;
}

private bool shutting_down = false;

private void read_loop () {
    var data_stream = new DataInputStream (stdout_stream);
    //    : LSP       CRLF (\r\n)
    data_stream.set_newline_type (DataStreamNewlineType.CR_LF);
```

```
    try {
        while (true) {
            size_t content_length = 0;

            // 1.
            string line;
            while ((line = data_stream.read_line (null)) != null && line != "") {
                if (line.has_prefix ("Content-Length:")) {
                    content_length = (size_t) uint64.parse (line.replace ("Content-Lengt
                }
            }

            if (content_length == 0)continue;

            // 2.         content_length
            uint8[] buffer = new uint8[content_length];
            size_t bytes_read;
            data_stream.read_all (buffer, out bytes_read);

            if (bytes_read > 0) {
                string json_data = (string) buffer;
                json_data = json_data.substring (0, (int) bytes_read);

                //                       (Main Loop)
                Idle.add (() => {
                    handle_incoming_message (json_data);
                    return false;
                });
            }
        }
    } catch (Error e) {
        warning ("LSP Read Error: %s", e.message);
    }
}

private void handle_incoming_message (string json_data) {
    try {
        var parser = new Json.Parser ();
        parser.load_from_data (json_data);
        var root = parser.get_root ().get_object ();

        //      :                (   'id')
        if (root.has_member ("id")) {
            int id = (int) root.get_int_member ("id");
            handle_response (id, root);
```

```
        }
        //      :           (   'method')
        else if (root.has_member ("method")) {
            string method = root.get_string_member ("method");
            handle_notification (method, root);
        }
    } catch (Error e) {
        print ("Failed to parse LSP JSON: %s", e.message);
    }
}

private void handle_response (int id, Json.Object root) {
    if (pending_callbacks.has_key (id)) {
        //            ,     request_completion
        response_data.set (id, root);

        var wrapper = pending_callbacks.get (id);
        if (wrapper != null) {
            pending_callbacks.unset (id);
            Idle.add ((owned) wrapper.callback);
        }
    }
}

private void handle_notification (string method, Json.Object root) {
    switch (method) {
    case "textDocument/publishDiagnostics" :
        handle_publish_diagnostics (root);
        break;
    case "$/progress" :
        break;
    case "window/showMessage" :
        break;
    }
}

private void handle_publish_diagnostics (Json.Object root) {
    var params = root.get_object_member ("params");
    var uri = params.get_string_member ("uri");
    var diag_list = params.get_array_member ("diagnostics");

    var diagnostics = new Gee.ArrayList<IdeLspDiagnostic> ();
    diag_list.foreach_element ((array, index, node) => {
        var obj = node.get_object ();
        var range = obj.get_object_member ("range");
        var start = range.get_object_member ("start");
```

```
        var end = range.get_object_member ("end");

        var d = new IdeLspDiagnostic ();
        d.message = obj.get_string_member ("message");
        d.severity = (int) obj.get_int_member ("severity");
        d.start_line = (int) start.get_int_member ("line");
        d.start_column = (int) start.get_int_member ("character");
        d.end_line = (int) end.get_int_member ("line");
        d.end_column = (int) end.get_int_member ("character");
        diagnostics.add (d);
    });

    diagnostics_received (uri, diagnostics);
}

public async void text_document_did_open (string uri, string language_id, int version, s
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("textDocument");
    builder.begin_object ();
    builder.set_member_name ("uri");
    builder.add_string_value (uri);
    builder.set_member_name ("languageId");
    builder.add_string_value (language_id);
    builder.set_member_name ("text");
    builder.add_string_value (content);
    builder.set_member_name ("version");
    builder.add_int_value (version);
    builder.end_object ();
    builder.end_object ();

    yield send_notification ("textDocument/didOpen", builder.get_root ());
}

public async void text_document_did_change (string uri, int version, string content, int
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("textDocument");
    builder.begin_object ();
    builder.set_member_name ("uri");
    builder.add_string_value (uri);
    builder.set_member_name ("version");
    builder.add_int_value (version);
    builder.end_object ();

    builder.set_member_name ("contentChanges");
```

```
    builder.begin_array ();
    builder.begin_object ();
    builder.set_member_name ("text");
    builder.add_string_value (content);
    builder.end_object ();
    builder.end_array ();
    builder.end_object ();

    message ("!LSP SEND UNREACHABLE");

    yield send_notification ("textDocument/didChange", builder.get_root ());
}

public async void send_did_change (string uri, int version, Gee.ArrayList<PendingChange>
    var json_changes = new Json.Array ();

    foreach (var c in changes) {
        var obj = new Json.Object ();
        var range = new Json.Object ();

        var start = new Json.Object ();
        start.set_int_member ("line", c.start_line);
        start.set_int_member ("character", c.start_char);

        var end = new Json.Object ();
        end.set_int_member ("line", c.end_line);
        end.set_int_member ("character", c.end_char);

        range.set_object_member ("start", start);
        range.set_object_member ("end", end);

        obj.set_object_member ("range", range);
        obj.set_string_member ("text", c.text);
        json_changes.add_object_element (obj);
    }

    //                didChange
    var params = new Json.Object ();
    var doc = new Json.Object ();
    doc.set_string_member ("uri", uri);
    doc.set_int_member ("version", version);

    params.set_object_member ("textDocument", doc);
    params.set_array_member ("contentChanges", json_changes);

    var node = new Json.Node (Json.NodeType.OBJECT);
```

```
    node.set_object (params);

    yield send_notification ("textDocument/didChange", node);
}

public async void text_document_did_close (string uri) {
    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("textDocument");
    builder.begin_object ();
    builder.set_member_name ("uri");
    builder.add_string_value (uri);
    builder.end_object ();
    builder.end_object ();

    yield send_notification ("textDocument/didClose", builder.get_root ());
}

public void shutdown () {
    shutting_down = true;
    if (process != null) {
        process.force_exit ();
    }
}

public async Gee.ArrayList<IdeLspLocation>? request_definition (string uri, int line, in
    int id = (int) next_request_id;

    var builder = new Json.Builder ();
    builder.begin_object ();
    builder.set_member_name ("textDocument");
    builder.begin_object ();
    builder.set_member_name ("uri");
    builder.add_string_value (uri);
    builder.end_object ();

    builder.set_member_name ("position");
    builder.begin_object ();
    builder.set_member_name ("line");
    builder.add_int_value (line);
    builder.set_member_name ("character");
    builder.add_int_value (character);
    builder.end_object ();
    builder.end_object ();

    var json = build_request_json ("textDocument/definition", builder.get_root ());
```

```
    //          CallbackWrapper
    pending_callbacks.set (id, new CallbackWrapper (request_definition.callback));
    send_message_sync (json);

    yield; //          handle_response

    var response = response_data.get (id);
    response_data.unset (id);

    if (response == null || !response.has_member ("result") || response.get_member ("res
        return null;
    }

    return parse_definition_result (response.get_member ("result"));
}

private Gee.ArrayList<IdeLspLocation> parse_definition_result (Json.Node node) {
    var locations = new Gee.ArrayList<IdeLspLocation> ();

    if (node.get_node_type () == Json.NodeType.ARRAY) {
        node.get_array ().foreach_element ((array, index, item_node) => {
            locations.add (parse_single_location (item_node.get_object ()));
        });
    } else if (node.get_node_type () == Json.NodeType.OBJECT) {
        locations.add (parse_single_location (node.get_object ()));
    }

    return locations;
}

private IdeLspLocation parse_single_location (Json.Object obj) {
    var loc = new IdeLspLocation ();
    loc.uri = obj.get_string_member ("uri");

    var range = obj.get_object_member ("range");
    var start = range.get_object_member ("start");
    var end = range.get_object_member ("end");

    loc.start_line = (int) start.get_int_member ("line");
    loc.start_column = (int) start.get_int_member ("character");
    loc.end_line = (int) end.get_int_member ("line");
    loc.end_column = (int) end.get_int_member ("character");

    return loc;
}
```

```
}
```

File: src/Services/TreeSitter/BaseTreeSitterHighlighter.vala

```
using Gtk;
using GtkSource;

[CCode (cname = "ts_tree_cursor_new_as_ptr")]
extern TreeSitter.TreeCursor ts_tree_cursor_new_as_ptr (TreeSitter.Node node);

private struct Iide.TsEdit {
    public uint32 start_byte;
    public uint32 old_end_byte;
    public uint32 new_end_byte;
    public TreeSitter.Point start_point;
    public TreeSitter.Point old_end_point;
    public TreeSitter.Point new_end_point;

    public TreeSitter.InputEdit to_ts_edit () {
        return TreeSitter.InputEdit () {
                    start_byte = this.start_byte,
                    old_end_byte = this.old_end_byte,
                    new_end_byte = this.new_end_byte,
                    start_point = this.start_point,
                    old_end_point = this.old_end_point,
                    new_end_point = this.new_end_point
        };
    }
}

public abstract class Iide.BaseTreeSitterHighlighter : Object {
    protected View view;
    protected Buffer buffer;
    private TreeSitter.Parser parser;
    private TreeSitter.Tree? tree = null;
    private uint debounce_source = 0;
    private const int DEBOUNCE_DELAY_MS = 50;

    protected abstract unowned TreeSitter.Language language ();

    private Gee.ArrayList<TsEdit?> pending_edits = new Gee.ArrayList<TsEdit?> ();

    protected BaseTreeSitterHighlighter (View view) {
        this.view = view;
        this.buffer = (Buffer) view.get_buffer ();
```

```
    parser = new TreeSitter.Parser ();
    parser.set_language (language ());

    buffer.delete_range.connect (on_delete_range);
    buffer.insert_text.connect (on_insert_text);

    buffer.notify["style-scheme"].connect_after (on_style_scheme_changed);

    Idle.add (() => {
        do_reparse ();
        return Source.REMOVE;
    });
}

private void on_style_scheme_changed () {
    apply_highlighting_full ();
}

private void calculate_text_stats (string text, out uint32 lines, out uint32 last_colum
    lines = 0;
    last_column = 0;
    int i = 0;
    unichar c;

    while (text.get_next_char (ref i, out c)) {
        if (c == '\n') {
            lines++;
            last_column = 0;
        } else {
            // Tree-sitter
            //
            last_column += (uint32) c.to_utf8 (null);
        }
    }
}

private void on_insert_text (TextIter iter, string text, int len_bytes) {
    var edit = TsEdit ();

    //             (            )
    TextIter start_buf;
    buffer.get_start_iter (out start_buf);
    edit.start_byte = (uint32) buffer.get_slice (start_buf, iter, false).length;

    edit.start_point = TreeSitter.Point () {
        row = (uint32) iter.get_line (),
```

```
        column = (uint32) iter.get_line_index ()
    };

    //       Tree-sitter -
    edit.old_end_byte = edit.start_byte;
    edit.old_end_point = edit.start_point;

    //      ,
    uint32 lines_added;
    uint32 last_line_bytes;
    calculate_text_stats (text, out lines_added, out last_line_bytes);

    edit.new_end_byte = edit.start_byte + (uint32) text.length;

    edit.new_end_point = TreeSitter.Point () {
        row = edit.start_point.row + lines_added,
        column = lines_added > 0 ? last_line_bytes : edit.start_point.column + last_line
    };

    pending_edits.add (edit);
    schedule_reparse ();
}

private void on_delete_range (TextIter start, TextIter end) {
    var edit = TsEdit ();

    TextIter start_buf;
    buffer.get_start_iter (out start_buf);
    edit.start_byte = (uint32) buffer.get_slice (start_buf, start, false).length;
    edit.old_end_byte = (uint32) buffer.get_slice (start_buf, end, false).length;
    edit.new_end_byte = edit.start_byte;

    edit.start_point = TreeSitter.Point () {
        row = (uint32) start.get_line (),
        column = (uint32) start.get_line_index ()
    };
    edit.old_end_point = TreeSitter.Point () {
        row = (uint32) end.get_line (),
        column = (uint32) end.get_line_index ()
    };
    edit.new_end_point = edit.start_point;

    pending_edits.add (edit);
    schedule_reparse ();
}
```

```
private void schedule_reparse () {
    if (debounce_source != 0) {
        GLib.Source.remove (debounce_source);
    }

    debounce_source = GLib.Timeout.add (DEBOUNCE_DELAY_MS, () => {
        do_reparse ();
        debounce_source = 0;
        return GLib.Source.REMOVE;
    });
}

private void do_reparse () {
    string full_text = this.buffer.text;

    // 1.             .       unowned,               /
    unowned TreeSitter.Tree? current_tree = this.tree;
    bool need_full_reparse = true;

    if (current_tree != null) {
        // 2.               .
        //    : edit()             "     "
        foreach (var e in pending_edits) {
            current_tree.edit (e.to_ts_edit ());
        }
        pending_edits.clear ();
        need_full_reparse = false;
    }

    // 3.             .
    //                         .
    //     :                           ,                ,
    //             (this.tree)                       .
    TreeSitter.Tree new_tree = parser.parse_string (current_tree, full_text.data);

    // 4.                    : current_tree (   )  new_tree (   ).
    if (!need_full_reparse) {
        var ranges = current_tree.get_changed_ranges (new_tree);
        this.tree = (owned) new_tree;
        apply_highlighting_incremental (ranges);
    } else {
        this.tree = (owned) new_tree;
        apply_highlighting_full ();
    }
}
```

```
private void apply_highlighting_incremental (TreeSitter.Range[] ranges) {
    if (ranges.length == 0)return;

    buffer.begin_user_action ();

    for (uint i = 0; i < ranges.length; i++) {
        var range = ranges[i];

        // 3.                  Tree-sitter        GTK
        TextIter start_iter, end_iter;
        //    :                            ,
        buffer.get_iter_at_line (out start_iter, (int) range.start_point.row);
        buffer.get_iter_at_line (out end_iter, (int) range.end_point.row);
        end_iter.forward_to_line_end ();

        // 4.
        buffer.remove_all_tags (start_iter, end_iter);

        // 5.                      (       Query    Cursor)
        //                  traverse         ,
        rehighlight_range (range);
    }

    buffer.end_user_action ();
}

private void rehighlight_range (TreeSitter.Range range) {
    var cursor = ts_tree_cursor_new_as_ptr (tree.root_node ());


    //
    highlight_nodes_in_range (cursor, range);
}

private void highlight_nodes_in_range (TreeSitter.TreeCursor cursor, TreeSitter.Range ra
    var node = cursor.current_node ();

    //                                -
    if (node.end_byte () < range.start_byte || node.start_byte () > range.end_byte) {
        return;
    }

    //             (              )
    highlight_node (node);

    //
```

```
    if (cursor.goto_first_child ()) {
        do {
            highlight_nodes_in_range (cursor, range);
        } while (cursor.goto_next_sibling ());
        cursor.goto_parent ();
    }
}

protected virtual void apply_highlighting_full () {
    if (tree == null)return;

    TextIter start, end;
    buffer.get_bounds (out start, out end);
    buffer.remove_all_tags (start, end);

    traverse_node (tree.root_node (), 0, null);

    view.queue_draw ();
}

private void traverse_node (TreeSitter.Node node, int depth, TreeSitter.Node? parent_noc
    highlight_node (node);

    for (uint i = 0; i < node.child_count (); i++) {
        traverse_node (node.child (i), depth + 1, node);
    }
}

protected abstract void highlight_node (TreeSitter.Node node);

protected void highlight_range (TreeSitter.Node node, string style_name) {
    TextIter s_iter, e_iter;
    get_iters_from_ts_node (buffer, node, out s_iter, out e_iter);

    apply_tag_to_range (s_iter, e_iter, style_name);
}

protected void apply_tag_to_range (TextIter start, TextIter end, string style_name) {
    var scheme = buffer.style_scheme;
    var style = scheme.get_style (style_name);

    if (style != null) {
        var tag = buffer.tag_table.lookup (style_name);
        if (tag == null) {
            tag = new Gtk.TextTag (style_name);
            buffer.tag_table.add (tag);
```

```
            }

            apply_style_to_tag (style, tag);
            buffer.apply_tag (tag, start, end);
        }
    }

    public static void get_iters_from_ts_node (TextBuffer buffer, TreeSitter.Node node,
                                               out TextIter start_iter, out TextIter end_ite
        buffer.get_iter_at_line (out start_iter, (int) node.start_point ().row);
        start_iter.set_line_index ((int) node.start_point ().column);

        buffer.get_iter_at_line (out end_iter, (int) node.end_point ().row);
        end_iter.set_line_index ((int) node.end_point ().column);
    }

    private void apply_style_to_tag (GtkSource.Style style, Gtk.TextTag tag) {
        if (style.foreground_set) {
            tag.foreground = style.foreground;
        }

        if (style.background_set) {
            tag.background = style.background;
        }

        if (style.bold_set) {
            tag.weight = style.bold ? Pango.Weight.BOLD : Pango.Weight.NORMAL;
        }

        if (style.italic_set) {
            tag.style = style.italic ? Pango.Style.ITALIC : Pango.Style.NORMAL;
        }
    }
}
```

File: src/Services/DocumentManager.vala

```
/*
 * documentmanager.vala
 *
 * Copyright 2026 kai
 *
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 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation, either version 3 of the License, or
 * (at your option) any later version.
```

```
 *
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 * You should have received a copy of the GNU General Public License
 * along with this program.  If not, see <https://www.gnu.org/licenses/>.
 *
 * SPDX-License-Identifier: GPL-3.0-or-later
 */

using Gee;
using GLib;
using Gtk;
using Panel;

public class Iide.DocumentManager : GLib.Object {
    public Window window;
    public Gee.HashMap<string, TextView> documents;
    private Iide.IdeLspManager lsp_manager;
    private string? current_workspace_root;

    private LoggerService logger = LoggerService.get_instance ();

    public DocumentManager (Window window) {
        this.window = window;
        documents = new Gee.HashMap<string, TextView> ();
        lsp_manager = Iide.IdeLspManager.get_instance ();

        lsp_manager.connect_diagnostics ((uri, diagnostics) => {
            var doc = documents.get (uri);
            if (doc != null) {
                var lsp_diagnostics = new Gee.ArrayList<IdeLspDiagnostic> ();
                foreach (var diag in diagnostics) {
                    var d = new IdeLspDiagnostic ();
                    d.severity = diag.severity;
                    d.message = diag.message;
                    d.start_line = diag.start_line;
                    d.start_column = diag.start_column;
                    d.end_line = diag.end_line;
                    d.end_column = diag.end_column;
                    lsp_diagnostics.add (d);
                }

                Idle.add (() => {
```

```
                doc.update_diagnostics (lsp_diagnostics);
                return false;
            });
        }
    });
}

public void set_workspace_root (string? root) {
    current_workspace_root = root;
}

public signal void document_opened (TextView document);
public signal void document_closed (string uri);

public Panel.Widget? open_document (GLib.File file, Panel.Position ? pos) {
    return open_document_with_selection (file, -1, -1, -1, pos);
}

public Panel.Widget? open_document_with_selection (GLib.File file, int line, int start_c
    string uri = file.get_uri ();

    logger.debug ("Doc", "Open document: " + uri + " / HAS_KEY: " + (documents.has_key (

    if (documents.has_key (uri)) {
        var widget = documents.get (uri);
        widget.raise ();
        widget.view_grab_focus ();
        if (line >= 0) {
            widget.select_and_scroll (line, start_col, end_col, false);
        }
        return widget;
    } else {
        var buffer = new GtkSource.Buffer (null);
        var source_file = new GtkSource.File ();
        source_file.location = file;
        var file_loader = new GtkSource.FileLoader (buffer, source_file);
        Iide.TextView? panel_widget = null;
        file_loader.load_async.begin (Priority.DEFAULT, null, null, (obj, res) => {
            try {
                file_loader.load_async.end (res);
                panel_widget = new Iide.TextView (file, buffer, window);

                panel_widget.notify["parent"].connect (() => {
                    if (panel_widget.parent == null) {
                        close_document (file);
                    }
```

```
                });

                panel_widget.text_changed.connect ((text) => {
                    lsp_manager.change_document.begin (uri, text);
                });

                panel_widget.buffer_saved.connect (() => {
                    string content = ((GtkSource.Buffer) panel_widget.text_view.buffer).
                    lsp_manager.change_document.begin (uri, content);
                });

                documents.set (uri, panel_widget);
                if (pos == null) {
                    window.grid.add (panel_widget);
                } else {
                    window.add_widget (panel_widget, pos);
                }
                panel_widget.raise ();
                panel_widget.view_grab_focus ();
                logger.debug ("Doc", "Document panel widget created: " + uri);

                if (line >= 0) {
                    // TODO: with timeout...
                    panel_widget.select_and_scroll (line, start_col, end_col, true);
                }

                string content = buffer.text;
                string? lang_id = lsp_manager.get_language_id_for_file (file);
                if (lang_id != null) {
                    lsp_manager.open_document.begin (uri, lang_id, content, current_work
                }
            } catch (Error e) {
                logger.error ("Doc", "Error Opening File", "Failed to read file %s: %s".
            }
        });
        return panel_widget;
    }
}



public bool close_document (GLib.File file) {
    string uri = file.get_uri ();
    if (documents.has_key (uri)) {
        documents.unset (uri);
        lsp_manager.close_document.begin (uri);
```

```
            document_closed (uri);
            return true;
        }
        return false;
    }

    public Panel.Widget? get_document_for_file (GLib.File file) {
        string uri = file.get_uri ();
        return documents.get (uri);
    }

    public bool is_file_open (GLib.File file) {
        return documents.has_key (file.get_uri ());
    }

    public Gee.ArrayList<string> get_open_document_uris () {
        var uris = new Gee.ArrayList<string> ();
        foreach (var uri in documents.keys) {
            uris.add (uri);
        }
        return uris;
    }

    public void open_document_by_uri (string uri, Iide.Window window) {
        var file = GLib.File.new_for_uri (uri);
        if (file.query_exists (null)) {
            open_document (file, null);
        }
    }
}
```

File: src/application.vala

```
/* application.vala
 *
 * Copyright 2026 kai
 *
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 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation, either version 3 of the License, or
 * (at your option) any later version.
 *
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 * GNU General Public License for more details.
```

```
 *
 * You should have received a copy of the GNU General Public License
 * along with this program.  If not, see <https://www.gnu.org/licenses/>.
 *
 * SPDX-License-Identifier: GPL-3.0-or-later
 */

public class Iide.Application : Adw.Application {
    private Iide.SettingsService settings;
    private Iide.ActionManager action_manager;

    public signal void zoom_changed (int zoom_level);
    public signal void minimap_changed (bool visible);

    public Application () {
        Object (
                application_id: "org.github.kai66673.iide",
                flags: ApplicationFlags.DEFAULT_FLAGS,
                resource_base_path: "/org/github/kai66673/iide"
        );
    }

    construct {
        settings = Iide.SettingsService.get_instance ();
        action_manager = Iide.ActionManager.get_instance ();

        register_builtin_actions ();
        apply_shortcuts ();
    }

    private void register_builtin_actions () {
        action_manager.register_action (new SaveAllAction (this));
        action_manager.register_action (new OpenProjectAction (this));
        action_manager.register_action (new PreferencesAction (this));
        action_manager.register_action (new ToggleMinimapAction (this));
        action_manager.register_action (new FuzzyFinderAction (this));
        action_manager.register_action (new SearchInFilesAction (this));
        action_manager.register_action (new ZoomInAction ());
        action_manager.register_action (new ZoomOutAction ());
        action_manager.register_action (new ZoomResetAction ());
        action_manager.register_action (new QuitAction ());
    }

    private void apply_shortcuts () {
        action_manager.apply_shortcuts_to_application (this);
```

```
        action_manager.get_all_actions ().foreach ((action) => {
            action.shortcut_changed.connect ((new_shortcut) => {
                if (new_shortcut != null && new_shortcut != "") {
                    this.set_accels_for_action ("app." + action.id, { new_shortcut });
                } else {
                    this.set_accels_for_action ("app." + action.id, {});
                }
            });
            return true;
        });
    }

    public Iide.ActionManager get_action_manager () {
        return action_manager;
    }

    public Iide.SettingsService get_settings () {
        return settings;
    }

    public override void activate () {
        base.activate ();

        var icon_theme = Gtk.IconTheme.get_for_display (Gdk.Display.get_default ());
        icon_theme.add_resource_path ("/org/github/kai66673/iide/icons");

        var css_provider = new Gtk.CssProvider ();
        css_provider.load_from_resource ("/org/github/kai66673/iide/style.css");
        Gtk.StyleContext.add_provider_for_display (
            Gdk.Display.get_default (),
            css_provider,
            Gtk.STYLE_PROVIDER_PRIORITY_APPLICATION
        );

        var win = this.active_window ?? new Iide.Window (this);
        win.present ();
    }
}

private class SaveAllAction : Iide.Action {
    private weak Iide.Application app;

    public SaveAllAction (Iide.Application app) {
        this.app = app;
    }
```

```
    public override string id { get { return "save"; } }
    public override string name { get { return _("Save All"); } }
    public override string? description { get { return _("Save all open documents"); } }
    public override string? icon_name { get { return "document-save-symbolic"; } }
    public override string? category { get { return "File"; } }

    public override bool can_execute () {
        return app?.active_window is Iide.Window;
    }

    public override void execute () {
        var win = app?.active_window as Iide.Window;
        win?.save_modified ();
    }
}

private class OpenProjectAction : Iide.Action {
    private weak Iide.Application app;

    public OpenProjectAction (Iide.Application app) {
        this.app = app;
    }

    public override string id { get { return "open_project"; } }
    public override string name { get { return _("Open Project"); } }
    public override string? description { get { return _("Open a project folder"); } }
    public override string? icon_name { get { return "folder-open-symbolic"; } }
    public override string? category { get { return "File"; } }

    public override bool can_execute () {
        return true;
    }

    public override void execute () {
        var win = app?.active_window as Iide.Window;
        win?.open_project_dialog ();
    }
}

private class PreferencesAction : Iide.Action {
    private weak Iide.Application app;

    public PreferencesAction (Iide.Application app) {
        this.app = app;
    }
```

```
    public override string id { get { return "preferences"; } }
    public override string name { get { return _("Preferences"); } }
    public override string? description { get { return _("Open preferences dialog"); } }
    public override string? icon_name { get { return "preferences-system-symbolic"; } }
    public override string? category { get { return "Application"; } }

    public override bool can_execute () {
        return true;
    }

    public override void execute () {
        var dialog = new Iide.PreferencesDialog ();
        dialog.set_transient_for (app?.active_window);
        dialog.present ();
    }
}

private class ToggleMinimapAction : Iide.Action {
    private weak Iide.Application app;

    public ToggleMinimapAction (Iide.Application app) {
        this.app = app;
        this.state = Iide.SettingsService.get_instance ().show_minimap;
    }

    public override string id { get { return "toggle_minimap"; } }
    public override string name { get { return _("Toggle Minimap"); } }
    public override string? description { get { return _("Show or hide the minimap"); } }
    public override string? icon_name { get { return "view-fullscreen-symbolic"; } }
    public override string? category { get { return "View"; } }
    public override bool is_toggle { get { return true; } }

    public override bool can_execute () {
        return true;
    }

    public override void execute () {
        var settings = Iide.SettingsService.get_instance ();
        state = !state;
        settings.show_minimap = state;
        app?.minimap_changed (state);
        state_changed (state);
        Iide.ActionManager.get_instance ().set_toggle_state (id, state);
    }
}
```

```
private class ZoomInAction : Iide.Action {
    private Iide.SettingsService settings;

    public override string id { get { return "zoom_in"; } }
    public override string name { get { return _("Zoom In"); } }
    public override string? description { get { return _("Increase editor font size"); } }
    public override string? icon_name { get { return "zoom-in-symbolic"; } }
    public override string? category { get { return "View"; } }

    public override bool can_execute () {
        settings = Iide.SettingsService.get_instance ();
        return settings.editor_font_size < FontSizeHelper.MAX_ZOOM_LEVEL;
    }

    public override void execute () {
        settings = Iide.SettingsService.get_instance ();
        var app = GLib.Application.get_default () as Iide.Application;
        settings.editor_font_size++;
        app?.zoom_changed (settings.editor_font_size);
    }
}

private class ZoomOutAction : Iide.Action {
    private Iide.SettingsService settings;

    public override string id { get { return "zoom_out"; } }
    public override string name { get { return _("Zoom Out"); } }
    public override string? description { get { return _("Decrease editor font size"); } }
    public override string? icon_name { get { return "zoom-out-symbolic"; } }
    public override string? category { get { return "View"; } }

    public override bool can_execute () {
        settings = Iide.SettingsService.get_instance ();
        return settings.editor_font_size > FontSizeHelper.MIN_ZOOM_LEVEL;
    }

    public override void execute () {
        settings = Iide.SettingsService.get_instance ();
        var app = GLib.Application.get_default () as Iide.Application;
        settings.editor_font_size--;
        app?.zoom_changed (settings.editor_font_size);
    }
}

private class ZoomResetAction : Iide.Action {
    private Iide.SettingsService settings;
```

```
    public override string id { get { return "zoom_reset"; } }
    public override string name { get { return _("Zoom Reset"); } }
    public override string? description { get { return _("Reset editor font size to default'
    public override string? icon_name { get { return "zoom-original-symbolic"; } }
    public override string? category { get { return "View"; } }

    public override bool can_execute () {
        return true;
    }

    public override void execute () {
        settings = Iide.SettingsService.get_instance ();
        var app = GLib.Application.get_default () as Iide.Application;
        settings.editor_font_size = FontSizeHelper.DEFAULT_ZOOM_LEVEL;
        app?.zoom_changed (settings.editor_font_size);
    }
}

private class QuitAction : Iide.Action {
    public override string id { get { return "quit"; } }
    public override string name { get { return _("Quit"); } }
    public override string? description { get { return _("Quit the application"); } }
    public override string? icon_name { get { return "application-exit-symbolic"; } }
    public override string? category { get { return "Application"; } }

    public override bool can_execute () {
        return true;
    }

    public override void execute () {
        var app = GLib.Application.get_default () as Iide.Application;
        app?.quit ();
    }
}

private class FuzzyFinderAction : Iide.Action {
    private weak Iide.Application app;

    public FuzzyFinderAction (Iide.Application app) {
        this.app = app;
    }

    public override string id { get { return "fuzzy_finder"; } }
    public override string name { get { return _("Quick Open"); } }
    public override string? description { get { return _("Open a file quickly by name"); } }
```

```
    public override string? icon_name { get { return "system-search-symbolic"; } }
    public override string? category { get { return "File"; } }

    public override bool can_execute () {
        return app?.active_window is Iide.Window;
    }

    public override void execute () {
        var win = app?.active_window as Iide.Window;
        if (win != null) {
            var dialog = new Iide.FuzzyFinderDialog (win, win.get_document_manager ());
            dialog.set_transient_for (win);
            dialog.present ();
        }
    }
}

private class SearchInFilesAction : Iide.Action {
    private weak Iide.Application app;

    public SearchInFilesAction (Iide.Application app) {
        this.app = app;
    }

    public override string id { get { return "search_in_files"; } }
    public override string name { get { return _("Search in Files"); } }
    public override string? description { get { return _("Search for text in all project fil
    public override string? icon_name { get { return "edit-find-symbolic"; } }
    public override string? category { get { return "Edit"; } }

    public override bool can_execute () {
        return app?.active_window is Iide.Window;
    }

    public override void execute () {
        var win = app?.active_window as Iide.Window;
        if (win != null) {
            var dialog = new Iide.SearchInFilesDialog (win, win.get_document_manager ());
            dialog.set_transient_for (win);
            dialog.present ();
        }
    }
}
```

File: src/Widgets/TextView/TextView.vala

```
/*
 * textdocument.vala
 *
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 *
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 * GNU General Public License for more details.
 *
 * You should have received a copy of the GNU General Public License
 * along with this program.  If not, see <https://www.gnu.org/licenses/>.
 *
 * SPDX-License-Identifier: GPL-3.0-or-later
 */



public class Iide.SaveDelegate : Panel.SaveDelegate {
    private Iide.TextView view;
    public SaveDelegate (Iide.TextView view) {
        Object ();
        this.view = view;

        var file = GLib.File.new_for_uri (view.uri);
        title = view.title;
        subtitle = file.get_path ();
    }

    public override bool save (GLib.Task task) {
        message ("Saving in delegate...");
        var result = view.save ();
        task.return_boolean (result);
        return result;
    }

    public override void close () {
        view.force_close ();
    }
```

```
    public override void discard () {
        view.force_close ();
    }
}

public class Iide.TextView : Panel.Widget {
    private SourceView source_view;
    private GtkSource.Map source_map;
    private FontZoomer font_zoomer;
    private Iide.SettingsService settings;

    public Window window;
    public string uri { get; private set; }
    public SourceView text_view { get { return source_view; } }

    public bool is_modified { get { return ((GtkSource.Buffer) source_view.buffer).get_modif

    public signal void text_changed (string text);
    public signal void buffer_saved ();

    public TextView (GLib.File file, GtkSource.Buffer buffer, Window window) {
        Object ();
        this.uri = file.get_uri ();
        this.window = window;
        this.settings = Iide.SettingsService.get_instance ();

        var adw_style_manager = Adw.StyleManager.get_default ();

        var style_manager = GtkSource.StyleSchemeManager.get_default ();
        if (adw_style_manager.color_scheme == Adw.ColorScheme.FORCE_LIGHT) {
            buffer.set_style_scheme (style_manager.get_scheme ("Adwaita"));
        } else {
            buffer.set_style_scheme (style_manager.get_scheme ("Adwaita-dark"));
        }

        adw_style_manager.notify["color-scheme"].connect (() => {
            if (adw_style_manager.color_scheme == Adw.ColorScheme.FORCE_LIGHT) {
                buffer.set_style_scheme (style_manager.get_scheme ("Adwaita"));
            } else {
                buffer.set_style_scheme (style_manager.get_scheme ("Adwaita-dark"));
            }
        });

        source_view = new SourceView (window, uri, buffer);
        icon_name = source_view.icon_name;
```

```
font_zoomer = new FontZoomer (source_view);

// Connect to application-level zoom and minimap changes
var app = GLib.Application.get_default () as Iide.Application;
if (app != null) {
    app.zoom_changed.connect ((level) => {
        font_zoomer.set_zoom_level (level);
    });
    app.minimap_changed.connect ((visible) => {
        toggle_minimap_visible (visible);
    });
}

// Connect to settings changes to apply to all open documents
settings.editor_setting_changed.connect ((key) => {
    switch (key) {
        case "editor-font-size":
            font_zoomer.set_zoom_level (settings.editor_font_size);
            break;
        case "show-minimap":
            toggle_minimap_visible (settings.show_minimap);
            break;
    }
});

source_map = new GtkSource.Map ();
source_map.set_view (source_view);
source_map.add_css_class ("textview-map");
source_map.visible = settings.show_minimap;

font_zoomer.zoom_changed.connect ((level) => {
    source_view.mark_renderer.set_icons_size (FontSizeHelper.get_size_for_zoom_level
});

var scroll = new Gtk.ScrolledWindow ();
scroll.hexpand = true;
scroll.vexpand = true;

scroll.set_child (source_view);

scroll.get_vadjustment ().bind_property ("value",
                                         source_map.get_vadjustment (), "value",
                                         BindingFlags.BIDIRECTIONAL | BindingFlags.S

var subbox = new Gtk.Box (Gtk.Orientation.HORIZONTAL, 0);
subbox.homogeneous = false;
```

```
        subbox.append (scroll);
        subbox.append (source_map);

        var font_map = Pango.CairoFontMap.get_default ();
        source_map.set_font_map (font_map);

        var box = new Gtk.Box (Gtk.Orientation.VERTICAL, 0);
        box.append (subbox);
        child = box;

        title = file.get_basename ();

        save_delegate = new Iide.SaveDelegate (this);
        modified = false;

        buffer.modified_changed.connect_after (() => {
            modified = buffer.get_modified ();
        });

        // buffer.changed.connect (() => {
        // text_changed (buffer.text);
        // });
    }

    public override void size_allocate (int width, int height, int baseline) {
        base.size_allocate (width, height, baseline);
        source_map.visible = settings.show_minimap && width >= 600;
    }

    public void toggle_minimap_visible (bool visible) {
        source_map.visible = visible;
    }

    public void view_grab_focus () {
        source_view.grab_focus ();
    }

    public bool save () {
        try {
            var text = source_view.buffer.text;
            var file = GLib.File.new_for_uri (uri);
            file.replace_contents (text.data, null, false, GLib.FileCreateFlags.NONE, null);
            ((GtkSource.Buffer) source_view.buffer).set_modified (false);
            buffer_saved ();
        } catch (Error e) {
            critical (e.message);
```

```
        }
        return true;
    }

    public void update_diagnostics (Gee.ArrayList<IdeLspDiagnostic> diagnostics) {
        var text_buffer = (Gtk.TextBuffer) source_view.buffer;

        text_buffer.begin_user_action ();

        LspDiagnosticsMark.clear_mark_attributes (source_view);

        int line_count = text_buffer.get_line_count ();

        foreach (var diag in diagnostics) {
            if (diag.start_line >= line_count) {
                continue;
            }

            Gtk.TextIter start_iter;
            text_buffer.get_iter_at_line (out start_iter, diag.start_line);

            var mark = new LspDiagnosticsMark.from_lsp_diagnostic (diag);
            text_buffer.add_mark (mark, start_iter); //
        }

        text_buffer.end_user_action ();
    }

    public void select_and_scroll (int line, int start_col, int end_col, bool is_new) {
        source_view.select_and_scroll (line, start_col, end_col, is_new);
    }
}
```

File: src/window.vala

```
/* window.vala
 *
 * Copyright 2026 kai
 *
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 * it under the terms of the GNU General Public License as published by
 * the Free Software Foundation, either version 3 of the License, or
 * (at your option) any later version.
 *
 * This program is distributed in the hope that it will be useful,
 * but WITHOUT ANY WARRANTY; without even the implied warranty of
```

```
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 * GNU General Public License for more details.
 *
 * You should have received a copy of the GNU General Public License
 * along with this program.  If not, see <https://www.gnu.org/licenses/>.
 *
 * SPDX-License-Identifier: GPL-3.0-or-later
 */

using Gtk;
using Adw;
using Panel;

public class Iide.Window : Panel.DocumentWorkspace {

    private Iide.DocumentManager document_manager;
    private Iide.ProjectManager project_manager;
    private Iide.SettingsService settings;

    public Window (Gtk.Application app) {
        Object (application: app);
        GtkSource.init ();
    }

    construct {
        settings = Iide.SettingsService.get_instance ();
        document_manager = new Iide.DocumentManager (this);
        project_manager = Iide.ProjectManager.get_instance ();
        document_manager.document_opened.connect ((widget) => {
            grid.add (widget);
            widget.raise ();
            widget.view_grab_focus ();
        });

        project_manager.project_opened.connect ((project_root) => {
            document_manager.set_workspace_root (project_root.get_uri ());
        });

        project_manager.project_closed.connect (() => {
            document_manager.set_workspace_root (null);
        });

        // Header
        var header = new Adw.HeaderBar ();
        var menu_button = new Gtk.MenuButton ();
        menu_button.icon_name = "open-menu-symbolic";
```

```
var menu = new GLib.Menu ();
menu.append (_("Open Project"), "app.open_project");
menu.append (_("Quick Open"), "app.fuzzy_finder");
menu.append (_("Search in Files"), "app.search_in_files");
menu.append (_("Save All"), "app.save");
menu.append (_("Preferences"), "app.preferences");
menu.append (_("About"), "app.about");
menu.append (_("Quit"), "app.quit");
menu_button.set_menu_model (menu);

header.pack_end (menu_button);

var panel_layout = settings.panel_layout;
if (panel_layout != null && panel_layout != "") {
    Iide.PanelLayoutHelper.deserialize_dock (panel_layout, dock);
} else {
    dock.reveal_start = settings.reveal_start_panel;
    dock.start_width = settings.panel_start_width;
    dock.reveal_end = settings.reveal_end_panel;
    dock.end_width = settings.panel_end_width;
    dock.reveal_bottom = settings.reveal_bottom_panel;
    dock.bottom_height = settings.panel_bottom_height;
}
var start_toggle_btn = new Panel.ToggleButton (dock, Panel.Area.START);
header.pack_start (start_toggle_btn);

var end_toggle_btn = new Panel.ToggleButton (dock, Panel.Area.END);
header.pack_end (end_toggle_btn);

set_titlebar (header);

// Theme switcher
var style_manager = Adw.StyleManager.get_default ();
style_manager.color_scheme = settings.color_scheme.to_adw_color_scheme ();

var theme_list = new Gtk.StringList ({ "System", "Light", "Dark" });
var expr = new Gtk.PropertyExpression (typeof (Gtk.StringObject), null, "string");
var theme_dropdown = new Gtk.DropDown (theme_list, expr) {
    selected = (uint) settings.color_scheme,
    tooltip_text = _("Color Scheme"),
    show_arrow = false
};

var factory = new Gtk.SignalListItemFactory ();
factory.setup.connect ((item) => {
```

```
    var list_item = item as Gtk.ListItem;
    var box = new Gtk.Box (Gtk.Orientation.HORIZONTAL, 6);
    var icon = new Gtk.Image ();
    var label = new Gtk.Label (null);
    label.xalign = 0;
    box.append (icon);
    box.append (label);
    list_item.set_child (box);
});
factory.bind.connect ((item) => {
    var list_item = item as Gtk.ListItem;
    var box = list_item.get_child () as Gtk.Box;
    var icon = box.get_first_child () as Gtk.Image;
    var label = icon.get_next_sibling () as Gtk.Label;
    var obj = list_item.get_item () as Gtk.StringObject;
    var text = obj.get_string ();
    label.set_label (text);

    string icon_name;
    switch (text) {
        case "System":
            icon_name = "weather-overcast-symbolic";
            break;
        case "Light":
            icon_name = "weather-clear-symbolic";
            break;
        case "Dark":
            icon_name = "weather-clear-night-symbolic";
            break;
        default:
            icon_name = "image-missing-symbolic";
            break;
    }
    icon.icon_name = icon_name;
});
theme_dropdown.set_factory (factory);

theme_dropdown.notify["selected"].connect (() => {
    var scheme = (ColorScheme) theme_dropdown.selected;
    settings.color_scheme = scheme;
    style_manager.color_scheme = scheme.to_adw_color_scheme ();
});
header.pack_end (theme_dropdown);

// statusbar (                  layout)
```

```
var panel_area_left = new Panel.Position ();
panel_area_left.area = Panel.Area.START;

var panel_widget_left1 = new Panel.Widget ();
panel_widget_left1.title = "Project Tree";
panel_widget_left1.icon_name = "folder-symbolic";

//      FileTreeView
var folder_view = new Iide.FileTreeView (null);
panel_widget_left1.child = folder_view;

//
project_manager.project_opened.connect ((project_root) => {
    folder_view.set_root_file (project_root);
});

project_manager.project_closed.connect (() => {
    folder_view.set_root_file (null);
});
panel_widget_left1.can_maximize = true;

var panel_widget_left2 = new Panel.Widget ();
panel_widget_left2.title = "LEFT 2";
panel_widget_left2.icon_name = "folder-symbolic";
panel_widget_left2.child = new Gtk.Label ("LEFT 2");
panel_widget_left2.can_maximize = true;

var panel_area_bottom = new Panel.Position ();
panel_area_bottom.area = Panel.Area.BOTTOM;

var panel_widget_bottom = new Panel.Widget ();
panel_widget_bottom.title = "Terminal";
panel_widget_bottom.icon_name = "utilities-terminal-symbolic";
panel_widget_bottom.child = new Iide.Terminal ();
panel_widget_bottom.can_maximize = true;

var panel_area_logs = new Panel.Position ();
panel_area_logs.area = Panel.Area.BOTTOM;

var panel_widget_logs = new Panel.Widget ();
panel_widget_logs.title = "Logs";
panel_widget_logs.icon_name = "document-properties-symbolic";
panel_widget_logs.child = new Iide.LogView ();
panel_widget_logs.can_maximize = true;

var panel_area_right = new Panel.Position ();
```

```
panel_area_right.area = Panel.Area.END;

var panel_widget_right = new Panel.Widget ();
panel_widget_right.title = "RIGHT";
panel_widget_right.icon_name = "folder-symbolic";
panel_widget_right.child = new Gtk.Label ("RIGHT");
panel_widget_right.can_maximize = true;

//                          layout
var dock_layout = settings.panel_layout;
if (dock_layout != null && dock_layout != "") {
    restore_dock_widgets (dock_layout,
                          panel_widget_left1, panel_widget_left2,
                          panel_widget_right, panel_widget_bottom, panel_widget_logs
} else {
    add_widget (panel_widget_left1, panel_area_left);
    add_widget (panel_widget_left2, panel_area_left);
    add_widget (panel_widget_right, panel_area_right);
    add_widget (panel_widget_bottom, panel_area_bottom);
    add_widget (panel_widget_logs, panel_area_logs);
}

//      toggle button    BOTTOM             layout
var bottom_toggle_btn = new Panel.ToggleButton (dock, Panel.Area.BOTTOM);
statusbar.add_suffix (1, bottom_toggle_btn);

Timeout.add (100, () => {
    var last_project_path = settings.current_project_path;

    var grid_data = settings.grid_layout;
    bool has_grid_docs = grid_data != null && grid_data != "";

    var open_docs = settings.open_documents;
    bool has_open_docs = open_docs.size > 0;

    if (last_project_path != null && last_project_path != "") {
        bool project_opened_handled = false;

        project_manager.project_opened.connect (() => {
            if (!project_opened_handled) {
                project_opened_handled = true;
                if (has_grid_docs) {
                    restore_grid_documents (grid_data);
                } else if (has_open_docs) {
                    foreach (var uri in open_docs) {
                        document_manager.open_document_by_uri (uri, this);
```

```
                    }
                }
            }
        });

        project_manager.open_project_by_path (last_project_path);

        Timeout.add (500, () => {
            if (!project_opened_handled) {
                project_opened_handled = true;
                if (has_grid_docs) {
                    restore_grid_documents (grid_data);
                } else if (has_open_docs) {
                    foreach (var uri in open_docs) {
                        document_manager.open_document_by_uri (uri, this);
                    }
                }
            }
            return Source.REMOVE;
        });
    } else {
        if (has_grid_docs) {
            restore_grid_documents (grid_data);
        } else if (has_open_docs) {
            foreach (var uri in open_docs) {
                document_manager.open_document_by_uri (uri, this);
            }
        }
    }

    return Source.REMOVE;
});

// Handle file activation to open documents
folder_view.file_activated.connect ((item) => {
    if (!item.is_directory) {
        document_manager.open_document (item.file, null);
    }
});

// Handle window close
this.close_request.connect (() => {
    save_window_settings ();
    settings.open_documents = document_manager.get_open_document_uris ();
    bool has_unsaved = false;
    foreach (var entry in document_manager.documents.entries) {
```

```
            if (entry.value is Iide.TextView) {
                var tv = (Iide.TextView) entry.value;
                if (tv.is_modified) {
                    has_unsaved = true;
                    break;
                }
            }
        }
        if (has_unsaved) {
            var dialog = new Adw.AlertDialog (_("Unsaved Changes"), _("You have unsaved
            dialog.add_response ("cancel", _("Cancel"));
            dialog.add_response ("discard", _("Discard"));
            dialog.add_response ("save", _("Save"));
            dialog.set_response_appearance ("save", Adw.ResponseAppearance.SUGGESTED);
            dialog.response.connect ((response) => {
                if (response == "save") {
                    foreach (var entry in document_manager.documents.entries) {
                        if (entry.value is Iide.TextView) {
                            var tv = (Iide.TextView) entry.value;
                            if (tv.is_modified) {
                                tv.save ();
                            }
                        }
                    }
                    settings.open_documents = document_manager.get_open_document_uris ()
                    this.destroy ();
                } else if (response == "discard") {
                    settings.open_documents = document_manager.get_open_document_uris ()
                    this.destroy ();
                }
            });
            dialog.present (this);
            return true;
        }
        return false;
    });
}

private void save_window_settings () {
    settings.panel_layout = Iide.PanelLayoutHelper.serialize_dock (dock);
    var grid_json = Iide.PanelLayoutHelper.serialize_grid (grid);
    settings.grid_layout = grid_json;

    bool maximized = false;
    var surface = this.get_surface ();
    if (surface != null) {
```

```
        var toplevel = surface as Gdk.Toplevel;
        if (toplevel != null) {
            var state = toplevel.get_state ();
            maximized = (state & Gdk.ToplevelState.MAXIMIZED) != 0;
        }
    }

    if (!maximized) {
        settings.window_width = (int) this.get_width ();
        settings.window_height = (int) this.get_height ();
    }
    settings.window_maximized = maximized;
}

private void restore_dock_widgets (string layout_data,
                                   Panel.Widget widget_left1,
                                   Panel.Widget widget_left2,
                                   Panel.Widget widget_right,
                                   Panel.Widget widget_bottom,
                                   Panel.Widget widget_logs) {
    var widgets = Iide.PanelLayoutHelper.parse_widgets (layout_data);

    Gee.HashMap<string, Panel.Widget> widget_map = new Gee.HashMap<string, Panel.Widget>
    widget_map.set ("Project Tree", widget_left1);
    widget_map.set ("LEFT 2", widget_left2);
    widget_map.set ("RIGHT", widget_right);
    widget_map.set ("Terminal", widget_bottom);
    widget_map.set ("Logs", widget_logs);
    widget_map.set ("BOTTOM", widget_bottom);

    if (widgets.size == 0) {
        var pos_left = new Panel.Position ();
        pos_left.area = Panel.Area.START;
        add_widget (widget_left1, pos_left);
        add_widget (widget_left2, pos_left);

        var pos_right = new Panel.Position ();
        pos_right.area = Panel.Area.END;
        add_widget (widget_right, pos_right);

        var pos_bottom = new Panel.Position ();
        pos_bottom.area = Panel.Area.BOTTOM;
        add_widget (widget_bottom, pos_bottom);

        var pos_logs = new Panel.Position ();
        pos_logs.area = Panel.Area.BOTTOM;
```

```
        add_widget (widget_logs, pos_logs);
        return;
    }

    foreach (var info in widgets) {
        var widget = widget_map.get (info.title);
        if (widget == null) {
            continue;
        }

        var pos = new Panel.Position ();
        pos.area = (Panel.Area) info.area;
        pos.column = info.column;
        pos.row = info.row;
        pos.depth = info.depth;

        add_widget (widget, pos);
    }
}

private async void restore_grid_documents_async (Gee.ArrayList<Iide.PanelLayoutHelper.Do
    uint last_col = 0;
    foreach (var doc_info in sorted_docs) {
        if (doc_info.column > last_col) {
            last_col = doc_info.column;
        }
    }

    for (uint i = 1; i <= last_col; i++) {
        grid.insert_column (i);
    }

    foreach (var doc_info in sorted_docs) {
        var file = GLib.File.new_for_uri (doc_info.uri);
        var pos = new Panel.Position ();
        pos.area = Panel.Area.CENTER;
        pos.column = doc_info.column;
        pos.row = doc_info.row;
        document_manager.open_document (file, pos);
    }
}

private void restore_grid_documents (string grid_data) {
    var docs = Iide.PanelLayoutHelper.parse_grid_documents (grid_data);
    if (docs.size == 0) {
        return;
```

```
        }

        var sorted_docs = new Gee.ArrayList<Iide.PanelLayoutHelper.DocumentInfo> ();
        foreach (var doc in docs) {
            sorted_docs.add (doc);
        }
        sorted_docs.sort ((a, b) => {
            int col_cmp = (int) (a.column - b.column);
            if (col_cmp != 0)return col_cmp;
            return (int) (a.row - b.row);
        });

        restore_grid_documents_async.begin (sorted_docs);
    }

    public void save_modified () {
        foreach (var uri in document_manager.documents.keys) {
            var widget = document_manager.documents[uri];
            if (widget is Iide.TextView) {
                var tv = widget as Iide.TextView;
                if (tv.is_modified) {
                    tv.save ();
                }
            }
        }
    }

    public void open_project_dialog () {
        project_manager.open_project_dialog.begin (this);
    }

    public Iide.DocumentManager get_document_manager () {
        return document_manager;
    }
}
```

File: src/meson.build

```
iide_sources = [
    'main.vala',
    'application.vala',
    'window.vala',
    'Widgets/TextView/TextView.vala',
    'Widgets/TextView/SourceView.vala',
    'Widgets/TextView/FontZoomer.vala',
    'Widgets/TextView/GutterMarkRenderer.vala',
```

```
    'Widgets/TextView/LspCompletionProvider.vala',
    'Widgets/ToolViews/ProjectView.vala',
    'Widgets/ToolViews/TerminalView.vala',
    'Widgets/ToolViews/LogView.vala',
    'Widgets/PreferencesDialog.vala',
    'Widgets/FuzzyFinderDialog.vala',
    'Widgets/SearchInFilesDialog.vala',
    'Services/Utils.vala',
    'Services/LoggerService.vala',
    'Services/DocumentManager.vala',
    'Services/IconProvider.vala',
    'Services/ProjectManager.vala',
    'Services/SettingsService.vala',
    'Services/PanelLayoutHelper.vala',
    'Services/LSP/LspClient.vala',
    'Services/LSP/IdeLspClient.vala',
    'Services/LSP/IdeLspService.vala',
    'Services/LSP/IdeLspManager.vala',
    'Services/LSP/LspDiagnosticsMark.vala',
    'Services/TreeSitter/TreeSitterManager.vala',
    'Services/TreeSitter/model/AstItem.vala',
    'Services/TreeSitter/model/AstModel.vala',
    'Services/TreeSitter/BaseTreeSitterHighlighter.vala',
    'Services/TreeSitter/python/PythonTreeSitterHighlighter.vala',
    'Services/TreeSitter/cpp/CppTreeSitterHighlighter.vala',
    'Services/TreeSitter/vala/ValaTreeSitterHighlighter.vala',
    'Services/Actions/Action.vala',
    'Services/Actions/ActionManager.vala',
    'Services/Actions/ShortcutSettings.vala',
]

iide_deps = [
    config_dep,
    dependency('gtk4'),
    dependency('gio-2.0', version: '>= 2.50'),
    dependency('libadwaita-1', version: '>= 1.4'),
    dependency('libpanel-1', version: '>= 1.7'),
    dependency('gtksourceview-5'),
    dependency('vte-2.91-gtk4', version: '>= 0.75.0'),
    dependency('gee-0.8'),
    dependency('json-glib-1.0'),
    dependency('jsonrpc-glib-1.0'),
]

iide_sources += gnome.compile_resources('iide-resources', 'iide.gresource.xml', c_name: 'iid
```

```
executable(
    'iide',
    iide_sources,
    dependencies: [iide_deps, tree_sitter_dep],
    include_directories: config_inc,
    install: true,
)
```